

# The European Cycling Risk Index: Methodology, Validation, and a City-Level Analysis of 219 Cities

Defensive Pedal Research

*Companion to Severity Without Frequency, Tiers, Not Ranks, and the public release Streets, Not Reputations. This report is the extensive record: every methodological decision, the full validation programme, and a data-generated profile of every indexed city.*

## ABSTRACT

We rank 219 European cities across 30 countries on the cycling risk of their street networks, using the per-road risk model that serves a production bicycle-routing service. Every public road in the coverage area — 67.3 million scored segments — contributes to three per-city dimensions: the risk of the street network itself (weight 0.40), the risk of 210 standardised simulated journeys (0.30), and population-weighted low-stress accessibility (0.30). The underlying per-road score was validated against 3.98 billion measured bicycle passages and 1,169 serious injuries across 13 European regions; at the city level the index predicts measured serious-injury rates (rate ratio 0.79 [0.71–0.89] per +10 index points,  $p < 0.001$ ) and correlates with bicycle commuting share at  $\rho = 0.500$  ( $n = 156$ ), meeting the pre-registered validation target. Ranks are published with 90% rank intervals from 2,000 method respecifications; 191 of 219 cities are confidently placed in one of three tiers. The three dimensions correlate (Spearman S–R 0.50, S–A -0.82, R–A -0.74) yet diverge informatively for individual cities; we analyse each dimension separately and profile every city.

## 1. Introduction

City cycling rankings are, almost without exception, composites of infrastructure inventories, usage statistics, and policy assessment — the Copenhagenize Index being the best-known example. Such composites answer the question “*which cities are doing well at cycling?*”. This index answers a narrower and more falsifiable one: “*how dangerous is the street network a rider actually encounters?*”. It uses no votes, no policy points, and no usage statistics as inputs; every number derives from the streets as mapped and modelled. The narrowing is deliberate:

it makes the index checkable against injury outcomes (Section 3), immune to advocacy, and sensitive only to physical change in the network.

The frame is every European city with a modelled motor-traffic layer: the union of the 117-city baseline published earlier and the Eurostat city-population top 200 within the 30 covered countries — 219 cities after removing four agglomeration artifacts and two Canary Islands cities that lie outside the map-data coverage entirely. Twenty cities are Romanian, a legacy of the pilot, and the frame therefore over-represents that country's higher-risk networks at the tail.

## 2. The street-level risk model

The unit of measurement is the individual road segment. Each of the 67.3 million segments receives a risk score assembled from factor families read from OpenStreetMap and from model-side overlays: legal and signed speed (including each country's statutory urban default where no sign is mapped), modelled motor-traffic volume from a capacity-constrained assignment model calibrated to observed congestion indices, lane count and heavy-vehicle access, the presence and grade of separation (track, lane, none; with mandatory-sidepath law respected), street class and function, lighting and surface, and junction/crossing context. Exact factor weights are proprietary to the routing product; consistent with the public product's disclosure policy, this report names factors and directions but not magnitudes. Two properties matter for what follows: the model is *identical in every city*, and it is the *production model* — the same scores plan real routes daily, so the index measures the network the service actually navigates.

Two model-side constructions deserve note. First, the traffic layer is a model, not a measurement: it reproduces congestion-calibrated demand over the road hierarchy, and validation against national count data shows it behaves as a road-class proxy rather than a volume gauge. It is retained because it is the largest outcome-supported component (Section 3) and because it is computed identically everywhere. Second, scores are exported once per model generation; this index was built from a single export whose only difference from the published generation is the extension of the traffic layer to the new cities — verified by reproducing the previous 117 cities' data byte-identically (112 of 117 exact; the remaining five differ only in traffic terms where a newly modelled neighbouring city shares border streets).

## 3. Validation of the risk algorithm

### 3.1 DESIGN

The validation programme avoids the two classic failure modes of crash-based city comparison: crash-per-capita metrics (confounded by exposure via the safety-in-numbers effect) and infrastructure inventories (unfalsifiable). Instead it joins permanent and campaign bicycle counters — which measure exposure directly — to geocoded crash registers, and asks whether the model's per-road score predicts *outcomes per passage*. The final panel comprises **2,034 counter sites in 13 regions of six countries, 3.98 billion measured passages, and 1,169 killed-or-seriously- injured (KSI) records.**

### 3.2 PRINCIPAL RESULTS

- **The score predicts severity, not frequency.** All-crash incidence is flat across the score ( $RR \approx 1.05$ , null), but serious injury per passage rises monotonically: KSI  $RR$  1.62 [1.28–2.06] above score 80, and riders on streets above 100 are seriously injured  $2.11\times$  [1.44–3.09] as often per passage as on streets below 30.
- **The signal survives road-class control on mid-block links** (KSI  $RR$  1.48 [1.02–2.13],  $p = 0.038$ ) but not at junctions, where road class explains the association — junction design is the model's largest known gap.
- **Factor-level audit.** Tested individually against 614 (later 1,035) KSI records: mandatory-sidepath carriageways ( $RR$  1.83), primary-road exposure (1.75), heavy-vehicle access (1.54), on-road stress (1.30) and motor-traffic volume (1.23) are confirmed with correct direction; surface smoothness and secondary-road exposure are well-powered nulls; the gradient family shows no detectable injury relationship (final  $RR$  1.10 [0.72–1.66]); one legacy factor (one-way bonus) was wrong-signed ( $RR$  1.80) and set to zero. Components the audit rejected are removed before city aggregation; production routing is unchanged.
- **Exposure-free replication.** In Munich (26,000 crashes) and Paris (BAAC), crash-type-disambiguated density places  $\sim 2\times$  more in-traffic cycling crashes on extreme-score roads (Munich  $1.95\times$ , Paris  $2.37\times$ ; junction crashes  $2.66\times/2.46\times$ ; facility falls  $0.39\times/0.42\times$ ) — replicated across two countries and coding systems.

### 3.3 CITY-LEVEL CRITERION

Aggregated to cities, 17 panel cities carry both an index value and a measured KSI rate per passage. The index predicts the rate:  $RR$  0.79 [0.71–0.89] per +10 index points ( $p < 0.001$ ); Spearman  $-0.32$ . With country fixed effects the association is null (1.00 [0.85–1.17]) — five countries of outcome data cannot separate a city from its national reporting and riding culture, and the panel's tier coverage is lopsided. This is reported as an inconclusive country-adjusted result, not as support for fine-grained rank claims.

### 3.4 CONVERGENT VALIDITY

Bicycle commuting share (Eurostat Urban Audit, 156 matched cities, some vintages dating to 2003) correlates with the index at  $\rho = 0.500$  — exactly the pre-registered target and on par with the best published city indices (Bike Score  $r = 0.52$ ; PeopleForBikes BNA  $\rho = 0.57$ ). Against the Copenhagenize Index 2025 — a deliberately different construct —  $\rho = 0.47$  over its 27 European top-30 cities, with 19 of 27 inside our top 50. In the two disagreements where independent injury data exists (Bordeaux, Zürich), the injury data sides with this index.

## 4. From roads to cities: the index

Three dimensions, weighted 0.40 / 0.30 / 0.30, each answering a distinct question a rider or an administration would ask. Weights follow the pilot's design (network first, experience and access equal); the respecification analysis below perturbs them.

### 4.1 S — STREET-NETWORK RISK (0.40)

*What it is:* the length-weighted mean comparable risk per street-km (0.6 of S) plus the share of street-km in the highest-risk band (0.4 of S). *Why it matters:* the mean is the exposure a rider cannot route around in daily use, and the tail is the component with the strongest validated severity claim — a city cannot bury a few lethal arterials inside a pleasant average. *Constructions:* streets with a parallel protected track (geometrically detected or legally implied by mandatory-sidepath tagging) are scored as the track — a road nobody may or would ride is not counted as ridden; roads above 80 km/h are excluded as bypass rather than urban fabric; city-specific manual overlays and outcome-rejected components are removed for comparability.

### 4.2 R — JOURNEY RISK (0.30)

*What it is:* 210 standardised trips per city — origins drawn from the population grid, destinations from amenity distributions — planned by the live production router and scored on the restricted per-road values. *Why it matters:* it measures the network as experienced, not as averaged: a city with a coherent low-risk sub-network scores far better here than its street average suggests (the "escape valve"), while a city whose calm streets do not connect cannot hide.

### 4.3 A — LOW-STRESS ACCESSIBILITY (0.30)

*What it is:* on a 1-km population grid, the share of daily destinations reachable over a *connected* low-stress network (0.7) plus the size of the largest connected low-stress component (0.3), averaged over three calibrated low-stress thresholds. *Why it matters:* risk that can be

avoided only by not going is not avoided; this dimension prices network connectivity, which the other two cannot see. Averaging across thresholds removes the cliff a single cut creates for cities whose street risk clusters near it.

#### 4.4 NORMALISATION, UNCERTAINTY, TIERS

Each raw input is normalised against goalposts at the 2.5th and 97.5th percentiles of the 219-city sample (clipping is reported, not hidden), the three dimensions are combined, and the entire method — weights, goalposts, aggregation — is respecified 2,000 times. The 90% rank interval accompanies every rank; the median city moves across 25 places, which is why tiers, not places, are the headline claim. 191 of 219 cities keep one tier in at least two-thirds of respecifications.

## 5. Results: the ranking

The full ranking, with per-dimension ranks. Overall rank is not a sufficient statistic — Section 6 shows why.

| #  | City             | Ctry | Index | Interval | S rank | R rank | A rank | Tier |
|----|------------------|------|-------|----------|--------|--------|--------|------|
| 1  | <i>Stockholm</i> | SE   | 98.0  | [1–7]    | 1      | 15     | 1      | 1    |
| 2  | <i>Helsinki</i>  | FI   | 97.0  | [1–9]    | 7      | 1      | 13     | 1    |
| 3  | <i>Antwerp</i>   | BE   | 96.4  | [2–13]   | 9      | 7      | 7      | 1    |
| 4  | <i>Haarlem</i>   | NL   | 96.3  | [1–11]   | 2      | 33     | 2      | 1    |
| 5  | <i>Leiden</i>    | NL   | 96.2  | [3–13]   | 3      | 37     | 3      | 1    |
| 6  | <i>Vantaa</i>    | FI   | 96.2  | [2–10]   | 13     | 2      | 10     | 1    |
| 7  | <i>The Hague</i> | NL   | 95.0  | [4–18]   | 6      | 23     | 8      | 1    |
| 8  | <i>Espoo</i>     | FI   | 94.7  | [3–14]   | 12     | 3      | 17     | 1    |
| 9  | <i>Amsterdam</i> | NL   | 94.2  | [5–17]   | 8      | 10     | 14     | 1    |
| 10 | <i>Barcelona</i> | ES   | 93.2  | [6–19]   | 20     | 21     | 4      | 1    |
| 11 | <i>Paris</i>     | FR   | 93.0  | [5–16]   | 16     | 25     | 5      | 1    |
| 12 | <i>Tampere</i>   | FI   | 92.5  | [4–17]   | 15     | 4      | 24     | 1    |
| 13 | <i>Eindhoven</i> | NL   | 92.4  | [9–23]   | 10     | 20     | 19     | 1    |
| 14 | <i>Utrecht</i>   | NL   | 91.8  | [11–22]  | 14     | 17     | 18     | 1    |
| 15 | <i>Brussels</i>  | BE   | 91.3  | [3–23]   | 4      | 74     | 6      | 1    |
| 16 | <i>Rotterdam</i> | NL   | 90.8  | [14–24]  | 21     | 19     | 16     | 1    |

| #  | City                   | Ctry | Index       | Interval | S rank | R rank | A rank | Tier |
|----|------------------------|------|-------------|----------|--------|--------|--------|------|
| 17 | <i>Copenhagen</i>      | DK   | <b>89.0</b> | [8–28]   | 5      | 35     | 41     | 1    |
| 18 | <i>San Sebastián</i>   | ES   | <b>88.8</b> | [17–29]  | 19     | 36     | 23     | 1    |
| 19 | <i>Turku</i>           | FI   | <b>88.4</b> | [6–24]   | 29     | 5      | 37     | 1    |
| 20 | <i>Hanover</i>         | DE   | <b>87.7</b> | [18–28]  | 25     | 26     | 26     | 1    |
| 21 | <i>Munich</i>          | DE   | <b>86.9</b> | [20–33]  | 22     | 41     | 25     | 1    |
| 22 | <i>Bremen</i>          | DE   | <b>86.4</b> | [18–36]  | 11     | 48     | 32     | 1    |
| 23 | <i>Ghent</i>           | BE   | <b>85.5</b> | [21–37]  | 17     | 45     | 33     | 1    |
| 24 | <i>Braunschweig</i>    | DE   | <b>84.1</b> | [23–36]  | 30     | 29     | 40     | 1    |
| 25 | <i>Bilbao</i>          | ES   | <b>83.8</b> | [13–44]  | 23     | 110    | 11     | 1    |
| 26 | <i>Almere</i>          | NL   | <b>83.6</b> | [11–36]  | 45     | 9      | 43     | 1    |
| 27 | <i>Lyon</i>            | FR   | <b>83.6</b> | [13–38]  | 46     | 65     | 9      | 1    |
| 28 | <i>Lübeck</i>          | DE   | <b>83.3</b> | [24–40]  | 24     | 27     | 48     | 1    |
| 29 | <i>Groningen</i>       | NL   | <b>82.6</b> | [26–37]  | 50     | 30     | 31     | 1    |
| 30 | <i>Montpellier</i>     | FR   | <b>82.4</b> | [25–42]  | 44     | 55     | 22     | 1    |
| 31 | <i>Toulouse</i>        | FR   | <b>82.1</b> | [27–44]  | 27     | 64     | 36     | 1    |
| 32 | <i>Grenoble</i>        | FR   | <b>80.0</b> | [27–48]  | 51     | 72     | 21     | 1    |
| 33 | <i>Tilburg</i>         | NL   | <b>79.6</b> | [31–46]  | 77     | 32     | 28     | 1    |
| 34 | <i>Strasbourg</i>      | FR   | <b>79.1</b> | [22–55]  | 115    | 18     | 20     | 1    |
| 35 | <i>Nantes</i>          | FR   | <b>79.0</b> | [27–52]  | 18     | 82     | 52     | 1    |
| 36 | <i>Málaga</i>          | ES   | <b>78.6</b> | [30–50]  | 28     | 69     | 46     | 1    |
| 37 | <i>Münster</i>         | DE   | <b>78.5</b> | [34–49]  | 41     | 57     | 42     | 1    |
| 38 | <i>Malmö</i>           | SE   | <b>78.3</b> | [12–57]  | 113    | 8      | 34     | 1    |
| 39 | <i>Seville</i>         | ES   | <b>78.0</b> | [24–55]  | 109    | 16     | 29     | 1    |
| 40 | <i>Bologna</i>         | IT   | <b>78.0</b> | [18–49]  | 71     | 13     | 50     | 1    |
| 41 | <i>Valencia</i>        | ES   | <b>77.8</b> | [15–67]  | 137    | 28     | 12     | 1    |
| 42 | <i>Tours</i>           | FR   | <b>76.7</b> | [32–65]  | 31     | 121    | 35     | 1    |
| 43 | <i>Valladolid</i>      | ES   | <b>76.4</b> | [37–60]  | 102    | 42     | 30     | 1    |
| 44 | <i>Freiburg</i>        | DE   | <b>76.1</b> | [37–58]  | 38     | 83     | 45     | 1    |
| 45 | <i>Vitoria-Gasteiz</i> | ES   | <b>75.7</b> | [39–56]  | 82     | 34     | 44     | 1    |
| 46 | <i>Sabadell</i>        | ES   | <b>75.5</b> | [24–69]  | 64     | 117    | 15     | 1    |
| 47 | <i>Zaragoza</i>        | ES   | <b>75.0</b> | [44–56]  | 61     | 58     | 49     | 1    |

| #  | City                 | Ctry | Index       | Interval | S rank     | R rank     | A rank     | Tier     |
|----|----------------------|------|-------------|----------|------------|------------|------------|----------|
| 48 | <i>Duisburg</i>      | DE   | <b>74.9</b> | [38–62]  | <b>34</b>  | <b>92</b>  | <b>55</b>  | <b>1</b> |
| 49 | <i>Vienna</i>        | AT   | <b>74.0</b> | [36–77]  | <b>129</b> | <b>22</b>  | <b>39</b>  | <b>1</b> |
| 50 | <i>Palma</i>         | ES   | <b>73.8</b> | [48–59]  | <b>63</b>  | <b>52</b>  | <b>54</b>  | <b>1</b> |
| 51 | <i>Berlin</i>        | DE   | <b>73.6</b> | [37–63]  | <b>32</b>  | <b>78</b>  | <b>67</b>  | <b>1</b> |
| 52 | <i>Mannheim</i>      | DE   | <b>73.6</b> | [40–73]  | <b>94</b>  | <b>86</b>  | <b>27</b>  | <b>1</b> |
| 53 | <i>Pamplona</i>      | ES   | <b>73.4</b> | [45–72]  | <b>100</b> | <b>60</b>  | <b>38</b>  | <b>1</b> |
| 54 | <i>Oulu</i>          | FI   | <b>71.8</b> | [10–81]  | <b>128</b> | <b>6</b>   | <b>63</b>  | <b>1</b> |
| 55 | <i>Córdoba</i>       | ES   | <b>71.7</b> | [44–70]  | <b>42</b>  | <b>104</b> | <b>61</b>  | <b>1</b> |
| 56 | <i>Lille</i>         | FR   | <b>71.1</b> | [51–79]  | <b>58</b>  | <b>113</b> | <b>47</b>  | <b>1</b> |
| 57 | <i>Bonn</i>          | DE   | <b>70.5</b> | [43–87]  | <b>36</b>  | <b>135</b> | <b>58</b>  | <b>1</b> |
| 58 | <i>Düsseldorf</i>    | DE   | <b>70.5</b> | [55–73]  | <b>60</b>  | <b>95</b>  | <b>59</b>  | <b>1</b> |
| 59 | <i>Magdeburg</i>     | DE   | <b>70.2</b> | [55–77]  | <b>86</b>  | <b>67</b>  | <b>57</b>  | <b>1</b> |
| 60 | <i>Maribor</i>       | SI   | <b>70.1</b> | [47–73]  | <b>52</b>  | <b>47</b>  | <b>100</b> | <b>1</b> |
| 61 | <i>Trondheim</i>     | NO   | <b>70.1</b> | [46–73]  | <b>65</b>  | <b>40</b>  | <b>94</b>  | <b>1</b> |
| 62 | <i>Oslo</i>          | NO   | <b>69.9</b> | [40–81]  | <b>33</b>  | <b>54</b>  | <b>125</b> | <b>1</b> |
| 63 | <i>Graz</i>          | AT   | <b>69.5</b> | [53–80]  | <b>92</b>  | <b>44</b>  | <b>69</b>  | <b>1</b> |
| 64 | <i>Augsburg</i>      | DE   | <b>69.4</b> | [54–74]  | <b>53</b>  | <b>88</b>  | <b>73</b>  | <b>1</b> |
| 65 | <i>Nancy</i>         | FR   | <b>69.0</b> | [38–108] | <b>26</b>  | <b>170</b> | <b>53</b>  | <b>1</b> |
| 66 | <i>Odense</i>        | DK   | <b>68.6</b> | [23–89]  | <b>122</b> | <b>14</b>  | <b>78</b>  | <b>1</b> |
| 67 | <i>Kiel</i>          | DE   | <b>68.3</b> | [62–82]  | <b>72</b>  | <b>96</b>  | <b>62</b>  | <b>1</b> |
| 68 | <i>Szeged</i>        | HU   | <b>68.3</b> | [63–75]  | <b>66</b>  | <b>76</b>  | <b>75</b>  | <b>1</b> |
| 69 | <i>Elche</i>         | ES   | <b>68.2</b> | [65–76]  | <b>68</b>  | <b>79</b>  | <b>74</b>  | <b>1</b> |
| 70 | <i>Gijón</i>         | ES   | <b>67.9</b> | [48–84]  | <b>40</b>  | <b>99</b>  | <b>96</b>  | <b>1</b> |
| 71 | <i>Cologne</i>       | DE   | <b>67.4</b> | [61–91]  | <b>91</b>  | <b>106</b> | <b>56</b>  | <b>1</b> |
| 72 | <i>Aarhus</i>        | DK   | <b>67.2</b> | [43–90]  | <b>101</b> | <b>24</b>  | <b>108</b> | <b>1</b> |
| 73 | <i>Murcia</i>        | ES   | <b>67.0</b> | [61–85]  | <b>56</b>  | <b>98</b>  | <b>89</b>  | <b>1</b> |
| 74 | <i>Oberhausen</i>    | DE   | <b>66.3</b> | [52–103] | <b>39</b>  | <b>141</b> | <b>80</b>  | <b>2</b> |
| 75 | <i>Rennes</i>        | FR   | <b>66.1</b> | [61–100] | <b>49</b>  | <b>127</b> | <b>82</b>  | <b>2</b> |
| 76 | <i>A Coruña</i>      | ES   | <b>65.7</b> | [61–101] | <b>116</b> | <b>105</b> | <b>51</b>  | <b>2</b> |
| 77 | <i>Białystok</i>     | PL   | <b>65.3</b> | [61–102] | <b>133</b> | <b>46</b>  | <b>71</b>  | <b>2</b> |
| 78 | <i>Halle (Saale)</i> | DE   | <b>65.1</b> | [70–94]  | <b>59</b>  | <b>107</b> | <b>101</b> | <b>2</b> |

| #   | City                        | Ctry | Index       | Interval | S rank | R rank | A rank | Tier |
|-----|-----------------------------|------|-------------|----------|--------|--------|--------|------|
| 79  | <i>Alicante</i>             | ES   | <b>65.0</b> | [76–89]  | 89     | 80     | 85     | 2    |
| 80  | <i>Wrocław</i>              | PL   | <b>65.0</b> | [66–99]  | 121    | 61     | 70     | 2    |
| 81  | <i>Luxembourg</i>           | LU   | <b>64.1</b> | [73–99]  | 112    | 87     | 65     | 2    |
| 82  | <i>Ljubljana</i>            | SI   | <b>63.9</b> | [76–96]  | 69     | 91     | 109    | 2    |
| 83  | <i>Karlsruhe</i>            | DE   | <b>63.7</b> | [76–106] | 75     | 126    | 72     | 2    |
| 84  | <i>Hamburg</i>              | DE   | <b>63.1</b> | [68–112] | 48     | 138    | 97     | 2    |
| 85  | <i>Uppsala</i>              | SE   | <b>62.5</b> | [18–131] | 172    | 11     | 83     | 2    |
| 86  | <i>Bergen</i>               | NO   | <b>62.4</b> | [70–113] | 55     | 97     | 133    | 2    |
| 87  | <i>Gdańsk</i>               | PL   | <b>61.4</b> | [88–103] | 95     | 100    | 106    | 2    |
| 88  | <i>Stavanger</i>            | NO   | <b>61.3</b> | [75–117] | 87     | 62     | 136    | 2    |
| 89  | <i>Krefeld</i>              | DE   | <b>61.1</b> | [81–118] | 104    | 136    | 64     | 2    |
| 90  | <i>Vigo</i>                 | ES   | <b>61.0</b> | [76–127] | 54     | 156    | 93     | 2    |
| 91  | <i>Almería</i>              | ES   | <b>60.8</b> | [91–108] | 93     | 111    | 102    | 2    |
| 92  | <i>Debrecen</i>             | HU   | <b>60.5</b> | [71–119] | 132    | 49     | 116    | 2    |
| 93  | <i>Łódź</i>                 | PL   | <b>60.4</b> | [88–113] | 108    | 73     | 117    | 2    |
| 94  | <i>Leipzig</i>              | DE   | <b>60.2</b> | [81–123] | 62     | 112    | 137    | 2    |
| 95  | <i>Rostock</i>              | DE   | <b>60.0</b> | [80–123] | 57     | 142    | 124    | 2    |
| 96  | <i>Toruń</i>                | PL   | <b>59.9</b> | [85–127] | 151    | 77     | 77     | 2    |
| 97  | <i>Angers</i>               | FR   | <b>59.9</b> | [73–133] | 47     | 160    | 110    | 2    |
| 98  | <i>Mainz</i>                | DE   | <b>59.7</b> | [88–127] | 97     | 144    | 76     | 2    |
| 99  | <i>Nuremberg</i>            | DE   | <b>59.7</b> | [95–116] | 103    | 122    | 90     | 2    |
| 100 | <i>Gelsenkirchen</i>        | DE   | <b>59.6</b> | [93–121] | 84     | 139    | 92     | 2    |
| 101 | <i>Terrassa</i>             | ES   | <b>59.6</b> | [85–126] | 124    | 129    | 68     | 2    |
| 102 | <i>Jerez de la Frontera</i> | ES   | <b>59.3</b> | [97–117] | 99     | 102    | 119    | 2    |
| 103 | <i>Tallinn</i>              | EE   | <b>59.3</b> | [53–130] | 136    | 31     | 145    | 2    |
| 104 | <i>Budapest</i>             | HU   | <b>59.1</b> | [78–124] | 138    | 53     | 121    | 2    |
| 105 | <i>Reykjavik</i>            | IS   | <b>59.0</b> | [89–129] | 150    | 90     | 81     | 2    |
| 106 | <i>Orléans</i>              | FR   | <b>58.9</b> | [98–124] | 105    | 134    | 88     | 2    |
| 107 | <i>Alkmaar</i>              | NL   | <b>58.8</b> | [80–140] | 164    | 75     | 66     | 2    |
| 108 | <i>Bern</i>                 | CH   | <b>58.7</b> | [67–144] | 43     | 181    | 98     | 2    |
| 109 | <i>Madrid</i>               | ES   | <b>58.6</b> | [82–127] | 139    | 59     | 127    | 2    |

| #   | City                    | Ctry | Index | Interval  | S rank | R rank | A rank | Tier |
|-----|-------------------------|------|-------|-----------|--------|--------|--------|------|
| 110 | <i>Clermont-Ferrand</i> | FR   | 58.6  | [93–128]  | 125    | 124    | 79     | 2    |
| 111 | <i>Warsaw</i>           | PL   | 58.2  | [67–137]  | 159    | 43     | 115    | 2    |
| 112 | <i>Geneva</i>           | CH   | 58.1  | [93–126]  | 74     | 133    | 128    | 2    |
| 113 | <i>Linz</i>             | AT   | 58.0  | [84–141]  | 163    | 70     | 84     | 2    |
| 114 | <i>Valletta</i>         | MT   | 57.9  | [65–151]  | 37     | 186    | 103    | 2    |
| 115 | <i>Granada</i>          | ES   | 57.9  | [90–140]  | 73     | 166    | 91     | 2    |
| 116 | <i>Bordeaux</i>         | FR   | 57.6  | [100–132] | 83     | 146    | 112    | 2    |
| 117 | <i>Venice</i>           | IT   | 57.2  | [62–142]  | 158    | 39     | 130    | 2    |
| 118 | <i>Aachen</i>           | DE   | 56.9  | [89–145]  | 67     | 171    | 105    | 2    |
| 119 | <i>Gdynia</i>           | PL   | 56.5  | [114–130] | 111    | 130    | 111    | 2    |
| 120 | <i>Florence</i>         | IT   | 56.1  | [88–137]  | 144    | 56     | 142    | 2    |
| 121 | <i>Poznań</i>           | PL   | 56.1  | [106–141] | 153    | 103    | 95     | 2    |
| 122 | <i>Frankfurt</i>        | DE   | 56.1  | [114–135] | 126    | 131    | 99     | 2    |
| 123 | <i>Liège</i>            | BE   | 56.0  | [91–147]  | 165    | 63     | 107    | 2    |
| 124 | <i>Oviedo</i>           | ES   | 55.9  | [66–157]  | 35     | 188    | 122    | 2    |
| 125 | <i>Mönchengladbach</i>  | DE   | 55.8  | [106–142] | 134    | 145    | 86     | 2    |
| 126 | <i>Lublin</i>           | PL   | 55.2  | [95–141]  | 143    | 66     | 146    | 2    |
| 127 | <i>Dortmund</i>         | DE   | 55.1  | [103–138] | 79     | 140    | 143    | 2    |
| 128 | <i>Gothenburg</i>       | SE   | 55.0  | [96–150]  | 166    | 68     | 114    | 2    |
| 129 | <i>Kraków</i>           | PL   | 54.8  | [114–136] | 106    | 114    | 144    | 2    |
| 130 | <i>Rouen</i>            | FR   | 54.5  | [90–159]  | 135    | 178    | 60     | 2    |
| 131 | <i>Dublin</i>           | IE   | 54.0  | [107–143] | 81     | 137    | 153    | 2    |
| 132 | <i>Bielefeld</i>        | DE   | 53.6  | [112–145] | 90     | 149    | 138    | 2    |
| 133 | <i>Stuttgart</i>        | DE   | 53.5  | [113–152] | 96     | 169    | 113    | 2    |
| 134 | <i>Bratislava</i>       | SK   | 53.0  | [119–143] | 142    | 108    | 141    | 2    |
| 135 | <i>Zurich</i>           | CH   | 52.3  | [120–153] | 98     | 165    | 129    | 2    |
| 136 | <i>Kassel</i>           | DE   | 51.5  | [109–168] | 85     | 189    | 104    | 2    |
| 137 | <i>Turin</i>            | IT   | 51.4  | [24–167]  | 194    | 12     | 118    | 2    |
| 138 | <i>Rzeszów</i>          | PL   | 51.0  | [135–150] | 148    | 125    | 131    | 2    |
| 139 | <i>Toulon</i>           | FR   | 51.0  | [118–155] | 88     | 159    | 151    | 2    |
| 140 | <i>Riga</i>             | LV   | 51.0  | [120–150] | 140    | 101    | 162    | 2    |

| #   | City                     | Ctry | Index | Interval  | S rank | R rank | A rank | Tier |
|-----|--------------------------|------|-------|-----------|--------|--------|--------|------|
| 141 | <i>Parma</i>             | IT   | 51.0  | [67–166]  | 186    | 38     | 120    | 2    |
| 142 | <i>Bydgoszcz</i>         | PL   | 51.0  | [120–155] | 160    | 94     | 135    | 2    |
| 143 | <i>Bergamo</i>           | IT   | 50.8  | [111–153] | 147    | 85     | 163    | 2    |
| 144 | <i>Dresden</i>           | DE   | 50.7  | [113–160] | 78     | 173    | 148    | 2    |
| 145 | <i>Reims</i>             | FR   | 50.1  | [133–160] | 120    | 167    | 126    | 2    |
| 146 | <i>Essen</i>             | DE   | 50.1  | [133–157] | 110    | 158    | 139    | 2    |
| 147 | <i>Basel</i>             | CH   | 50.0  | [125–164] | 155    | 168    | 87     | 3    |
| 148 | <i>Szczecin</i>          | PL   | 48.9  | [137–159] | 162    | 115    | 134    | 3    |
| 149 | <i>Bochum</i>            | DE   | 48.1  | [121–168] | 80     | 180    | 155    | 3    |
| 150 | <i>Cartagena</i>         | ES   | 47.7  | [135–165] | 107    | 175    | 150    | 3    |
| 151 | <i>Zagreb</i>            | HR   | 47.7  | [136–164] | 119    | 123    | 181    | 3    |
| 152 | <i>Le Havre</i>          | FR   | 47.6  | [144–159] | 127    | 143    | 166    | 3    |
| 153 | <i>Prague</i>            | CZ   | 47.5  | [143–158] | 141    | 132    | 169    | 3    |
| 154 | <i>Lausanne</i>          | CH   | 47.1  | [112–175] | 70     | 197    | 149    | 3    |
| 155 | <i>Pescara</i>           | IT   | 47.1  | [128–162] | 169    | 89     | 164    | 3    |
| 156 | <i>Vilnius</i>           | LT   | 46.6  | [149–162] | 130    | 148    | 167    | 3    |
| 157 | <i>Verona</i>            | IT   | 46.4  | [124–166] | 174    | 81     | 158    | 3    |
| 158 | <i>Alcalá de Henares</i> | ES   | 46.3  | [139–173] | 114    | 185    | 132    | 3    |
| 159 | <i>Nice</i>              | FR   | 45.2  | [145–165] | 168    | 118    | 159    | 3    |
| 160 | <i>Brescia</i>           | IT   | 44.6  | [103–176] | 185    | 50     | 160    | 3    |
| 161 | <i>Wuppertal</i>         | DE   | 43.8  | [147–177] | 117    | 187    | 152    | 3    |
| 162 | <i>Erfurt</i>            | DE   | 43.7  | [158–169] | 154    | 157    | 156    | 3    |
| 163 | <i>Częstochowa</i>       | PL   | 42.9  | [135–182] | 191    | 109    | 123    | 3    |
| 164 | <i>Katowice</i>          | PL   | 42.8  | [159–169] | 149    | 152    | 176    | 3    |
| 165 | <i>Marseille</i>         | FR   | 42.4  | [159–171] | 145    | 154    | 177    | 3    |
| 166 | <i>Radom</i>             | PL   | 42.4  | [150–175] | 182    | 116    | 154    | 3    |
| 167 | <i>Chemnitz</i>          | DE   | 41.5  | [157–176] | 131    | 177    | 173    | 3    |
| 168 | <i>Prato</i>             | IT   | 40.7  | [125–185] | 195    | 71     | 161    | 3    |
| 169 | <i>Brno</i>              | CZ   | 40.1  | [168–175] | 170    | 155    | 168    | 3    |
| 170 | <i>Thessaloniki</i>      | GR   | 39.9  | [134–187] | 76     | 194    | 198    | 3    |
| 171 | <i>Kaunas</i>            | LT   | 39.8  | [165–176] | 177    | 147    | 165    | 3    |

| #   | City               | Ctry | Index | Interval  | S rank | R rank | A rank | Tier |
|-----|--------------------|------|-------|-----------|--------|--------|--------|------|
| 172 | <i>Plovdiv</i>     | BG   | 39.7  | [110–187] | 197    | 51     | 178    | 3    |
| 173 | <i>Ostrava</i>     | CZ   | 39.3  | [145–181] | 184    | 93     | 183    | 3    |
| 174 | <i>Košice</i>      | SK   | 39.1  | [166–176] | 156    | 162    | 182    | 3    |
| 175 | <i>Genoa</i>       | IT   | 37.5  | [157–191] | 118    | 199    | 180    | 3    |
| 176 | <i>Wiesbaden</i>   | DE   | 37.1  | [165–188] | 171    | 191    | 147    | 3    |
| 177 | <i>Milan</i>       | IT   | 36.6  | [162–184] | 181    | 120    | 192    | 3    |
| 178 | <i>Varna</i>       | BG   | 36.3  | [169–183] | 157    | 176    | 189    | 3    |
| 179 | <i>Cluj-Napoca</i> | RO   | 36.1  | [171–183] | 161    | 179    | 185    | 3    |
| 180 | <i>Padua</i>       | IT   | 34.7  | [176–187] | 183    | 163    | 171    | 3    |
| 181 | <i>Lisbon</i>      | PT   | 34.2  | [147–193] | 205    | 84     | 175    | 3    |
| 182 | <i>Cork</i>        | IE   | 33.0  | [179–189] | 173    | 182    | 184    | 3    |
| 183 | <i>Porto</i>       | PT   | 32.0  | [169–196] | 146    | 201    | 197    | 3    |
| 184 | <i>Sofia</i>       | BG   | 31.8  | [182–189] | 175    | 174    | 201    | 3    |
| 185 | <i>Rome</i>        | IT   | 31.5  | [176–193] | 176    | 151    | 214    | 3    |
| 186 | <i>Dijon</i>       | FR   | 31.2  | [174–196] | 189    | 190    | 140    | 3    |
| 187 | <i>Charleroi</i>   | BE   | 31.0  | [178–192] | 188    | 150    | 188    | 3    |
| 188 | <i>Naples</i>      | IT   | 29.0  | [173–200] | 152    | 206    | 196    | 3    |
| 189 | <i>Trieste</i>     | IT   | 28.8  | [188–194] | 180    | 183    | 195    | 3    |
| 190 | <i>Palermo</i>     | IT   | 28.7  | [186–196] | 179    | 172    | 210    | 3    |
| 191 | <i>Sintra</i>      | PT   | 28.2  | [181–198] | 167    | 193    | 212    | 3    |
| 192 | <i>Cagliari</i>    | IT   | 27.6  | [184–196] | 200    | 161    | 187    | 3    |
| 193 | <i>Arad</i>        | RO   | 27.0  | [187–197] | 192    | 164    | 200    | 3    |
| 194 | <i>Satu Mare</i>   | RO   | 26.8  | [170–208] | 216    | 128    | 157    | 3    |
| 195 | <i>Oradea</i>      | RO   | 26.4  | [172–205] | 209    | 119    | 190    | 3    |
| 196 | <i>Nicosia</i>     | CY   | 25.7  | [193–198] | 187    | 184    | 199    | 3    |
| 197 | <i>Targu Mures</i> | RO   | 23.9  | [189–201] | 203    | 192    | 170    | 3    |
| 198 | <i>Messina</i>     | IT   | 23.8  | [169–209] | 123    | 213    | 215    | 3    |
| 199 | <i>Brasov</i>      | RO   | 23.1  | [191–201] | 193    | 203    | 172    | 3    |
| 200 | <i>Burgas</i>      | BG   | 21.8  | [189–209] | 208    | 153    | 209    | 3    |
| 201 | <i>Sibiu</i>       | RO   | 21.7  | [198–201] | 198    | 196    | 191    | 3    |
| 202 | <i>Timisoara</i>   | RO   | 19.7  | [200–204] | 201    | 195    | 203    | 3    |

| #   | City             | Ctry | Index | Interval  | S rank | R rank | A rank | Tier |
|-----|------------------|------|-------|-----------|--------|--------|--------|------|
| 203 | <i>Constanta</i> | RO   | 18.2  | [202–207] | 202    | 200    | 202    | 3    |
| 204 | <i>Athens</i>    | GR   | 17.2  | [203–208] | 204    | 204    | 193    | 3    |
| 205 | <i>Braga</i>     | PT   | 17.0  | [200–210] | 190    | 205    | 208    | 3    |
| 206 | <i>Bari</i>      | IT   | 16.8  | [203–210] | 199    | 202    | 211    | 3    |
| 207 | <i>Bucharest</i> | RO   | 16.0  | [201–209] | 206    | 210    | 179    | 3    |
| 208 | <i>Baia Mare</i> | RO   | 15.6  | [201–210] | 196    | 211    | 204    | 3    |
| 209 | <i>Catania</i>   | IT   | 15.3  | [193–214] | 178    | 214    | 216    | 3    |
| 210 | <i>Bacau</i>     | RO   | 13.1  | [203–212] | 213    | 198    | 194    | 3    |
| 211 | <i>Split</i>     | HR   | 11.9  | [203–214] | 207    | 215    | 174    | 3    |
| 212 | <i>Pitesti</i>   | RO   | 10.9  | [207–213] | 214    | 207    | 186    | 3    |
| 213 | <i>Craiova</i>   | RO   | 9.2   | [211–214] | 210    | 208    | 207    | 3    |
| 214 | <i>Galati</i>    | RO   | 7.0   | [212–215] | 211    | 209    | 217    | 3    |
| 215 | <i>Iasi</i>      | RO   | 3.3   | [215–218] | 215    | 212    | 218    | 3    |
| 216 | <i>Botosani</i>  | RO   | 2.0   | [214–219] | 217    | 216    | 205    | 3    |
| 217 | <i>Buzau</i>     | RO   | 1.7   | [215–219] | 218    | 217    | 206    | 3    |
| 218 | <i>Ploiesti</i>  | RO   | 1.6   | [215–219] | 212    | 218    | 219    | 3    |
| 219 | <i>Braila</i>    | RO   | 0.4   | [218–219] | 219    | 219    | 213    | 3    |

### 5.1 COUNTRY PATTERNS

Mean index by country (countries with  $\geq 4$  indexed cities):

| Country   | Cities | Mean index | Rank span |
|-----------|--------|------------|-----------|
| <i>FI</i> | 6      | 90.1       | 2–54      |
| <i>NL</i> | 11     | 87.4       | 4–107     |
| <i>SE</i> | 4      | 73.5       | 1–128     |
| <i>BE</i> | 5      | 72.0       | 3–187     |
| <i>ES</i> | 27     | 69.0       | 10–158    |
| <i>NO</i> | 4      | 65.9       | 61–88     |
| <i>FR</i> | 22     | 64.5       | 11–186    |
| <i>DE</i> | 40     | 63.4       | 20–176    |
| <i>PL</i> | 16     | 54.5       | 77–166    |
| <i>CH</i> | 5      | 53.2       | 108–154   |

| Country   | Cities    | Mean index  | Rank span |
|-----------|-----------|-------------|-----------|
| <i>IT</i> | <b>21</b> | <b>39.7</b> | 40–209    |
| <i>BG</i> | <b>4</b>  | <b>32.4</b> | 172–200   |
| <i>PT</i> | <b>4</b>  | <b>27.9</b> | 181–205   |
| <i>RO</i> | <b>20</b> | <b>15.2</b> | 179–219   |

## 6. The dimensions, separately

### 6.1 STREET-NETWORK RISK (S)

Across the 219 cities the mean street risk per km averages 59.0 (median 58.1, IQR 53.3–64.0). The dimension is dominated by speed regime and modelled traffic: cities with Dutch-level signed-30 coverage occupy the strong end regardless of country, and compact cities whose arterials thread the municipality sit at the weak end.

| City                     | Ctry | Value       | Overall rank |
|--------------------------|------|-------------|--------------|
| <b>STRONGEST FIFTEEN</b> |      |             |              |
| <i>Brussels</i>          | BE   | <b>41.1</b> | <b>15</b>    |
| <i>Haarlem</i>           | NL   | <b>42.3</b> | <b>4</b>     |
| <i>Bremen</i>            | DE   | <b>42.9</b> | <b>22</b>    |
| <i>Copenhagen</i>        | DK   | <b>43.5</b> | <b>17</b>    |
| <i>Stockholm</i>         | SE   | <b>44.2</b> | <b>1</b>     |
| <i>Leiden</i>            | NL   | <b>45.0</b> | <b>5</b>     |
| <i>Barcelona</i>         | ES   | <b>45.2</b> | <b>10</b>    |
| <i>Nantes</i>            | FR   | <b>45.9</b> | <b>35</b>    |
| <i>Antwerp</i>           | BE   | <b>46.5</b> | <b>3</b>     |
| <i>The Hague</i>         | NL   | <b>46.6</b> | <b>7</b>     |
| <i>Paris</i>             | FR   | <b>46.6</b> | <b>11</b>    |
| <i>Eindhoven</i>         | NL   | <b>46.8</b> | <b>13</b>    |
| <i>Helsinki</i>          | FI   | <b>46.9</b> | <b>2</b>     |
| <i>Ghent</i>             | BE   | <b>47.0</b> | <b>23</b>    |
| <i>Bilbao</i>            | ES   | <b>47.0</b> | <b>25</b>    |
| <b>WEAKEST FIFTEEN</b>   |      |             |              |
| <i>Craiova</i>           | RO   | <b>73.3</b> | <b>213</b>   |

| City             | Ctry | Value | Overall rank |
|------------------|------|-------|--------------|
| <i>Timisoara</i> | RO   | 73.8  | 202          |
| <i>Split</i>     | HR   | 73.8  | 211          |
| <i>Oradea</i>    | RO   | 74.6  | 195          |
| <i>Burgas</i>    | BG   | 74.7  | 200          |
| <i>Ploiesti</i>  | RO   | 75.2  | 218          |
| <i>Baia Mare</i> | RO   | 75.9  | 208          |
| <i>Bacau</i>     | RO   | 76.1  | 210          |
| <i>Iasi</i>      | RO   | 77.2  | 215          |
| <i>Pitesti</i>   | RO   | 77.6  | 212          |
| <i>Galati</i>    | RO   | 78.4  | 214          |
| <i>Buzau</i>     | RO   | 79.7  | 217          |
| <i>Satu Mare</i> | RO   | 82.4  | 194          |
| <i>Botosani</i>  | RO   | 83.7  | 216          |
| <i>Braila</i>    | RO   | 84.6  | 219          |

## 6.2 JOURNEY RISK (R)

Simulated journeys average 15.3 across cities (median 11.4, IQR -0.7–29.0). Values are on the model's raw comparable scale, which is negative where a journey rides mostly bonus-carrying protected infrastructure — the strongest cities' typical trip scores below zero. R rewards coherent low-risk sub-networks; its divergence from S is the single most informative cross-dimensional signal (Section 7).

| City                     | Ctry | Value | Overall rank |
|--------------------------|------|-------|--------------|
| <b>STRONGEST FIFTEEN</b> |      |       |              |
| <i>Tampere</i>           | FI   | -30.6 | 12           |
| <i>Helsinki</i>          | FI   | -28.9 | 2            |
| <i>Espoo</i>             | FI   | -27.6 | 8            |
| <i>Turku</i>             | FI   | -25.9 | 19           |
| <i>Vantaa</i>            | FI   | -25.6 | 6            |
| <i>Oulu</i>              | FI   | -22.1 | 54           |
| <i>Antwerp</i>           | BE   | -18.6 | 3            |
| <i>Malmö</i>             | SE   | -18.5 | 38           |

| City                   | Ctry | Value        | Overall rank |
|------------------------|------|--------------|--------------|
| <i>Almere</i>          | NL   | <b>-18.4</b> | <b>26</b>    |
| <i>Amsterdam</i>       | NL   | <b>-16.2</b> | <b>9</b>     |
| <i>Uppsala</i>         | SE   | <b>-16.2</b> | <b>85</b>    |
| <i>Turin</i>           | IT   | <b>-14.9</b> | <b>137</b>   |
| <i>Bologna</i>         | IT   | <b>-14.7</b> | <b>40</b>    |
| <i>Odense</i>          | DK   | <b>-14.5</b> | <b>66</b>    |
| <i>Stockholm</i>       | SE   | <b>-14.4</b> | <b>1</b>     |
| <b>WEAKEST FIFTEEN</b> |      |              |              |
| <i>Braga</i>           | PT   | <b>54.9</b>  | <b>205</b>   |
| <i>Naples</i>          | IT   | <b>57.6</b>  | <b>188</b>   |
| <i>Pitesti</i>         | RO   | <b>57.6</b>  | <b>212</b>   |
| <i>Craiova</i>         | RO   | <b>58.1</b>  | <b>213</b>   |
| <i>Galati</i>          | RO   | <b>58.4</b>  | <b>214</b>   |
| <i>Bucharest</i>       | RO   | <b>60.0</b>  | <b>207</b>   |
| <i>Baia Mare</i>       | RO   | <b>62.0</b>  | <b>208</b>   |
| <i>Iasi</i>            | RO   | <b>63.1</b>  | <b>215</b>   |
| <i>Messina</i>         | IT   | <b>71.4</b>  | <b>198</b>   |
| <i>Buzau</i>           | RO   | <b>74.0</b>  | <b>217</b>   |
| <i>Catania</i>         | IT   | <b>75.5</b>  | <b>209</b>   |
| <i>Braila</i>          | RO   | <b>75.7</b>  | <b>219</b>   |
| <i>Ploiesti</i>        | RO   | <b>76.0</b>  | <b>218</b>   |
| <i>Botosani</i>        | RO   | <b>78.3</b>  | <b>216</b>   |
| <i>Split</i>           | HR   | <b>85.6</b>  | <b>211</b>   |

### 6.3 LOW-STRESS ACCESSIBILITY (A)

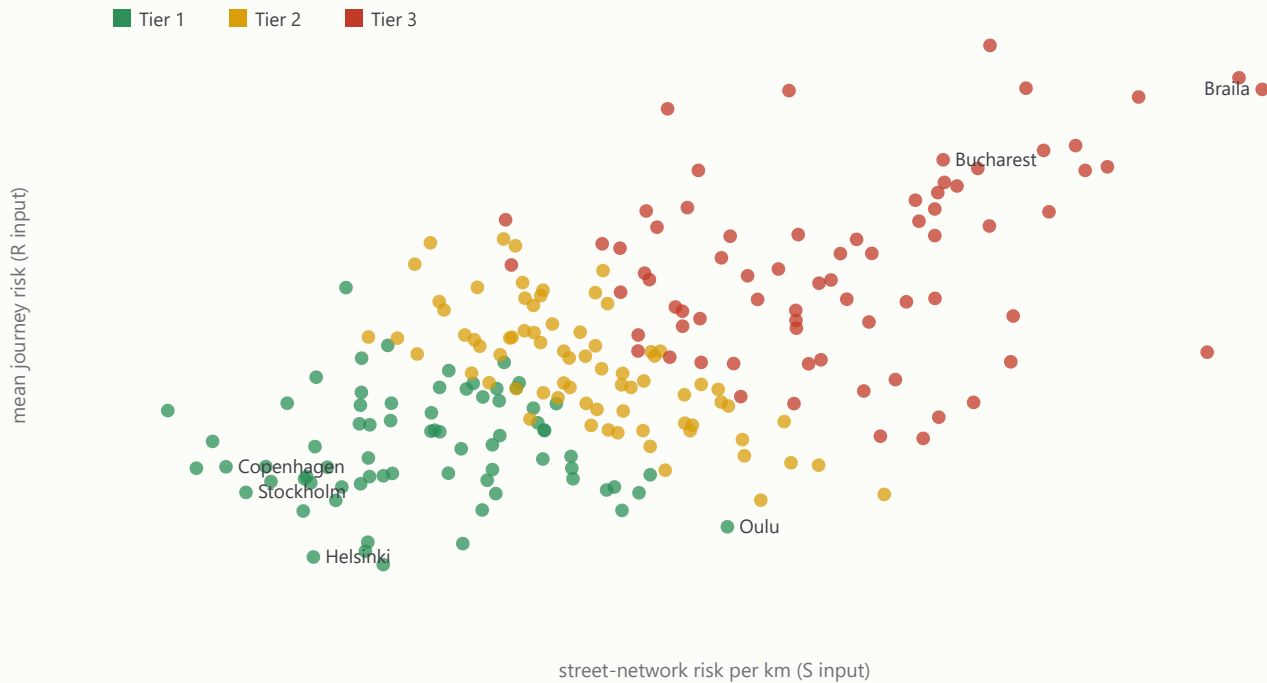
The composite accessibility input averages 0.414 (median 0.410, IQR 0.285–0.538); on average 49% of residents are locked to high-stress streets at the central threshold. A is the harshest dimension: connectivity is multiplicative, and most cities' calm streets form islands.

| City                     | Ctry | Value        | Overall rank |
|--------------------------|------|--------------|--------------|
| <b>STRONGEST FIFTEEN</b> |      |              |              |
| <i>Haarlem</i>           | NL   | <b>0.841</b> | <b>4</b>     |

| City                   | Ctry | Value        | Overall rank |
|------------------------|------|--------------|--------------|
| <i>Brussels</i>        | BE   | <b>0.787</b> | <b>15</b>    |
| <i>Stockholm</i>       | SE   | <b>0.774</b> | <b>1</b>     |
| <i>Leiden</i>          | NL   | <b>0.761</b> | <b>5</b>     |
| <i>Paris</i>           | FR   | <b>0.755</b> | <b>11</b>    |
| <i>Barcelona</i>       | ES   | <b>0.723</b> | <b>10</b>    |
| <i>Antwerp</i>         | BE   | <b>0.703</b> | <b>3</b>     |
| <i>Lyon</i>            | FR   | <b>0.698</b> | <b>27</b>    |
| <i>The Hague</i>       | NL   | <b>0.698</b> | <b>7</b>     |
| <i>Vantaa</i>          | FI   | <b>0.697</b> | <b>6</b>     |
| <i>Bilbao</i>          | ES   | <b>0.696</b> | <b>25</b>    |
| <i>Valencia</i>        | ES   | <b>0.691</b> | <b>41</b>    |
| <i>Helsinki</i>        | FI   | <b>0.682</b> | <b>2</b>     |
| <i>Amsterdam</i>       | NL   | <b>0.674</b> | <b>9</b>     |
| <i>Sabadell</i>        | ES   | <b>0.674</b> | <b>46</b>    |
| <b>WEAKEST FIFTEEN</b> |      |              |              |
| <i>Botosani</i>        | RO   | <b>0.169</b> | <b>216</b>   |
| <i>Buzau</i>           | RO   | <b>0.165</b> | <b>217</b>   |
| <i>Craiova</i>         | RO   | <b>0.159</b> | <b>213</b>   |
| <i>Braga</i>           | PT   | <b>0.155</b> | <b>205</b>   |
| <i>Burgas</i>          | BG   | <b>0.151</b> | <b>200</b>   |
| <i>Palermo</i>         | IT   | <b>0.150</b> | <b>190</b>   |
| <i>Bari</i>            | IT   | <b>0.149</b> | <b>206</b>   |
| <i>Sintra</i>          | PT   | <b>0.144</b> | <b>191</b>   |
| <i>Braila</i>          | RO   | <b>0.138</b> | <b>219</b>   |
| <i>Rome</i>            | IT   | <b>0.125</b> | <b>185</b>   |
| <i>Ploiesti</i>        | RO   | <b>0.124</b> | <b>218</b>   |
| <i>Iasi</i>            | RO   | <b>0.116</b> | <b>215</b>   |
| <i>Messina</i>         | IT   | <b>0.113</b> | <b>198</b>   |
| <i>Galati</i>          | RO   | <b>0.112</b> | <b>214</b>   |
| <i>Catania</i>         | IT   | <b>0.104</b> | <b>209</b>   |

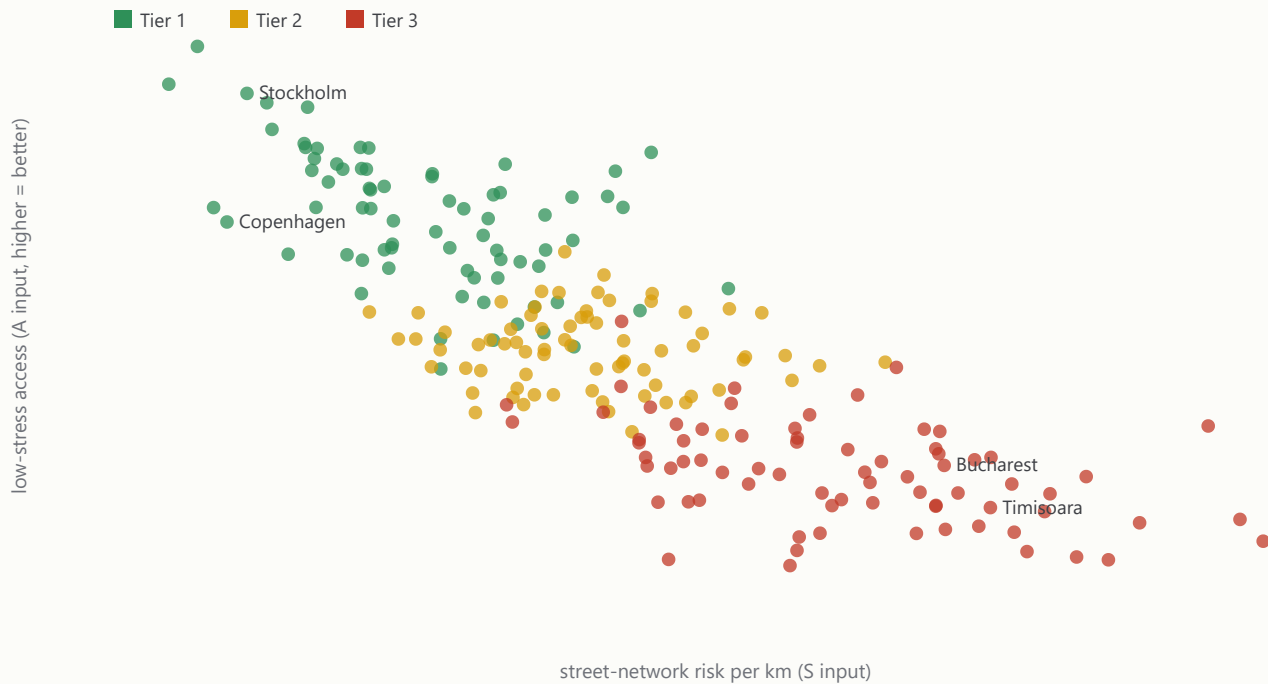
## 7. Cross-dimensional analysis

The dimensions correlate (Spearman S-R 0.50, S-A -0.82, R-A -0.74) — safe streets, safe journeys and connected access tend to come together — but the exceptions carry the diagnosis.



**Figure 1.** Street risk vs journey risk. Cities far below the diagonal trend own an "escape valve": a connected low-risk sub-network that journeys can use even where the average street is hostile. Cities on or above it offer the router no refuge.

The escape-valve gap (street risk minus journey risk) ranges from 85.5 (Oulu) to -11.8 (Split). The fifteen largest escape valves: Oulu, Turin, Uppsala, Tampere, Turku, Malmö, Espoo, Helsinki, Parma, Vantaa, Odense, Venice, Brescia, Seville, Strasbourg. The fifteen smallest: Baia Mare, Porto, Kassel, Bucharest, Genoa, Oviedo, Braila, Lausanne, Buzau, Botosani, Naples, Ploiesti, Catania, Messina, Split. The second list is a policy target in itself: these are the cities where building a coherent sub-network would move the experienced risk fastest.



**Figure 2.** Street risk vs low-stress access. The vertical spread at fixed street risk is connectivity: cities with identical averages differ several-fold in how much of daily life their calm streets can actually reach.

## 8. City profiles

One data-generated profile per city, alphabetical. Dimension ranks are out of 219; "escape-valve rank" orders the street-minus-journey gap (1 = largest escape valve). Factor names follow the public sign-only disclosure policy.

---

76. A CORUÑA ES · INDEX 65.7 · INTERVAL 61–101 · TIER 2 (MIDDLE) · TIER STABILITY 67%

A Coruña holds an uneven profile — strongest on low-stress access (rank 51), weakest on street-network risk (rank 116). Streets: mean comparable risk 56.9 per km over 489 street-km (dimension rank 116/219), with 12.9% of the network in the highest-risk band. Journeys: 210 standardised trips average 10.0 (rank 105/219; 90th percentile 42.7). Access: composite low-stress accessibility 0.549 (rank 51/219); 0% of 259,172 residents are locked to high-stress streets at the central threshold, against 450 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, Incline, Heavy-vehicle access. Leading protections: Signed ≤30 km/h zone, Cycleway/Sidewalk.

---

118. AACHEN DE · INDEX 56.9 · INTERVAL 89–145 · TIER 2 (MIDDLE) · TIER STABILITY 95%

Aachen holds an uneven profile — strongest on street-network risk (rank 67), weakest on journey risk (rank 171). Streets: mean comparable risk 53.4 per km over 1,063 street-km (dimension rank 67/219), with 9.4% of the network in the highest-risk band. Journeys: 210 standardised trips average 31.4 (rank 171/219; 90th percentile 50.8). Access: composite low-stress accessibility 0.417 (rank 105/219); 54% of 214,663 residents are locked to high-stress streets at the central threshold, against 860 km of low-stress network. Leading risk

drivers on its streets: Motor-traffic volume, Incline, On-road stress (speed × lanes). Leading protections: Signed ≤30 km/h zone, Bicycle Designated.

---

72. AARHUS DK · INDEX 67.2 · INTERVAL 43–90 · TIER 1 (LOWER ASSESSED RISK) · TIER STABILITY 64%

Aarhus holds an uneven profile — strongest on journey risk (rank 24), weakest on low-stress access (rank 108). Streets: mean comparable risk 57.2 per km over 3,096 street-km (dimension rank 101/219), with 10.1% of the network in the highest-risk band. Journeys: 210 standardised trips average -11.4 (rank 24/219; 90th percentile 4.0). Access: composite low-stress accessibility 0.414 (rank 108/219); 85% of 335,337 residents are locked to high-stress streets at the central threshold, against 1,630 km of low-stress network. Leading risk drivers on its streets: On-road stress (speed × lanes), Score cap, 50 km/h limit. Leading protections: Separated cycle track, Cycleway/Sidewalk. Notes: straddles a tier boundary under respecification.

---

158. ALCALÁ DE HENARES ES · INDEX 46.3 · INTERVAL 139–173 · TIER 3 (HIGHER ASSESSED RISK) · TIER STABILITY 77%

Alcalá de Henares holds an uneven profile — strongest on street-network risk (rank 114), weakest on journey risk (rank 185). Streets: mean comparable risk 59.1 per km over 549 street-km (dimension rank 114/219), with 10.2% of the network in the highest-risk band. Journeys: 210 standardised trips average 40.2 (rank 185/219; 90th percentile 56.8). Access: composite low-stress accessibility 0.358 (rank 132/219); 19% of 193,162 residents are locked to high-stress streets at the central threshold, against 397 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, Incline, Heavy-vehicle access. Leading protections: Signed ≤30 km/h zone, Cycleway/Sidewalk.

---

79. ALICANTE ES · INDEX 65.0 · INTERVAL 76–89 · TIER 2 (MIDDLE) · TIER STABILITY 73%

Alicante holds a balanced profile. Streets: mean comparable risk 57.8 per km over 1,085 street-km (dimension rank 89/219), with 8.0% of the network in the highest-risk band. Journeys: 210 standardised trips average 5.5 (rank 80/219; 90th percentile 33.0). Access: composite low-stress accessibility 0.457 (rank 85/219); 8% of 401,574 residents are locked to high-stress streets at the central threshold, against 1,163 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, Incline, Heavy-vehicle access. Leading protections: Cycleway/Sidewalk, Separated cycle track.

---

107. ALKMAAR NL · INDEX 58.8 · INTERVAL 80–140 · TIER 2 (MIDDLE) · TIER STABILITY 94%

Alkmaar holds an uneven profile — strongest on low-stress access (rank 66), weakest on street-network risk (rank 164). Streets: mean comparable risk 58.2 per km over 599 street-km (dimension rank 164/219), with 20.2% of the network in the highest-risk band. Journeys: 210 standardised trips average 4.1 (rank 75/219; 90th percentile 34.3). Access: composite low-stress accessibility 0.492 (rank 66/219); 14% of 105,253 residents are locked to high-stress streets at the central threshold, against 435 km of low-stress network. Leading risk drivers on its streets: On-road stress (speed × lanes), Score cap, Heavy-vehicle access. Leading protections: Cycleway/Sidewalk, Separated cycle track.

---

26. ALMERE NL · INDEX 83.6 · INTERVAL 11–36 · TIER 1 (LOWER ASSESSED RISK) · TIER STABILITY 100%

Almere holds a balanced profile. Streets: mean comparable risk 53.6 per km over 1,040 street-km (dimension rank 45/219), with 5.0% of the network in the highest-risk band. Journeys: 210 standardised trips average -18.4 (rank 9/219; 90th percentile -11.3). Access: composite low-stress accessibility 0.573 (rank 43/219); 27% of 232,828 residents are locked to high-stress streets at the central threshold, against 1,142 km of low-stress network. Leading risk drivers on its streets: Score cap, Motor-traffic volume, Heavy-vehicle access. Leading protections: Cycleway/Sidewalk, Separated cycle track. Notes: top-ten on at least one dimension.

---

91. ALMERÍA ES · INDEX 60.8 · INTERVAL 91–108 · TIER 2 (MIDDLE) · TIER STABILITY 99%

Almería holds a balanced profile. Streets: mean comparable risk 59.2 per km over 781 street-km (dimension rank 93/219), with 6.7% of the network in the highest-risk band. Journeys: 210 standardised trips average 12.2 (rank 111/219; 90th percentile 41.4). Access: composite low-stress accessibility 0.423 (rank 102/219); 14% of 230,082 residents are locked to high-stress streets at the central threshold, against 1,035 km of low-stress network. Leading risk drivers on its streets: Incline, Motor-traffic volume, Heavy-vehicle access. Leading protections: Cycleway/Sidewalk, Separated cycle track.

---

9. AMSTERDAM NL · INDEX 94.2 · INTERVAL 5–17 · TIER 1 (LOWER ASSESSED RISK) · TIER STABILITY 100%

Amsterdam holds a balanced profile. Streets: mean comparable risk 47.8 per km over 1,675 street-km (dimension rank 8/219), with 2.3% of the network in the highest-risk band. Journeys: 210 standardised trips average -16.2 (rank 10/219; 90th percentile -5.5). Access: composite low-stress accessibility 0.674 (rank 14/219); 3% of 774,206 residents are locked to high-stress streets at the central threshold, against 2,180 km of low-stress network. Leading risk drivers on its streets: Score cap, Motor-traffic volume, Heavy-vehicle access. Leading protections: Cycleway/Sidewalk, Separated cycle track. Notes: top-ten overall; top-ten on at least one dimension.

---

97. ANGERS FR · INDEX 59.9 · INTERVAL 73–133 · TIER 2 (MIDDLE) · TIER STABILITY 84%

Angers holds an uneven profile — strongest on street-network risk (rank 47), weakest on journey risk (rank 160). Streets: mean comparable risk 51.9 per km over 647 street-km (dimension rank 47/219), with 7.6% of the network in the highest-risk band. Journeys: 210 standardised trips average 28.2 (rank 160/219; 90th percentile 42.2). Access: composite low-stress accessibility 0.410 (rank 110/219); 5% of 145,109 residents are locked to high-stress streets at the central threshold, against 356 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, On-road stress (speed × lanes), 50 km/h limit. Leading protections: Signed ≤30 km/h zone, Signed bike network.

---

3. ANTWERP BE · INDEX 96.4 · INTERVAL 2–13 · TIER 1 (LOWER ASSESSED RISK) · TIER STABILITY 100%

Antwerp holds a balanced profile. Streets: mean comparable risk 46.5 per km over 1,220 street-km (dimension rank 9/219), with 3.9% of the network in the highest-risk band. Journeys: 210 standardised trips average -18.6 (rank 7/219; 90th percentile -8.7). Access: composite low-stress accessibility 0.703 (rank 7/219); 9% of 538,620 residents are locked to high-stress streets at the central threshold, against 1,514 km of low-stress network. Leading risk drivers on its streets: Score cap, Motor-traffic volume, On-road stress (speed × lanes). Leading

protections: Cycleway/Sidewalk, Separated cycle track. Notes: top-ten overall; top-ten on at least one dimension.

---

**193. ARAD RO** · INDEX 27.0 · INTERVAL 187–197 · TIER 3 (HIGHER ASSESSED RISK) · TIER STABILITY 100%

Arad holds a balanced profile. Streets: mean comparable risk 71.6 per km over 732 street-km (dimension rank 192/219), with 17.7% of the network in the highest-risk band. Journeys: 210 standardised trips average 28.9 (rank 164/219; 90th percentile 58.3). Access: composite low-stress accessibility 0.189 (rank 200/219); 99% of 128,074 residents are locked to high-stress streets at the central threshold, against 132 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, On-road stress (speed × lanes), 50 km/h limit. Leading protections: Cycleway/Sidewalk, Separated cycle track.

---

**204. ATHENS GR** · INDEX 17.2 · INTERVAL 203–208 · TIER 3 (HIGHER ASSESSED RISK) · TIER STABILITY 100%

Athens holds a balanced profile. Streets: mean comparable risk 72.5 per km over 980 street-km (dimension rank 204/219), with 20.7% of the network in the highest-risk band. Journeys: 210 standardised trips average 54.1 (rank 204/219; 90th percentile 65.3). Access: composite low-stress accessibility 0.207 (rank 193/219); 99% of 622,088 residents are locked to high-stress streets at the central threshold, against 155 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, On-road stress (speed × lanes), 50 km/h limit. Leading protections: Pedestrian Zone, Residential street.

---

**64. AUGSBURG DE** · INDEX 69.4 · INTERVAL 54–74 · TIER 1 (LOWER ASSESSED RISK) · TIER STABILITY 84%

Augsburg holds a balanced profile. Streets: mean comparable risk 53.6 per km over 1,164 street-km (dimension rank 53/219), with 6.8% of the network in the highest-risk band. Journeys: 210 standardised trips average 6.9 (rank 88/219; 90th percentile 27.4). Access: composite low-stress accessibility 0.477 (rank 73/219); 72% of 300,974 residents are locked to high-stress streets at the central threshold, against 1,028 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, Track Grade 4/5, On-road stress (speed × lanes). Leading protections: Bicycle Designated, Signed ≤30 km/h zone.

---

**210. BACAU RO** · INDEX 13.1 · INTERVAL 203–212 · TIER 3 (HIGHER ASSESSED RISK) · TIER STABILITY 100%

Bacau holds a balanced profile. Streets: mean comparable risk 76.1 per km over 390 street-km (dimension rank 213/219), with 25.5% of the network in the highest-risk band. Journeys: 210 standardised trips average 48.3 (rank 198/219; 90th percentile 81.0). Access: composite low-stress accessibility 0.206 (rank 194/219); 100% of 116,669 residents are locked to high-stress streets at the central threshold, against 38 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, On-road stress (speed × lanes), 50 km/h limit. Leading protections: Residential street, Service Road.

---

**208. BAIA MARE RO** · INDEX 15.6 · INTERVAL 201–210 · TIER 3 (HIGHER ASSESSED RISK) · TIER STABILITY 100%

Baia Mare holds a balanced profile. Streets: mean comparable risk 75.9 per km over 416 street-km (dimension rank 196/219), with 13.2% of the network in the highest-risk band. Journeys: 210 standardised trips average

62.0 (rank 211/219; 90th percentile 88.7). Access: composite low-stress accessibility 0.180 (rank 204/219); 88% of 109,100 residents are locked to high-stress streets at the central threshold, against 471 km of low-stress network. Its routed journeys score close to (or above) its street average — the router finds little refuge from the ordinary street network (escape-valve rank 205 of 219). Leading risk drivers on its streets: Incline, Motor-traffic volume, On-road stress (speed × lanes). Leading protections: Path, Car-Free Zone.

---

**10. BARCELONA ES · INDEX 93.2 · INTERVAL 6–19 · TIER 1 (LOWER ASSESSED RISK) · TIER STABILITY 100%**

Barcelona holds a balanced profile. Streets: mean comparable risk 45.2 per km over 1,303 street-km (dimension rank 20/219), with 7.9% of the network in the highest-risk band. Journeys: 210 standardised trips average -12.0 (rank 21/219; 90th percentile 0.2). Access: composite low-stress accessibility 0.723 (rank 4/219); 5% of 1,744,231 residents are locked to high-stress streets at the central threshold, against 1,596 km of low-stress network. Leading risk drivers on its streets: Incline, Score cap, Heavy-vehicle access. Leading protections: Cycleway/Sidewalk, Signed ≤30 km/h zone. Notes: top-ten overall; top-ten on at least one dimension.

---

**206. BARI IT · INDEX 16.8 · INTERVAL 203–210 · TIER 3 (HIGHER ASSESSED RISK) · TIER STABILITY 100%**

Bari holds a balanced profile. Streets: mean comparable risk 70.8 per km over 1,153 street-km (dimension rank 199/219), with 20.2% of the network in the highest-risk band. Journeys: 210 standardised trips average 50.9 (rank 202/219; 90th percentile 81.3). Access: composite low-stress accessibility 0.149 (rank 211/219); 98% of 301,196 residents are locked to high-stress streets at the central threshold, against 183 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, Heavy-vehicle access, On-road stress (speed × lanes). Leading protections: Cycleway/Sidewalk, Separated cycle track.

---

**147. BASEL CH · INDEX 50.0 · INTERVAL 125–164 · TIER 3 (HIGHER ASSESSED RISK) · TIER STABILITY 51%**

Basel holds an uneven profile — strongest on low-stress access (rank 87), weakest on journey risk (rank 168). Streets: mean comparable risk 59.1 per km over 368 street-km (dimension rank 155/219), with 16.6% of the network in the highest-risk band. Journeys: 210 standardised trips average 30.4 (rank 168/219; 90th percentile 46.4). Access: composite low-stress accessibility 0.451 (rank 87/219); 0% of 161,619 residents are locked to high-stress streets at the central threshold, against 184 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, On-road stress (speed × lanes), 50 km/h limit. Leading protections: Signed ≤30 km/h zone, Signed bike network. Notes: straddles a tier boundary under respecification.

---

**143. BERGAMO IT · INDEX 50.8 · INTERVAL 111–153 · TIER 2 (MIDDLE) · TIER STABILITY 65%**

Bergamo holds an uneven profile — strongest on journey risk (rank 85), weakest on low-stress access (rank 163). Streets: mean comparable risk 63.1 per km over 419 street-km (dimension rank 147/219), with 10.2% of the network in the highest-risk band. Journeys: 209 standardised trips average 5.8 (rank 85/219; 90th percentile 28.9). Access: composite low-stress accessibility 0.289 (rank 163/219); 88% of 115,714 residents are locked to high-stress streets at the central threshold, against 195 km of low-stress network. Leading risk drivers on its streets: Score cap, Heavy-vehicle access, Motor-traffic volume. Leading protections: Cycleway/Sidewalk, Separated cycle track. Notes: straddles a tier boundary under respecification.

86. BERGEN NO · INDEX 62.4 · INTERVAL 70–113 · TIER 2 (MIDDLE) · TIER STABILITY 70%

Bergen holds an uneven profile — strongest on street-network risk (rank 55), weakest on low-stress access (rank 133). Streets: mean comparable risk 55.0 per km over 2,586 street-km (dimension rank 55/219), with 5.4% of the network in the highest-risk band. Journeys: 210 standardised trips average 8.8 (rank 97/219; 90th percentile 43.4). Access: composite low-stress accessibility 0.355 (rank 133/219); 20% of 282,215 residents are locked to high-stress streets at the central threshold, against 2,728 km of low-stress network. Leading risk drivers on its streets: Incline, Motor-traffic volume, On-road stress (speed × lanes). Leading protections: Cycleway/Sidewalk, Separated cycle track.

---

51. BERLIN DE · INDEX 73.6 · INTERVAL 37–63 · TIER 1 (LOWER ASSESSED RISK) · TIER STABILITY 98%

Berlin holds an uneven profile — strongest on street-network risk (rank 32), weakest on journey risk (rank 78). Streets: mean comparable risk 48.8 per km over 8,764 street-km (dimension rank 32/219), with 7.7% of the network in the highest-risk band. Journeys: 209 standardised trips average 5.1 (rank 78/219; 90th percentile 22.4). Access: composite low-stress accessibility 0.490 (rank 67/219); 27% of 3,235,142 residents are locked to high-stress streets at the central threshold, against 7,004 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, Score cap, Track Grade 4/5. Leading protections: Signed ≤30 km/h zone, Separated cycle track.

---

108. BERN CH · INDEX 58.7 · INTERVAL 67–144 · TIER 2 (MIDDLE) · TIER STABILITY 79%

Bern holds an uneven profile — strongest on street-network risk (rank 43), weakest on journey risk (rank 181). Streets: mean comparable risk 51.0 per km over 521 street-km (dimension rank 43/219), with 7.5% of the network in the highest-risk band. Journeys: 210 standardised trips average 36.6 (rank 181/219; 90th percentile 47.9). Access: composite low-stress accessibility 0.425 (rank 98/219); 10% of 119,340 residents are locked to high-stress streets at the central threshold, against 474 km of low-stress network. Leading risk drivers on its streets: Incline, Motor-traffic volume, On-road stress (speed × lanes). Leading protections: Signed ≤30 km/h zone, Car-Free Zone.

---

77. BIAŁYSTOK PL · INDEX 65.3 · INTERVAL 61–102 · TIER 2 (MIDDLE) · TIER STABILITY 58%

Białystok holds an uneven profile — strongest on journey risk (rank 46), weakest on street-network risk (rank 133). Streets: mean comparable risk 60.3 per km over 1,067 street-km (dimension rank 133/219), with 11.1% of the network in the highest-risk band. Journeys: 210 standardised trips average -4.1 (rank 46/219; 90th percentile 13.7). Access: composite low-stress accessibility 0.479 (rank 71/219); 87% of 286,740 residents are locked to high-stress streets at the central threshold, against 541 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, On-road stress (speed × lanes), 50 km/h limit. Leading protections: Bicycle Designated, Cycleway/Sidewalk. Notes: straddles a tier boundary under respecification.

---

132. BIELEFELD DE · INDEX 53.6 · INTERVAL 112–145 · TIER 2 (MIDDLE) · TIER STABILITY 84%

Bielefeld holds an uneven profile — strongest on street-network risk (rank 90), weakest on journey risk (rank 149). Streets: mean comparable risk 56.4 per km over 2,189 street-km (dimension rank 90/219), with 9.5% of

the network in the highest-risk band. Journeys: 210 standardised trips average 23.2 (rank 149/219; 90th percentile 44.5). Access: composite low-stress accessibility 0.346 (rank 138/219); 79% of 326,589 residents are locked to high-stress streets at the central threshold, against 1,273 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, On-road stress (speed × lanes), Incline. Leading protections: Signed ≤30 km/h zone, Bicycle Designated.

---

25. **BILBAO ES** · INDEX 83.8 · INTERVAL 13–44 · TIER 1 (LOWER ASSESSED RISK) · TIER STABILITY 100%

Bilbao holds an uneven profile — strongest on low-stress access (rank 11), weakest on journey risk (rank 110). Streets: mean comparable risk 47.0 per km over 400 street-km (dimension rank 23/219), with 7.4% of the network in the highest-risk band. Journeys: 210 standardised trips average 11.4 (rank 110/219; 90th percentile 36.5). Access: composite low-stress accessibility 0.696 (rank 11/219); 0% of 356,270 residents are locked to high-stress streets at the central threshold, against 498 km of low-stress network. Leading risk drivers on its streets: Incline, Heavy-vehicle access, Motor-traffic volume. Leading protections: Signed ≤30 km/h zone, Pedestrian Zone.

---

149. **BOCHUM DE** · INDEX 48.1 · INTERVAL 121–168 · TIER 3 (HIGHER ASSESSED RISK) · TIER STABILITY 56%

Bochum holds an uneven profile — strongest on street-network risk (rank 80), weakest on journey risk (rank 180). Streets: mean comparable risk 54.8 per km over 1,887 street-km (dimension rank 80/219), with 10.2% of the network in the highest-risk band. Journeys: 210 standardised trips average 36.4 (rank 180/219; 90th percentile 53.2). Access: composite low-stress accessibility 0.308 (rank 155/219); 74% of 341,599 residents are locked to high-stress streets at the central threshold, against 1,088 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, On-road stress (speed × lanes), Incline. Leading protections: Signed ≤30 km/h zone, Bicycle Designated. Notes: straddles a tier boundary under respecification.

---

40. **BOLOGNA IT** · INDEX 78.0 · INTERVAL 18–49 · TIER 1 (LOWER ASSESSED RISK) · TIER STABILITY 100%

Bologna holds an uneven profile — strongest on journey risk (rank 13), weakest on street-network risk (rank 71). Streets: mean comparable risk 54.2 per km over 1,184 street-km (dimension rank 71/219), with 9.5% of the network in the highest-risk band. Journeys: 210 standardised trips average -14.7 (rank 13/219; 90th percentile 2.1). Access: composite low-stress accessibility 0.551 (rank 50/219); 7% of 365,705 residents are locked to high-stress streets at the central threshold, against 801 km of low-stress network. Leading risk drivers on its streets: Score cap, Motor-traffic volume, Incline. Leading protections: Cycleway/Sidewalk, Separated cycle track.

---

57. **BONN DE** · INDEX 70.5 · INTERVAL 43–87 · TIER 1 (LOWER ASSESSED RISK) · TIER STABILITY 88%

Bonn holds an uneven profile — strongest on street-network risk (rank 36), weakest on journey risk (rank 135). Streets: mean comparable risk 49.9 per km over 1,144 street-km (dimension rank 36/219), with 7.5% of the network in the highest-risk band. Journeys: 210 standardised trips average 18.5 (rank 135/219; 90th percentile 36.1). Access: composite low-stress accessibility 0.526 (rank 58/219); 13% of 300,622 residents are locked to high-stress streets at the central threshold, against 998 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, Incline, On-road stress (speed × lanes). Leading protections: Signed ≤30 km/h zone, Bicycle Designated.

116. BORDEAUX FR · INDEX 57.6 · INTERVAL 100–132 · TIER 2 (MIDDLE) · TIER STABILITY 99%

Bordeaux holds an uneven profile — strongest on street-network risk (rank 83), weakest on journey risk (rank 146). Streets: mean comparable risk 55.3 per km over 858 street-km (dimension rank 83/219), with 10.1% of the network in the highest-risk band. Journeys: 210 standardised trips average 21.7 (rank 146/219; 90th percentile 42.6). Access: composite low-stress accessibility 0.407 (rank 112/219); 19% of 267,216 residents are locked to high-stress streets at the central threshold, against 366 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, On-road stress (speed × lanes), 50 km/h limit. Leading protections: Signed bike network, Signed ≤30 km/h zone.

---

216. BOTOSANI RO · INDEX 2.0 · INTERVAL 214–219 · TIER 3 (HIGHER ASSESSED RISK) · TIER STABILITY 100%

Botosani holds a balanced profile. Streets: mean comparable risk 83.7 per km over 271 street-km (dimension rank 217/219), with 35.4% of the network in the highest-risk band. Journeys: 210 standardised trips average 78.3 (rank 216/219; 90th percentile 90.9). Access: composite low-stress accessibility 0.169 (rank 205/219); 97% of 91,625 residents are locked to high-stress streets at the central threshold, against 39 km of low-stress network. Its routed journeys score close to (or above) its street average — the router finds little refuge from the ordinary street network (escape-valve rank 214 of 219). Leading risk drivers on its streets: Motor-traffic volume, On-road stress (speed × lanes), 50 km/h limit. Leading protections: Residential street, Pedestrian Zone.

---

205. BRAGA PT · INDEX 17.0 · INTERVAL 200–210 · TIER 3 (HIGHER ASSESSED RISK) · TIER STABILITY 100%

Braga holds a balanced profile. Streets: mean comparable risk 72.0 per km over 1,389 street-km (dimension rank 190/219), with 17.1% of the network in the highest-risk band. Journeys: 210 standardised trips average 54.9 (rank 205/219; 90th percentile 86.5). Access: composite low-stress accessibility 0.155 (rank 208/219); 97% of 187,521 residents are locked to high-stress streets at the central threshold, against 434 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, Incline, On-road stress (speed × lanes). Leading protections: Residential street, Cycleway/Sidewalk.

---

219. BRAILA RO · INDEX 0.4 · INTERVAL 218–219 · TIER 3 (HIGHER ASSESSED RISK) · TIER STABILITY 100%

Braila holds a balanced profile. Streets: mean comparable risk 84.6 per km over 400 street-km (dimension rank 219/219), with 39.7% of the network in the highest-risk band. Journeys: 209 standardised trips average 75.7 (rank 219/219; 90th percentile 92.8). Access: composite low-stress accessibility 0.138 (rank 213/219); 96% of 151,827 residents are locked to high-stress streets at the central threshold, against 46 km of low-stress network. Its routed journeys score close to (or above) its street average — the router finds little refuge from the ordinary street network (escape-valve rank 211 of 219). Leading risk drivers on its streets: Motor-traffic volume, On-road stress (speed × lanes), 50 km/h limit. Leading protections: Residential street, Signed ≤30 km/h zone.

---

199. BRASOV RO · INDEX 23.1 · INTERVAL 191–201 · TIER 3 (HIGHER ASSESSED RISK) · TIER STABILITY 100%

Brasov holds a balanced profile. Streets: mean comparable risk 71.7 per km over 888 street-km (dimension rank 193/219), with 17.6% of the network in the highest-risk band. Journeys: 210 standardised trips average 52.7 (rank 203/219; 90th percentile 76.4). Access: composite low-stress accessibility 0.263 (rank 172/219); 83% of 235,694 residents are locked to high-stress streets at the central threshold, against 410 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, On-road stress (speed × lanes), Incline. Leading protections: Signed ≤30 km/h zone, Service Road.

---

134. BRATISLAVA SK · INDEX 53.0 · INTERVAL 119–143 · TIER 2 (MIDDLE) · TIER STABILITY 82%

Bratislava holds a balanced profile. Streets: mean comparable risk 60.0 per km over 2,096 street-km (dimension rank 142/219), with 12.3% of the network in the highest-risk band. Journeys: 210 standardised trips average 10.5 (rank 108/219; 90th percentile 43.4). Access: composite low-stress accessibility 0.344 (rank 141/219); 58% of 392,119 residents are locked to high-stress streets at the central threshold, against 1,374 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, Track Grade 4/5, Incline. Leading protections: Cycleway/Sidewalk, Signed ≤30 km/h zone.

---

24. BRAUNSCHWEIG DE · INDEX 84.1 · INTERVAL 23–36 · TIER 1 (LOWER ASSESSED RISK) · TIER STABILITY 100%

Braunschweig holds a balanced profile. Streets: mean comparable risk 50.1 per km over 1,102 street-km (dimension rank 30/219), with 5.9% of the network in the highest-risk band. Journeys: 210 standardised trips average -10.2 (rank 29/219; 90th percentile 8.5). Access: composite low-stress accessibility 0.593 (rank 40/219); 20% of 231,178 residents are locked to high-stress streets at the central threshold, against 1,176 km of low-stress network. Leading risk drivers on its streets: Score cap, Motor-traffic volume, Track Grade 4/5. Leading protections: Bicycle Designated, Signed ≤30 km/h zone.

---

22. BREMEN DE · INDEX 86.4 · INTERVAL 18–36 · TIER 1 (LOWER ASSESSED RISK) · TIER STABILITY 100%

Bremen holds a balanced profile. Streets: mean comparable risk 42.9 per km over 2,429 street-km (dimension rank 11/219), with 5.9% of the network in the highest-risk band. Journeys: 210 standardised trips average -3.0 (rank 48/219; 90th percentile 19.9). Access: composite low-stress accessibility 0.612 (rank 32/219); 31% of 550,217 residents are locked to high-stress streets at the central threshold, against 1,821 km of low-stress network. Leading risk drivers on its streets: 50 km/h limit, Heavy-vehicle access, Motor-traffic volume. Leading protections: Signed ≤30 km/h zone, Bicycle Designated.

---

160. BRESCIA IT · INDEX 44.6 · INTERVAL 103–176 · TIER 3 (HIGHER ASSESSED RISK) · TIER STABILITY 62%

Brescia holds an uneven profile — strongest on journey risk (rank 50), weakest on street-network risk (rank 185). Streets: mean comparable risk 71.1 per km over 862 street-km (dimension rank 185/219), with 15.2% of the network in the highest-risk band. Journeys: 210 standardised trips average -2.3 (rank 50/219; 90th percentile 22.2). Access: composite low-stress accessibility 0.298 (rank 160/219); 81% of 203,969 residents are locked to high-stress streets at the central threshold, against 473 km of low-stress network. Its routed journeys score far below its street average — a working low-risk sub-network the router can exploit (escape-valve rank 13 of 219). Leading risk drivers on its streets: Motor-traffic volume, On-road stress (speed × lanes), Score cap.

Leading protections: Cycleway/Sidewalk, Separated cycle track. Notes: straddles a tier boundary under respecification.

---

**169. BRNO CZ · INDEX 40.1 · INTERVAL 168–175 · TIER 3 (HIGHER ASSESSED RISK) · TIER STABILITY 100%**

Brno holds a balanced profile. Streets: mean comparable risk 66.1 per km over 1,567 street-km (dimension rank 170/219), with 12.3% of the network in the highest-risk band. Journeys: 210 standardised trips average 26.3 (rank 155/219; 90th percentile 50.4). Access: composite low-stress accessibility 0.279 (rank 168/219); 72% of 358,325 residents are locked to high-stress streets at the central threshold, against 1,264 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, Incline, On-road stress (speed × lanes). Leading protections: Signed ≤30 km/h zone, Cycleway/Sidewalk.

---

**15. BRUSSELS BE · INDEX 91.3 · INTERVAL 3–23 · TIER 1 (LOWER ASSESSED RISK) · TIER STABILITY 100%**

Brussels holds an uneven profile — strongest on street-network risk (rank 4), weakest on journey risk (rank 74). Streets: mean comparable risk 41.1 per km over 2,110 street-km (dimension rank 4/219), with 3.1% of the network in the highest-risk band. Journeys: 210 standardised trips average 3.9 (rank 74/219; 90th percentile 20.9). Access: composite low-stress accessibility 0.787 (rank 6/219); 0% of 1,270,904 residents are locked to high-stress streets at the central threshold, against 2,120 km of low-stress network. Leading risk drivers on its streets: Score cap, Incline, Heavy-vehicle access. Leading protections: Signed ≤30 km/h zone, Cycleway/Sidewalk. Notes: top-ten on at least one dimension.

---

**207. BUCHAREST RO · INDEX 16.0 · INTERVAL 201–209 · TIER 3 (HIGHER ASSESSED RISK) · TIER STABILITY 100%**

Bucharest holds a balanced profile. Streets: mean comparable risk 71.9 per km over 3,751 street-km (dimension rank 206/219), with 23.3% of the network in the highest-risk band. Journeys: 210 standardised trips average 60.0 (rank 210/219; 90th percentile 83.3). Access: composite low-stress accessibility 0.246 (rank 179/219); 78% of 1,834,401 residents are locked to high-stress streets at the central threshold, against 1,304 km of low-stress network. Its routed journeys score close to (or above) its street average — the router finds little refuge from the ordinary street network (escape-valve rank 208 of 219). Leading risk drivers on its streets: Motor-traffic volume, On-road stress (speed × lanes), Bucharest City. Leading protections: Signed ≤30 km/h zone, Residential street.

---

**104. BUDAPEST HU · INDEX 59.1 · INTERVAL 78–124 · TIER 2 (MIDDLE) · TIER STABILITY 87%**

Budapest holds an uneven profile — strongest on journey risk (rank 53), weakest on street-network risk (rank 138). Streets: mean comparable risk 59.0 per km over 5,448 street-km (dimension rank 138/219), with 13.2% of the network in the highest-risk band. Journeys: 210 standardised trips average -1.1 (rank 53/219; 90th percentile 27.5). Access: composite low-stress accessibility 0.386 (rank 121/219); 38% of 1,849,171 residents are locked to high-stress streets at the central threshold, against 3,074 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, On-road stress (speed × lanes), 50 km/h limit. Leading protections: Signed ≤30 km/h zone, Signed bike network.

200. BURGAS BG · INDEX 21.8 · INTERVAL 189–209 · TIER 3 (HIGHER ASSESSED RISK) · TIER STABILITY 100%

Burgas holds an uneven profile — strongest on journey risk (rank 153), weakest on low-stress access (rank 209). Streets: mean comparable risk 74.7 per km over 896 street-km (dimension rank 208/219), with 21.1% of the network in the highest-risk band. Journeys: 210 standardised trips average 25.0 (rank 153/219; 90th percentile 68.2). Access: composite low-stress accessibility 0.151 (rank 209/219); 87% of 144,584 residents are locked to high-stress streets at the central threshold, against 343 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, On-road stress (speed × lanes), Track Grade 4/5. Leading protections: Cycleway/Sidewalk, Separated cycle track.

---

217. BUZAU RO · INDEX 1.7 · INTERVAL 215–219 · TIER 3 (HIGHER ASSESSED RISK) · TIER STABILITY 100%

Buzau holds a balanced profile. Streets: mean comparable risk 79.7 per km over 349 street-km (dimension rank 218/219), with 30.8% of the network in the highest-risk band. Journeys: 210 standardised trips average 74.0 (rank 217/219; 90th percentile 90.0). Access: composite low-stress accessibility 0.165 (rank 206/219); 99% of 105,005 residents are locked to high-stress streets at the central threshold, against 46 km of low-stress network. Its routed journeys score close to (or above) its street average — the router finds little refuge from the ordinary street network (escape-valve rank 213 of 219). Leading risk drivers on its streets: Motor-traffic volume, On-road stress (speed × lanes), 50 km/h limit. Leading protections: Signed ≤30 km/h zone, Service Road.

---

142. BYDGOSZCZ PL · INDEX 51.0 · INTERVAL 120–155 · TIER 2 (MIDDLE) · TIER STABILITY 64%

Bydgoszcz holds an uneven profile — strongest on journey risk (rank 94), weakest on street-network risk (rank 160). Streets: mean comparable risk 63.0 per km over 1,494 street-km (dimension rank 160/219), with 13.7% of the network in the highest-risk band. Journeys: 210 standardised trips average 8.6 (rank 94/219; 90th percentile 32.3). Access: composite low-stress accessibility 0.353 (rank 135/219); 92% of 339,941 residents are locked to high-stress streets at the central threshold, against 741 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, On-road stress (speed × lanes), Track Grade 4/5. Leading protections: Cycleway/Sidewalk, Separated cycle track. Notes: straddles a tier boundary under respecification.

---

192. CAGLIARI IT · INDEX 27.6 · INTERVAL 184–196 · TIER 3 (HIGHER ASSESSED RISK) · TIER STABILITY 100%

Cagliari holds a balanced profile. Streets: mean comparable risk 70.5 per km over 637 street-km (dimension rank 200/219), with 21.5% of the network in the highest-risk band. Journeys: 210 standardised trips average 28.2 (rank 161/219; 90th percentile 72.0). Access: composite low-stress accessibility 0.230 (rank 187/219); 100% of 134,639 residents are locked to high-stress streets at the central threshold, against 194 km of low-stress network. Leading risk drivers on its streets: On-road stress (speed × lanes), Motor-traffic volume, Heavy-vehicle access. Leading protections: Cycleway/Sidewalk, Separated cycle track.

---

150. CARTAGENA ES · INDEX 47.7 · INTERVAL 135–165 · TIER 3 (HIGHER ASSESSED RISK) · TIER STABILITY 66%

Cartagena holds an uneven profile — strongest on street-network risk (rank 107), weakest on journey risk (rank 175). Streets: mean comparable risk 60.3 per km over 2,051 street-km (dimension rank 107/219), with 7.8% of the network in the highest-risk band. Journeys: 210 standardised trips average 33.2 (rank 175/219; 90th percentile 58.9). Access: composite low-stress accessibility 0.329 (rank 150/219); 25% of 247,214 residents are locked to high-stress streets at the central threshold, against 1,561 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, Incline, Heavy-vehicle access. Leading protections: Signed  $\leq 30$  km/h zone, Cycleway/Sidewalk. Notes: straddles a tier boundary under respecification.

---

**209. CATANIA IT** · INDEX 15.3 · INTERVAL 193–214 · TIER 3 (HIGHER ASSESSED RISK) · TIER STABILITY 100%

Catania holds a balanced profile. Streets: mean comparable risk 65.8 per km over 1,369 street-km (dimension rank 178/219), with 16.1% of the network in the highest-risk band. Journeys: 210 standardised trips average 75.5 (rank 214/219; 90th percentile 96.4). Access: composite low-stress accessibility 0.104 (rank 216/219); 100% of 286,809 residents are locked to high-stress streets at the central threshold, against 200 km of low-stress network. Its routed journeys score close to (or above) its street average — the router finds little refuge from the ordinary street network (escape-valve rank 217 of 219). Leading risk drivers on its streets: Motor-traffic volume, Heavy-vehicle access, On-road stress (speed  $\times$  lanes). Leading protections: Residential street, Service Road.

---

**187. CHARLEROI BE** · INDEX 31.0 · INTERVAL 178–192 · TIER 3 (HIGHER ASSESSED RISK) · TIER STABILITY 100%

Charleroi holds a balanced profile. Streets: mean comparable risk 69.0 per km over 1,041 street-km (dimension rank 188/219), with 19.6% of the network in the highest-risk band. Journeys: 210 standardised trips average 23.7 (rank 150/219; 90th percentile 56.8). Access: composite low-stress accessibility 0.222 (rank 188/219); 90% of 207,385 residents are locked to high-stress streets at the central threshold, against 290 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, On-road stress (speed  $\times$  lanes), 50 km/h limit. Leading protections: Cycleway/Sidewalk, Separated cycle track.

---

**167. CHEMNITZ DE** · INDEX 41.5 · INTERVAL 157–176 · TIER 3 (HIGHER ASSESSED RISK) · TIER STABILITY 96%

Chemnitz holds an uneven profile — strongest on street-network risk (rank 131), weakest on journey risk (rank 177). Streets: mean comparable risk 60.1 per km over 1,644 street-km (dimension rank 131/219), with 11.2% of the network in the highest-risk band. Journeys: 210 standardised trips average 34.6 (rank 177/219; 90th percentile 54.5). Access: composite low-stress accessibility 0.257 (rank 173/219); 87% of 228,532 residents are locked to high-stress streets at the central threshold, against 922 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, Incline, On-road stress (speed  $\times$  lanes). Leading protections: Signed  $\leq 30$  km/h zone, Service Road.

---

**110. CLERMONT-FERRAND FR** · INDEX 58.6 · INTERVAL 93–128 · TIER 2 (MIDDLE) · TIER STABILITY 100%

Clermont-Ferrand holds an uneven profile — strongest on low-stress access (rank 79), weakest on street-network risk (rank 125). Streets: mean comparable risk 57.7 per km over 629 street-km (dimension rank

125/219), with 13.3% of the network in the highest-risk band. Journeys: 210 standardised trips average 16.0 (rank 124/219; 90th percentile 37.6). Access: composite low-stress accessibility 0.465 (rank 79/219); 9% of 158,380 residents are locked to high-stress streets at the central threshold, against 358 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, Heavy-vehicle access, Incline. Leading protections: Signed  $\leq 30$  km/h zone, Cycleway/Sidewalk.

---

**179. CLUJ-NAPOCA RO** · INDEX 36.1 · INTERVAL 171–183 · TIER 3 (HIGHER ASSESSED RISK) · TIER STABILITY 100%

Cluj-Napoca holds a balanced profile. Streets: mean comparable risk 65.4 per km over 928 street-km (dimension rank 161/219), with 11.0% of the network in the highest-risk band. Journeys: 210 standardised trips average 35.5 (rank 179/219; 90th percentile 59.2). Access: composite low-stress accessibility 0.233 (rank 185/219); 82% of 315,238 residents are locked to high-stress streets at the central threshold, against 491 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, Incline, On-road stress (speed  $\times$  lanes). Leading protections: Car-Free Zone, Cycleway/Sidewalk.

---

**71. COLOGNE DE** · INDEX 67.4 · INTERVAL 61–91 · TIER 1 (LOWER ASSESSED RISK) · TIER STABILITY 60%

Cologne holds an uneven profile — strongest on low-stress access (rank 56), weakest on journey risk (rank 106). Streets: mean comparable risk 55.1 per km over 3,506 street-km (dimension rank 91/219), with 11.2% of the network in the highest-risk band. Journeys: 210 standardised trips average 10.1 (rank 106/219; 90th percentile 26.4). Access: composite low-stress accessibility 0.535 (rank 56/219); 45% of 1,049,893 residents are locked to high-stress streets at the central threshold, against 2,497 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, On-road stress (speed  $\times$  lanes), 50 km/h limit. Leading protections: Bicycle Designated, Signed  $\leq 30$  km/h zone. Notes: straddles a tier boundary under respecification.

---

**203. CONSTANTA RO** · INDEX 18.2 · INTERVAL 202–207 · TIER 3 (HIGHER ASSESSED RISK) · TIER STABILITY 100%

Constanta holds a balanced profile. Streets: mean comparable risk 71.6 per km over 946 street-km (dimension rank 202/219), with 21.1% of the network in the highest-risk band. Journeys: 210 standardised trips average 49.0 (rank 200/219; 90th percentile 71.6). Access: composite low-stress accessibility 0.188 (rank 202/219); 99% of 267,291 residents are locked to high-stress streets at the central threshold, against 180 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, On-road stress (speed  $\times$  lanes), 50 km/h limit. Leading protections: Signed  $\leq 30$  km/h zone, Service Road.

---

**17. COPENHAGEN DK** · INDEX 89.0 · INTERVAL 8–28 · TIER 1 (LOWER ASSESSED RISK) · TIER STABILITY 100%

Copenhagen holds a balanced profile. Streets: mean comparable risk 43.5 per km over 1,370 street-km (dimension rank 5/219), with 3.2% of the network in the highest-risk band. Journeys: 210 standardised trips average -8.7 (rank 35/219; 90th percentile 9.1). Access: composite low-stress accessibility 0.592 (rank 41/219); 2% of 646,742 residents are locked to high-stress streets at the central threshold, against 751 km of low-stress network. Leading risk drivers on its streets: Score cap, 50 km/h limit, On-road stress (speed  $\times$  lanes). Leading protections: Separated cycle track, Cycleway/Sidewalk. Notes: top-ten on at least one dimension.

182. CORK IE · INDEX 33.0 · INTERVAL 179–189 · TIER 3 (HIGHER ASSESSED RISK) · TIER STABILITY 100%

Cork holds a balanced profile. Streets: mean comparable risk 63.1 per km over 1,908 street-km (dimension rank 173/219), with 17.4% of the network in the highest-risk band. Journeys: 210 standardised trips average 38.0 (rank 182/219; 90th percentile 62.5). Access: composite low-stress accessibility 0.236 (rank 184/219); 96% of 193,838 residents are locked to high-stress streets at the central threshold, against 535 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, On-road stress (speed × lanes), 50 km/h limit. Leading protections: Cycleway/Sidewalk, Signed ≤30 km/h zone.

---

213. CRAIOVA RO · INDEX 9.2 · INTERVAL 211–214 · TIER 3 (HIGHER ASSESSED RISK) · TIER STABILITY 100%

Craiova holds a balanced profile. Streets: mean comparable risk 73.3 per km over 834 street-km (dimension rank 210/219), with 29.3% of the network in the highest-risk band. Journeys: 210 standardised trips average 58.1 (rank 208/219; 90th percentile 84.6). Access: composite low-stress accessibility 0.159 (rank 207/219); 98% of 238,912 residents are locked to high-stress streets at the central threshold, against 140 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, On-road stress (speed × lanes), 50 km/h limit. Leading protections: Signed ≤30 km/h zone, Residential street.

---

163. CZĘSTOCHOWA PL · INDEX 42.9 · INTERVAL 135–182 · TIER 3 (HIGHER ASSESSED RISK) · TIER STABILITY 83%

Częstochowa holds an uneven profile — strongest on journey risk (rank 109), weakest on street-network risk (rank 191). Streets: mean comparable risk 70.0 per km over 1,042 street-km (dimension rank 191/219), with 19.4% of the network in the highest-risk band. Journeys: 210 standardised trips average 10.8 (rank 109/219; 90th percentile 34.5). Access: composite low-stress accessibility 0.385 (rank 123/219); 94% of 215,456 residents are locked to high-stress streets at the central threshold, against 382 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, On-road stress (speed × lanes), 50 km/h limit. Leading protections: Bicycle Designated, Cycle Infra Car-Free.

---

55. CÓRDOBA ES · INDEX 71.7 · INTERVAL 44–70 · TIER 1 (LOWER ASSESSED RISK) · TIER STABILITY 97%

Córdoba holds an uneven profile — strongest on street-network risk (rank 42), weakest on journey risk (rank 104). Streets: mean comparable risk 53.3 per km over 1,640 street-km (dimension rank 42/219), with 4.9% of the network in the highest-risk band. Journeys: 210 standardised trips average 10.0 (rank 104/219; 90th percentile 35.7). Access: composite low-stress accessibility 0.512 (rank 61/219); 10% of 343,726 residents are locked to high-stress streets at the central threshold, against 2,390 km of low-stress network. Leading risk drivers on its streets: Incline, Heavy-vehicle access, Motor-traffic volume. Leading protections: Cycleway/Sidewalk, Signed ≤30 km/h zone.

---

92. DEBRECEN HU · INDEX 60.5 · INTERVAL 71–119 · TIER 2 (MIDDLE) · TIER STABILITY 83%

Debrecen holds an uneven profile — strongest on journey risk (rank 49), weakest on street-network risk (rank 132). Streets: mean comparable risk 64.0 per km over 1,208 street-km (dimension rank 132/219), with 6.9% of

the network in the highest-risk band. Journeys: 210 standardised trips average -2.6 (rank 49/219; 90th percentile 20.6). Access: composite low-stress accessibility 0.396 (rank 116/219); 69% of 260,634 residents are locked to high-stress streets at the central threshold, against 654 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, On-road stress (speed × lanes), 50 km/h limit. Leading protections: Cycleway/Sidewalk, Separated cycle track.

---

**186. DIJON FR · INDEX 31.2 · INTERVAL 174–196 · TIER 3 (HIGHER ASSESSED RISK) · TIER STABILITY 100%**

Dijon holds an uneven profile — strongest on low-stress access (rank 140), weakest on journey risk (rank 190). Streets: mean comparable risk 68.5 per km over 584 street-km (dimension rank 189/219), with 20.7% of the network in the highest-risk band. Journeys: 210 standardised trips average 42.2 (rank 190/219; 90th percentile 74.2). Access: composite low-stress accessibility 0.346 (rank 140/219); 16% of 143,194 residents are locked to high-stress streets at the central threshold, against 216 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, On-road stress (speed × lanes), 50 km/h limit. Leading protections: Cycleway/Sidewalk, Separated cycle track.

---

**127. DORTMUND DE · INDEX 55.1 · INTERVAL 103–138 · TIER 2 (MIDDLE) · TIER STABILITY 90%**

Dortmund holds an uneven profile — strongest on street-network risk (rank 79), weakest on low-stress access (rank 143). Streets: mean comparable risk 54.8 per km over 2,853 street-km (dimension rank 79/219), with 10.1% of the network in the highest-risk band. Journeys: 210 standardised trips average 20.3 (rank 140/219; 90th percentile 42.2). Access: composite low-stress accessibility 0.342 (rank 143/219); 64% of 581,910 residents are locked to high-stress streets at the central threshold, against 1,808 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, On-road stress (speed × lanes), 50 km/h limit. Leading protections: Signed ≤30 km/h zone, Bicycle Designated.

---

**144. DRESDEN DE · INDEX 50.7 · INTERVAL 113–160 · TIER 2 (MIDDLE) · TIER STABILITY 62%**

Dresden holds an uneven profile — strongest on street-network risk (rank 78), weakest on journey risk (rank 173). Streets: mean comparable risk 55.2 per km over 2,318 street-km (dimension rank 78/219), with 9.6% of the network in the highest-risk band. Journeys: 210 standardised trips average 32.5 (rank 173/219; 90th percentile 50.9). Access: composite low-stress accessibility 0.332 (rank 148/219); 73% of 520,280 residents are locked to high-stress streets at the central threshold, against 1,816 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, Track Grade 4/5, Incline. Leading protections: Signed ≤30 km/h zone, Car-Free Zone. Notes: straddles a tier boundary under respecification.

---

**131. DUBLIN IE · INDEX 54.0 · INTERVAL 107–143 · TIER 2 (MIDDLE) · TIER STABILITY 84%**

Dublin holds an uneven profile — strongest on street-network risk (rank 81), weakest on low-stress access (rank 153). Streets: mean comparable risk 53.3 per km over 1,947 street-km (dimension rank 81/219), with 12.1% of the network in the highest-risk band. Journeys: 210 standardised trips average 19.7 (rank 137/219; 90th percentile 45.6). Access: composite low-stress accessibility 0.321 (rank 153/219); 36% of 533,623 residents are locked to high-stress streets at the central threshold, against 1,147 km of low-stress network. Leading risk

drivers on its streets: Motor-traffic volume, On-road stress (speed × lanes), Heavy-vehicle access. Leading protections: Signed ≤30 km/h zone, Cycleway/Sidewalk.

---

48. DUISBURG DE · INDEX 74.9 · INTERVAL 38–62 · TIER 1 (LOWER ASSESSED RISK) · TIER STABILITY 100%

Duisburg holds an uneven profile — strongest on street-network risk (rank 34), weakest on journey risk (rank 92). Streets: mean comparable risk 48.8 per km over 1,827 street-km (dimension rank 34/219), with 8.1% of the network in the highest-risk band. Journeys: 210 standardised trips average 7.9 (rank 92/219; 90th percentile 26.0). Access: composite low-stress accessibility 0.538 (rank 55/219); 12% of 467,099 residents are locked to high-stress streets at the central threshold, against 1,513 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, On-road stress (speed × lanes), 50 km/h limit. Leading protections: Signed ≤30 km/h zone, Bicycle Designated.

---

58. DÜSSELDORF DE · INDEX 70.5 · INTERVAL 55–73 · TIER 1 (LOWER ASSESSED RISK) · TIER STABILITY 96%

Düsseldorf holds a balanced profile. Streets: mean comparable risk 53.0 per km over 1,941 street-km (dimension rank 60/219), with 8.5% of the network in the highest-risk band. Journeys: 210 standardised trips average 8.7 (rank 95/219; 90th percentile 23.5). Access: composite low-stress accessibility 0.523 (rank 59/219); 53% of 610,480 residents are locked to high-stress streets at the central threshold, against 1,518 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, On-road stress (speed × lanes), 50 km/h limit. Leading protections: Bicycle Designated, Signed ≤30 km/h zone.

---

13. EINDHOVEN NL · INDEX 92.4 · INTERVAL 9–23 · TIER 1 (LOWER ASSESSED RISK) · TIER STABILITY 100%

Eindhoven holds a balanced profile. Streets: mean comparable risk 46.8 per km over 995 street-km (dimension rank 10/219), with 3.5% of the network in the highest-risk band. Journeys: 210 standardised trips average -12.3 (rank 20/219; 90th percentile 1.4). Access: composite low-stress accessibility 0.665 (rank 19/219); 9% of 221,413 residents are locked to high-stress streets at the central threshold, against 1,084 km of low-stress network. Leading risk drivers on its streets: Score cap, Heavy-vehicle access, Motor-traffic volume. Leading protections: Cycleway/Sidewalk, Separated cycle track. Notes: top-ten on at least one dimension.

---

69. ELCHE ES · INDEX 68.2 · INTERVAL 65–76 · TIER 1 (LOWER ASSESSED RISK) · TIER STABILITY 73%

Elche holds a balanced profile. Streets: mean comparable risk 56.6 per km over 1,771 street-km (dimension rank 68/219), with 6.0% of the network in the highest-risk band. Journeys: 210 standardised trips average 5.4 (rank 79/219; 90th percentile 44.8). Access: composite low-stress accessibility 0.478 (rank 74/219); 7% of 260,857 residents are locked to high-stress streets at the central threshold, against 1,364 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, Heavy-vehicle access, Incline. Leading protections: Cycleway/Sidewalk, Signed ≤30 km/h zone.

---

162. ERFURT DE · INDEX 43.7 · INTERVAL 158–169 · TIER 3 (HIGHER ASSESSED RISK) · TIER STABILITY 98%

Erfurt holds a balanced profile. Streets: mean comparable risk 61.3 per km over 1,182 street-km (dimension rank 154/219), with 13.9% of the network in the highest-risk band. Journeys: 210 standardised trips average

27.0 (rank 157/219; 90th percentile 49.9). Access: composite low-stress accessibility 0.304 (rank 156/219); 80% of 202,557 residents are locked to high-stress streets at the central threshold, against 948 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, Track Grade 4/5, Incline. Leading protections: Signed  $\leq 30$  km/h zone, Bicycle Designated.

---

8. ESPOO FI · INDEX 94.7 · INTERVAL 3–14 · TIER 1 (LOWER ASSESSED RISK) · TIER STABILITY 100%

Espoo holds a balanced profile. Streets: mean comparable risk 49.0 per km over 2,122 street-km (dimension rank 12/219), with 1.8% of the network in the highest-risk band. Journeys: 210 standardised trips average -27.6 (rank 3/219; 90th percentile -11.2). Access: composite low-stress accessibility 0.667 (rank 17/219); 11% of 293,108 residents are locked to high-stress streets at the central threshold, against 2,846 km of low-stress network. Its routed journeys score far below its street average — a working low-risk sub-network the router can exploit (escape-valve rank 7 of 219). Leading risk drivers on its streets: Score cap, Incline, MTB  $\geq 1$ . Leading protections: Cycleway/Sidewalk, Separated cycle track. Notes: top-ten overall; top-ten on at least one dimension.

---

146. ESSEN DE · INDEX 50.1 · INTERVAL 133–157 · TIER 2 (MIDDLE) · TIER STABILITY 57%

Essen holds an uneven profile — strongest on street-network risk (rank 110), weakest on journey risk (rank 158). Streets: mean comparable risk 55.7 per km over 2,360 street-km (dimension rank 110/219), with 13.7% of the network in the highest-risk band. Journeys: 210 standardised trips average 27.4 (rank 158/219; 90th percentile 44.4). Access: composite low-stress accessibility 0.346 (rank 139/219); 46% of 547,081 residents are locked to high-stress streets at the central threshold, against 1,598 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, Incline, On-road stress (speed  $\times$  lanes). Leading protections: Signed  $\leq 30$  km/h zone, Bicycle Designated. Notes: straddles a tier boundary under respecification.

---

120. FLORENCE IT · INDEX 56.1 · INTERVAL 88–137 · TIER 2 (MIDDLE) · TIER STABILITY 85%

Florence holds an uneven profile — strongest on journey risk (rank 56), weakest on street-network risk (rank 144). Streets: mean comparable risk 61.9 per km over 1,095 street-km (dimension rank 144/219), with 11.0% of the network in the highest-risk band. Journeys: 210 standardised trips average -0.6 (rank 56/219; 90th percentile 22.2). Access: composite low-stress accessibility 0.344 (rank 142/219); 79% of 351,064 residents are locked to high-stress streets at the central threshold, against 537 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, Heavy-vehicle access, Incline. Leading protections: Cycleway/Sidewalk, Separated cycle track.

---

122. FRANKFURT DE · INDEX 56.1 · INTERVAL 114–135 · TIER 2 (MIDDLE) · TIER STABILITY 100%

Frankfurt holds a balanced profile. Streets: mean comparable risk 56.9 per km over 2,025 street-km (dimension rank 126/219), with 14.3% of the network in the highest-risk band. Journeys: 210 standardised trips average 17.2 (rank 131/219; 90th percentile 35.1). Access: composite low-stress accessibility 0.424 (rank 99/219); 22% of 708,360 residents are locked to high-stress streets at the central threshold, against 1,645 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, Track Grade 4/5, On-road stress (speed  $\times$  lanes). Leading protections: Signed  $\leq 30$  km/h zone, Bicycle Designated.

44. FREIBURG DE · INDEX 76.1 · INTERVAL 37–58 · TIER 1 (LOWER ASSESSED RISK) · TIER STABILITY 100%

Freiburg holds an uneven profile — strongest on street-network risk (rank 38), weakest on journey risk (rank 83). Streets: mean comparable risk 50.0 per km over 881 street-km (dimension rank 38/219), with 7.8% of the network in the highest-risk band. Journeys: 210 standardised trips average 5.5 (rank 83/219; 90th percentile 29.0). Access: composite low-stress accessibility 0.560 (rank 45/219); 23% of 195,183 residents are locked to high-stress streets at the central threshold, against 1,236 km of low-stress network. Leading risk drivers on its streets: Track Grade 4/5, Incline, Motor-traffic volume. Leading protections: Car-Free Zone, Bicycle Designated.

---

214. GALATI RO · INDEX 7.0 · INTERVAL 212–215 · TIER 3 (HIGHER ASSESSED RISK) · TIER STABILITY 100%

Galati holds a balanced profile. Streets: mean comparable risk 78.4 per km over 662 street-km (dimension rank 211/219), with 22.6% of the network in the highest-risk band. Journeys: 210 standardised trips average 58.4 (rank 209/219; 90th percentile 86.5). Access: composite low-stress accessibility 0.112 (rank 217/219); 96% of 212,138 residents are locked to high-stress streets at the central threshold, against 103 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, On-road stress (speed × lanes), 50 km/h limit. Leading protections: Service Road, Cycleway/Sidewalk.

---

87. GDAŃSK PL · INDEX 61.4 · INTERVAL 88–103 · TIER 2 (MIDDLE) · TIER STABILITY 96%

Gdańsk holds a balanced profile. Streets: mean comparable risk 57.1 per km over 2,219 street-km (dimension rank 95/219), with 9.2% of the network in the highest-risk band. Journeys: 210 standardised trips average 9.1 (rank 100/219; 90th percentile 38.0). Access: composite low-stress accessibility 0.417 (rank 106/219); 80% of 441,548 residents are locked to high-stress streets at the central threshold, against 1,400 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, Incline, Track Grade 4/5. Leading protections: Signed ≤30 km/h zone, Bicycle Designated.

---

119. GDYNIA PL · INDEX 56.5 · INTERVAL 114–130 · TIER 2 (MIDDLE) · TIER STABILITY 99%

Gdynia holds a balanced profile. Streets: mean comparable risk 60.7 per km over 939 street-km (dimension rank 111/219), with 8.0% of the network in the highest-risk band. Journeys: 210 standardised trips average 17.2 (rank 130/219; 90th percentile 41.2). Access: composite low-stress accessibility 0.409 (rank 111/219); 64% of 238,919 residents are locked to high-stress streets at the central threshold, against 818 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, Incline, Track Grade 4/5. Leading protections: Signed ≤30 km/h zone, Car-Free Zone.

---

100. GELSENKIRCHEN DE · INDEX 59.6 · INTERVAL 93–121 · TIER 2 (MIDDLE) · TIER STABILITY 100%

Gelsenkirchen holds an uneven profile — strongest on street-network risk (rank 84), weakest on journey risk (rank 139). Streets: mean comparable risk 54.7 per km over 1,092 street-km (dimension rank 84/219), with 11.0% of the network in the highest-risk band. Journeys: 210 standardised trips average 20.1 (rank 139/219; 90th percentile 39.3). Access: composite low-stress accessibility 0.439 (rank 92/219); 50% of 239,785 residents are locked to high-stress streets at the central threshold, against 730 km of low-stress network. Leading risk

drivers on its streets: Motor-traffic volume, On-road stress (speed × lanes), 50 km/h limit. Leading protections: Bicycle Designated, Signed ≤30 km/h zone.

---

112. GENEVA CH · INDEX 58.1 · INTERVAL 93–126 · TIER 2 (MIDDLE) · TIER STABILITY 97%

Geneva holds an uneven profile — strongest on street-network risk (rank 74), weakest on journey risk (rank 133). Streets: mean comparable risk 53.5 per km over 299 street-km (dimension rank 74/219), with 10.9% of the network in the highest-risk band. Journeys: 210 standardised trips average 18.3 (rank 133/219; 90th percentile 35.4). Access: composite low-stress accessibility 0.381 (rank 128/219); 0% of 202,402 residents are locked to high-stress streets at the central threshold, against 166 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, On-road stress (speed × lanes), 50 km/h limit. Leading protections: Signed bike network, Separated cycle track.

---

175. GENOA IT · INDEX 37.5 · INTERVAL 157–191 · TIER 3 (HIGHER ASSESSED RISK) · TIER STABILITY 97%

Genoa holds an uneven profile — strongest on street-network risk (rank 118), weakest on journey risk (rank 199). Streets: mean comparable risk 60.1 per km over 1,739 street-km (dimension rank 118/219), with 9.4% of the network in the highest-risk band. Journeys: 210 standardised trips average 48.5 (rank 199/219; 90th percentile 77.8). Access: composite low-stress accessibility 0.245 (rank 180/219); 81% of 546,709 residents are locked to high-stress streets at the central threshold, against 1,203 km of low-stress network. Its routed journeys score close to (or above) its street average — the router finds little refuge from the ordinary street network (escape-valve rank 209 of 219). Leading risk drivers on its streets: Incline, Motor-traffic volume, Heavy-vehicle access. Leading protections: Pedestrian Zone, Residential street.

---

23. GHENT BE · INDEX 85.5 · INTERVAL 21–37 · TIER 1 (LOWER ASSESSED RISK) · TIER STABILITY 100%

Ghent holds a balanced profile. Streets: mean comparable risk 47.0 per km over 1,392 street-km (dimension rank 17/219), with 5.8% of the network in the highest-risk band. Journeys: 210 standardised trips average -4.2 (rank 45/219; 90th percentile 14.2). Access: composite low-stress accessibility 0.612 (rank 33/219); 17% of 259,126 residents are locked to high-stress streets at the central threshold, against 1,014 km of low-stress network. Leading risk drivers on its streets: Score cap, Motor-traffic volume, On-road stress (speed × lanes). Leading protections: Cycleway/Sidewalk, Separated cycle track.

---

70. GIJÓN ES · INDEX 67.9 · INTERVAL 48–84 · TIER 1 (LOWER ASSESSED RISK) · TIER STABILITY 69%

Gijón holds an uneven profile — strongest on street-network risk (rank 40), weakest on journey risk (rank 99). Streets: mean comparable risk 51.9 per km over 1,113 street-km (dimension rank 40/219), with 5.9% of the network in the highest-risk band. Journeys: 210 standardised trips average 9.0 (rank 99/219; 90th percentile 41.2). Access: composite low-stress accessibility 0.426 (rank 96/219); 8% of 273,456 residents are locked to high-stress streets at the central threshold, against 910 km of low-stress network. Leading risk drivers on its streets: Incline, Heavy-vehicle access, Motor-traffic volume. Leading protections: Signed ≤30 km/h zone, Cycleway/Sidewalk.

---

128. GOTHENBURG SE · INDEX 55.0 · INTERVAL 96–150 · TIER 2 (MIDDLE) · TIER STABILITY 87%

Gothenburg holds an uneven profile — strongest on journey risk (rank 68), weakest on street-network risk (rank 166). Streets: mean comparable risk 65.6 per km over 3,751 street-km (dimension rank 166/219), with 12.0% of the network in the highest-risk band. Journeys: 208 standardised trips average 1.4 (rank 68/219; 90th percentile 20.3). Access: composite low-stress accessibility 0.402 (rank 114/219); 82% of 563,683 residents are locked to high-stress streets at the central threshold, against 2,161 km of low-stress network. Leading risk drivers on its streets: On-road stress (speed × lanes), Motor-traffic volume, Incline. Leading protections: Bicycle Designated, Cycle Infra Car-Free.

---

115. GRANADA ES · INDEX 57.9 · INTERVAL 90–140 · TIER 2 (MIDDLE) · TIER STABILITY 98%

Granada holds an uneven profile — strongest on street-network risk (rank 73), weakest on journey risk (rank 166). Streets: mean comparable risk 56.0 per km over 631 street-km (dimension rank 73/219), with 7.7% of the network in the highest-risk band. Journeys: 210 standardised trips average 29.6 (rank 166/219; 90th percentile 55.8). Access: composite low-stress accessibility 0.440 (rank 91/219); 10% of 259,109 residents are locked to high-stress streets at the central threshold, against 740 km of low-stress network. Leading risk drivers on its streets: Incline, Heavy-vehicle access, Motor-traffic volume. Leading protections: Signed ≤30 km/h zone, Pedestrian Zone.

---

63. GRAZ AT · INDEX 69.5 · INTERVAL 53–80 · TIER 1 (LOWER ASSESSED RISK) · TIER STABILITY 82%

Graz holds an uneven profile — strongest on journey risk (rank 44), weakest on street-network risk (rank 92). Streets: mean comparable risk 52.8 per km over 1,276 street-km (dimension rank 92/219), with 13.9% of the network in the highest-risk band. Journeys: 210 standardised trips average -4.7 (rank 44/219; 90th percentile 22.4). Access: composite low-stress accessibility 0.486 (rank 69/219); 25% of 302,143 residents are locked to high-stress streets at the central threshold, against 966 km of low-stress network. Leading risk drivers on its streets: Legal Dismount, Motor-traffic volume, Incline. Leading protections: Signed ≤30 km/h zone, Cycleway/Sidewalk.

---

32. GRENOBLE FR · INDEX 80.0 · INTERVAL 27–48 · TIER 1 (LOWER ASSESSED RISK) · TIER STABILITY 100%

Grenoble holds an uneven profile — strongest on low-stress access (rank 21), weakest on journey risk (rank 72). Streets: mean comparable risk 51.6 per km over 326 street-km (dimension rank 51/219), with 9.0% of the network in the highest-risk band. Journeys: 210 standardised trips average 3.4 (rank 72/219; 90th percentile 29.1). Access: composite low-stress accessibility 0.660 (rank 21/219); 0% of 135,756 residents are locked to high-stress streets at the central threshold, against 248 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, Score cap, Heavy-vehicle access. Leading protections: Signed ≤30 km/h zone, Cycleway/Sidewalk.

---

29. GRONINGEN NL · INDEX 82.6 · INTERVAL 26–37 · TIER 1 (LOWER ASSESSED RISK) · TIER STABILITY 100%

Groningen holds a balanced profile. Streets: mean comparable risk 52.3 per km over 1,055 street-km (dimension rank 50/219), with 7.8% of the network in the highest-risk band. Journeys: 210 standardised trips average -10.1 (rank 30/219; 90th percentile 3.3). Access: composite low-stress accessibility 0.622 (rank 31/219); 10% of 230,666 residents are locked to high-stress streets at the central threshold, against 1,064 km of low-

stress network. Leading risk drivers on its streets: Score cap, On-road stress (speed × lanes), Heavy-vehicle access. Leading protections: Cycleway/Sidewalk, Separated cycle track.

---

**4. HAARLEM NL · INDEX 96.3 · INTERVAL 1–11 · TIER 1 (LOWER ASSESSED RISK) · TIER STABILITY 100%**

Haarlem holds a balanced profile. Streets: mean comparable risk 42.3 per km over 415 street-km (dimension rank 2/219), with 1.8% of the network in the highest-risk band. Journeys: 210 standardised trips average -9.0 (rank 33/219; 90th percentile 4.3). Access: composite low-stress accessibility 0.841 (rank 2/219); 0% of 147,840 residents are locked to high-stress streets at the central threshold, against 517 km of low-stress network. Leading risk drivers on its streets: Score cap, Heavy-vehicle access, Motor-traffic volume. Leading protections: Cycleway/Sidewalk, Separated cycle track. Notes: top-ten overall; top-ten on at least one dimension.

---

**78. HALLE (SAALE) DE · INDEX 65.1 · INTERVAL 70–94 · TIER 2 (MIDDLE) · TIER STABILITY 61%**

Halle (Saale) holds an uneven profile — strongest on street-network risk (rank 59), weakest on journey risk (rank 107). Streets: mean comparable risk 53.9 per km over 944 street-km (dimension rank 59/219), with 7.3% of the network in the highest-risk band. Journeys: 210 standardised trips average 10.1 (rank 107/219; 90th percentile 29.2). Access: composite low-stress accessibility 0.424 (rank 101/219); 42% of 216,860 residents are locked to high-stress streets at the central threshold, against 668 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, On-road stress (speed × lanes), 50 km/h limit. Leading protections: Signed ≤30 km/h zone, Cycleway/Sidewalk. Notes: straddles a tier boundary under respecification.

---

**84. HAMBURG DE · INDEX 63.1 · INTERVAL 68–112 · TIER 2 (MIDDLE) · TIER STABILITY 74%**

Hamburg holds an uneven profile — strongest on street-network risk (rank 48), weakest on journey risk (rank 138). Streets: mean comparable risk 50.3 per km over 6,052 street-km (dimension rank 48/219), with 9.7% of the network in the highest-risk band. Journeys: 210 standardised trips average 20.1 (rank 138/219; 90th percentile 40.4). Access: composite low-stress accessibility 0.425 (rank 97/219); 47% of 1,673,225 residents are locked to high-stress streets at the central threshold, against 3,774 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, On-road stress (speed × lanes), 50 km/h limit. Leading protections: Signed ≤30 km/h zone, Bicycle Designated.

---

**20. HANOVER DE · INDEX 87.7 · INTERVAL 18–28 · TIER 1 (LOWER ASSESSED RISK) · TIER STABILITY 100%**

Hanover holds a balanced profile. Streets: mean comparable risk 49.1 per km over 1,845 street-km (dimension rank 25/219), with 5.3% of the network in the highest-risk band. Journeys: 210 standardised trips average -10.9 (rank 26/219; 90th percentile 2.1). Access: composite low-stress accessibility 0.637 (rank 26/219); 15% of 482,592 residents are locked to high-stress streets at the central threshold, against 1,752 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, Score cap, On-road stress (speed × lanes). Leading protections: Bicycle Designated, Signed ≤30 km/h zone.

---

**2. HELSINKI FI · INDEX 97.0 · INTERVAL 1–9 · TIER 1 (LOWER ASSESSED RISK) · TIER STABILITY 100%**

Helsinki holds a balanced profile. Streets: mean comparable risk 46.9 per km over 2,030 street-km (dimension rank 7/219), with 0.9% of the network in the highest-risk band. Journeys: 210 standardised trips average -28.9 (rank 1/219; 90th percentile -22.7). Access: composite low-stress accessibility 0.682 (rank 13/219); 1% of 662,727 residents are locked to high-stress streets at the central threshold, against 2,865 km of low-stress network. Its routed journeys score far below its street average — a working low-risk sub-network the router can exploit (escape-valve rank 8 of 219). Leading risk drivers on its streets: Score cap, Incline, MTB  $\geq 1$ . Leading protections: Cycleway/Sidewalk, Separated cycle track. Notes: top-ten overall; top-ten on at least one dimension.

---

215. IASI RO · INDEX 3.3 · INTERVAL 215–218 · TIER 3 (HIGHER ASSESSED RISK) · TIER STABILITY 100%

Iasi holds a balanced profile. Streets: mean comparable risk 77.2 per km over 796 street-km (dimension rank 215/219), with 26.1% of the network in the highest-risk band. Journeys: 210 standardised trips average 63.1 (rank 212/219; 90th percentile 87.0). Access: composite low-stress accessibility 0.116 (rank 218/219); 97% of 267,550 residents are locked to high-stress streets at the central threshold, against 141 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, On-road stress (speed  $\times$  lanes), 50 km/h limit. Leading protections: Residential street, Service Road.

---

102. JEREZ DE LA FRONTERA ES · INDEX 59.3 · INTERVAL 97–117 · TIER 2 (MIDDLE) · TIER STABILITY 98%

Jerez de la Frontera holds a balanced profile. Streets: mean comparable risk 59.2 per km over 1,281 street-km (dimension rank 99/219), with 7.8% of the network in the highest-risk band. Journeys: 210 standardised trips average 9.7 (rank 102/219; 90th percentile 46.1). Access: composite low-stress accessibility 0.391 (rank 119/219); 10% of 233,562 residents are locked to high-stress streets at the central threshold, against 1,330 km of low-stress network. Leading risk drivers on its streets: Incline, Motor-traffic volume, Heavy-vehicle access. Leading protections: Signed  $\leq 30$  km/h zone, Cycleway/Sidewalk.

---

83. KARLSRUHE DE · INDEX 63.7 · INTERVAL 76–106 · TIER 2 (MIDDLE) · TIER STABILITY 91%

Karlsruhe holds an uneven profile — strongest on low-stress access (rank 72), weakest on journey risk (rank 126). Streets: mean comparable risk 54.3 per km over 1,225 street-km (dimension rank 75/219), with 10.1% of the network in the highest-risk band. Journeys: 210 standardised trips average 16.4 (rank 126/219; 90th percentile 32.7). Access: composite low-stress accessibility 0.478 (rank 72/219); 52% of 328,610 residents are locked to high-stress streets at the central threshold, against 1,252 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, Track Grade 4/5, Incline. Leading protections: Signed  $\leq 30$  km/h zone, Car-Free Zone.

---

136. KASSEL DE · INDEX 51.5 · INTERVAL 109–168 · TIER 2 (MIDDLE) · TIER STABILITY 69%

Kassel holds an uneven profile — strongest on street-network risk (rank 85), weakest on journey risk (rank 189). Streets: mean comparable risk 54.5 per km over 988 street-km (dimension rank 85/219), with 11.3% of the network in the highest-risk band. Journeys: 210 standardised trips average 42.3 (rank 189/219; 90th percentile 63.3). Access: composite low-stress accessibility 0.419 (rank 104/219); 47% of 184,814 residents are locked to high-stress streets at the central threshold, against 778 km of low-stress network. Its routed journeys score

close to (or above) its street average — the router finds little refuge from the ordinary street network (escape-valve rank 207 of 219). Leading risk drivers on its streets: Motor-traffic volume, Incline, On-road stress (speed × lanes). Leading protections: Signed ≤30 km/h zone, Bicycle Designated.

---

164. KATOWICE PL · INDEX 42.8 · INTERVAL 159–169 · TIER 3 (HIGHER ASSESSED RISK) · TIER STABILITY 94%

Katowice holds a balanced profile. Streets: mean comparable risk 62.3 per km over 1,327 street-km (dimension rank 149/219), with 11.4% of the network in the highest-risk band. Journeys: 210 standardised trips average 24.4 (rank 152/219; 90th percentile 52.7). Access: composite low-stress accessibility 0.253 (rank 176/219); 91% of 287,559 residents are locked to high-stress streets at the central threshold, against 641 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, On-road stress (speed × lanes), 50 km/h limit. Leading protections: Service Road, Bicycle Designated.

---

171. KAUNAS LT · INDEX 39.8 · INTERVAL 165–176 · TIER 3 (HIGHER ASSESSED RISK) · TIER STABILITY 99%

Kaunas holds a balanced profile. Streets: mean comparable risk 66.1 per km over 1,877 street-km (dimension rank 177/219), with 15.2% of the network in the highest-risk band. Journeys: 210 standardised trips average 22.3 (rank 147/219; 90th percentile 48.2). Access: composite low-stress accessibility 0.285 (rank 165/219); 92% of 239,169 residents are locked to high-stress streets at the central threshold, against 451 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, On-road stress (speed × lanes), 50 km/h limit. Leading protections: Bicycle Designated, Service Road.

---

67. KIEL DE · INDEX 68.3 · INTERVAL 62–82 · TIER 1 (LOWER ASSESSED RISK) · TIER STABILITY 76%

Kiel holds a balanced profile. Streets: mean comparable risk 54.2 per km over 954 street-km (dimension rank 72/219), with 9.6% of the network in the highest-risk band. Journeys: 210 standardised trips average 8.8 (rank 96/219; 90th percentile 31.8). Access: composite low-stress accessibility 0.512 (rank 62/219); 48% of 234,447 residents are locked to high-stress streets at the central threshold, against 711 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, On-road stress (speed × lanes), 50 km/h limit. Leading protections: Bicycle Designated, Signed ≤30 km/h zone.

---

174. KOŠICE SK · INDEX 39.1 · INTERVAL 166–176 · TIER 3 (HIGHER ASSESSED RISK) · TIER STABILITY 100%

Košice holds a balanced profile. Streets: mean comparable risk 64.6 per km over 998 street-km (dimension rank 156/219), with 11.3% of the network in the highest-risk band. Journeys: 210 standardised trips average 28.8 (rank 162/219; 90th percentile 59.8). Access: composite low-stress accessibility 0.241 (rank 182/219); 87% of 247,633 residents are locked to high-stress streets at the central threshold, against 734 km of low-stress network. Leading risk drivers on its streets: Incline, Track Grade 4/5, Motor-traffic volume. Leading protections: Cycleway/Sidewalk, Separated cycle track.

---

129. KRAKÓW PL · INDEX 54.8 · INTERVAL 114–136 · TIER 2 (MIDDLE) · TIER STABILITY 89%

Kraków holds a balanced profile. Streets: mean comparable risk 58.4 per km over 3,271 street-km (dimension rank 106/219), with 9.6% of the network in the highest-risk band. Journeys: 210 standardised trips average 13.2 (rank 114/219; 90th percentile 32.5). Access: composite low-stress accessibility 0.336 (rank 144/219); 87% of 722,627 residents are locked to high-stress streets at the central threshold, against 1,423 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, On-road stress (speed × lanes), 50 km/h limit. Leading protections: Signed ≤30 km/h zone, Bicycle Designated.

---

89. KREFELD DE · INDEX 61.1 · INTERVAL 81–118 · TIER 2 (MIDDLE) · TIER STABILITY 100%

Krefeld holds an uneven profile — strongest on low-stress access (rank 64), weakest on journey risk (rank 136). Streets: mean comparable risk 55.9 per km over 1,167 street-km (dimension rank 104/219), with 12.2% of the network in the highest-risk band. Journeys: 210 standardised trips average 19.1 (rank 136/219; 90th percentile 36.9). Access: composite low-stress accessibility 0.493 (rank 64/219); 31% of 204,885 residents are locked to high-stress streets at the central threshold, against 743 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, On-road stress (speed × lanes), 50 km/h limit. Leading protections: Bicycle Designated, Signed ≤30 km/h zone.

---

154. LAUSANNE CH · INDEX 47.1 · INTERVAL 112–175 · TIER 3 (HIGHER ASSESSED RISK) · TIER STABILITY 57%

Lausanne holds an uneven profile — strongest on street-network risk (rank 70), weakest on journey risk (rank 197). Streets: mean comparable risk 54.5 per km over 457 street-km (dimension rank 70/219), with 8.9% of the network in the highest-risk band. Journeys: 210 standardised trips average 46.6 (rank 197/219; 90th percentile 58.6). Access: composite low-stress accessibility 0.332 (rank 149/219); 80% of 110,030 residents are locked to high-stress streets at the central threshold, against 332 km of low-stress network. Its routed journeys score close to (or above) its street average — the router finds little refuge from the ordinary street network (escape-valve rank 212 of 219). Leading risk drivers on its streets: Incline, On-road stress (speed × lanes), Motor-traffic volume. Leading protections: Signed ≤30 km/h zone, Car-Free Zone. Notes: straddles a tier boundary under respecification.

---

152. LE HAVRE FR · INDEX 47.6 · INTERVAL 144–159 · TIER 3 (HIGHER ASSESSED RISK) · TIER STABILITY 60%

Le Havre holds a balanced profile. Streets: mean comparable risk 59.8 per km over 916 street-km (dimension rank 127/219), with 11.0% of the network in the highest-risk band. Journeys: 210 standardised trips average 20.8 (rank 143/219; 90th percentile 42.2). Access: composite low-stress accessibility 0.283 (rank 166/219); 99% of 160,810 residents are locked to high-stress streets at the central threshold, against 218 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, On-road stress (speed × lanes), 50 km/h limit. Leading protections: Cycleway/Sidewalk, Separated cycle track. Notes: straddles a tier boundary under respecification.

---

5. LEIDEN NL · INDEX 96.2 · INTERVAL 3–13 · TIER 1 (LOWER ASSESSED RISK) · TIER STABILITY 100%

Leiden holds a balanced profile. Streets: mean comparable risk 45.0 per km over 323 street-km (dimension rank 3/219), with 1.8% of the network in the highest-risk band. Journeys: 210 standardised trips average -8.7

(rank 37/219; 90th percentile 10.4). Access: composite low-stress accessibility 0.761 (rank 3/219); 0% of 111,679 residents are locked to high-stress streets at the central threshold, against 406 km of low-stress network. Leading risk drivers on its streets: Score cap, Motor-traffic volume, On-road stress (speed × lanes). Leading protections: Cycleway/Sidewalk, Separated cycle track. Notes: top-ten overall; top-ten on at least one dimension.

---

**94. LEIPZIG DE · INDEX 60.2 · INTERVAL 81–123 · TIER 2 (MIDDLE) · TIER STABILITY 85%**

Leipzig holds an uneven profile — strongest on street-network risk (rank 62), weakest on low-stress access (rank 137). Streets: mean comparable risk 53.2 per km over 2,296 street-km (dimension rank 62/219), with 8.7% of the network in the highest-risk band. Journeys: 210 standardised trips average 12.2 (rank 112/219; 90th percentile 33.1). Access: composite low-stress accessibility 0.349 (rank 137/219); 60% of 500,066 residents are locked to high-stress streets at the central threshold, against 1,506 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, On-road stress (speed × lanes), 50 km/h limit. Leading protections: Signed ≤30 km/h zone, Bicycle Designated.

---

**56. LILLE FR · INDEX 71.1 · INTERVAL 51–79 · TIER 1 (LOWER ASSESSED RISK) · TIER STABILITY 97%**

Lille holds an uneven profile — strongest on low-stress access (rank 47), weakest on journey risk (rank 113). Streets: mean comparable risk 52.3 per km over 607 street-km (dimension rank 58/219), with 8.9% of the network in the highest-risk band. Journeys: 210 standardised trips average 12.8 (rank 113/219; 90th percentile 32.6). Access: composite low-stress accessibility 0.555 (rank 47/219); 13% of 236,876 residents are locked to high-stress streets at the central threshold, against 381 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, Score cap, On-road stress (speed × lanes). Leading protections: Signed ≤30 km/h zone, Cycleway/Sidewalk.

---

**113. LINZ AT · INDEX 58.0 · INTERVAL 84–141 · TIER 2 (MIDDLE) · TIER STABILITY 93%**

Linz holds an uneven profile — strongest on journey risk (rank 70), weakest on street-network risk (rank 163). Streets: mean comparable risk 55.5 per km over 864 street-km (dimension rank 163/219), with 23.0% of the network in the highest-risk band. Journeys: 210 standardised trips average 2.0 (rank 70/219; 90th percentile 19.9). Access: composite low-stress accessibility 0.459 (rank 84/219); 66% of 189,574 residents are locked to high-stress streets at the central threshold, against 589 km of low-stress network. Leading risk drivers on its streets: Legal Dismount, Motor-traffic volume, Incline. Leading protections: Signed ≤30 km/h zone, Bicycle Designated.

---

**181. LISBON PT · INDEX 34.2 · INTERVAL 147–193 · TIER 3 (HIGHER ASSESSED RISK) · TIER STABILITY 92%**

Lisbon holds an uneven profile — strongest on journey risk (rank 84), weakest on street-network risk (rank 205). Streets: mean comparable risk 73.1 per km over 1,514 street-km (dimension rank 205/219), with 21.4% of the network in the highest-risk band. Journeys: 210 standardised trips average 5.7 (rank 84/219; 90th percentile 40.8). Access: composite low-stress accessibility 0.254 (rank 175/219); 95% of 532,174 residents are locked to high-stress streets at the central threshold, against 600 km of low-stress network. Leading risk

drivers on its streets: Motor-traffic volume, On-road stress (speed × lanes), 50 km/h limit. Leading protections: Cycleway/Sidewalk, Separated cycle track.

---

123. LIÈGE BE · INDEX 56.0 · INTERVAL 91–147 · TIER 2 (MIDDLE) · TIER STABILITY 89%

Liège holds an uneven profile — strongest on journey risk (rank 63), weakest on street-network risk (rank 165). Streets: mean comparable risk 62.0 per km over 742 street-km (dimension rank 165/219), with 16.0% of the network in the highest-risk band. Journeys: 210 standardised trips average 0.6 (rank 63/219; 90th percentile 39.2). Access: composite low-stress accessibility 0.416 (rank 107/219); 29% of 187,949 residents are locked to high-stress streets at the central threshold, against 453 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, On-road stress (speed × lanes), Score cap. Leading protections: Cycleway/Sidewalk, Separated cycle track.

---

82. LJUBLJANA SI · INDEX 63.9 · INTERVAL 76–96 · TIER 2 (MIDDLE) · TIER STABILITY 73%

Ljubljana holds an uneven profile — strongest on street-network risk (rank 69), weakest on low-stress access (rank 109). Streets: mean comparable risk 56.0 per km over 1,621 street-km (dimension rank 69/219), with 6.7% of the network in the highest-risk band. Journeys: 210 standardised trips average 7.9 (rank 91/219; 90th percentile 32.1). Access: composite low-stress accessibility 0.411 (rank 109/219); 82% of 302,716 residents are locked to high-stress streets at the central threshold, against 1,381 km of low-stress network. Leading risk drivers on its streets: Incline, Track Grade 4/5, Motor-traffic volume. Leading protections: Bicycle Designated, Signed ≤30 km/h zone.

---

126. LUBLIN PL · INDEX 55.2 · INTERVAL 95–141 · TIER 2 (MIDDLE) · TIER STABILITY 85%

Lublin holds an uneven profile — strongest on journey risk (rank 66), weakest on low-stress access (rank 146). Streets: mean comparable risk 61.7 per km over 1,605 street-km (dimension rank 143/219), with 11.0% of the network in the highest-risk band. Journeys: 210 standardised trips average 1.0 (rank 66/219; 90th percentile 28.8). Access: composite low-stress accessibility 0.335 (rank 146/219); 86% of 325,425 residents are locked to high-stress streets at the central threshold, against 775 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, On-road stress (speed × lanes), 50 km/h limit. Leading protections: Cycleway/Sidewalk, Separated cycle track.

---

81. LUXEMBOURG LU · INDEX 64.1 · INTERVAL 73–99 · TIER 2 (MIDDLE) · TIER STABILITY 87%

Luxembourg holds an uneven profile — strongest on low-stress access (rank 65), weakest on street-network risk (rank 112). Streets: mean comparable risk 56.6 per km over 453 street-km (dimension rank 112/219), with 12.6% of the network in the highest-risk band. Journeys: 210 standardised trips average 6.7 (rank 87/219; 90th percentile 29.2). Access: composite low-stress accessibility 0.492 (rank 65/219); 2% of 121,571 residents are locked to high-stress streets at the central threshold, against 452 km of low-stress network. Leading risk drivers on its streets: Incline, Motor-traffic volume, On-road stress (speed × lanes). Leading protections: Signed ≤30 km/h zone, Bicycle Designated.

---

27. LYON FR · INDEX 83.6 · INTERVAL 13–38 · TIER 1 (LOWER ASSESSED RISK) · TIER STABILITY 100%

Lyon holds an uneven profile — strongest on low-stress access (rank 9), weakest on journey risk (rank 65). Streets: mean comparable risk 48.8 per km over 787 street-km (dimension rank 46/219), with 10.9% of the network in the highest-risk band. Journeys: 210 standardised trips average 0.9 (rank 65/219; 90th percentile 18.7). Access: composite low-stress accessibility 0.698 (rank 9/219); 1% of 527,759 residents are locked to high-stress streets at the central threshold, against 611 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, Score cap, Dismount. Leading protections: Signed  $\leq 30$  km/h zone, Cycleway/Sidewalk. Notes: top-ten on at least one dimension.

---

28. LÜBECK DE · INDEX 83.3 · INTERVAL 24–40 · TIER 1 (LOWER ASSESSED RISK) · TIER STABILITY 100%

Lübeck holds a balanced profile. Streets: mean comparable risk 49.7 per km over 1,044 street-km (dimension rank 24/219), with 4.6% of the network in the highest-risk band. Journeys: 210 standardised trips average -10.7 (rank 27/219; 90th percentile 8.8). Access: composite low-stress accessibility 0.552 (rank 48/219); 23% of 208,540 residents are locked to high-stress streets at the central threshold, against 915 km of low-stress network. Leading risk drivers on its streets: Score cap, Motor-traffic volume, On-road stress (speed  $\times$  lanes). Leading protections: Cycleway/Sidewalk, Bicycle Designated.

---

109. MADRID ES · INDEX 58.6 · INTERVAL 82–127 · TIER 2 (MIDDLE) · TIER STABILITY 88%

Madrid holds an uneven profile — strongest on journey risk (rank 59), weakest on street-network risk (rank 139). Streets: mean comparable risk 60.0 per km over 4,840 street-km (dimension rank 139/219), with 12.2% of the network in the highest-risk band. Journeys: 209 standardised trips average -0.6 (rank 59/219; 90th percentile 26.7). Access: composite low-stress accessibility 0.382 (rank 127/219); 4% of 3,742,851 residents are locked to high-stress streets at the central threshold, against 4,107 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, Incline, On-road stress (speed  $\times$  lanes). Leading protections: Signed  $\leq 30$  km/h zone, Cycleway/Sidewalk.

---

59. MAGDEBURG DE · INDEX 70.2 · INTERVAL 55–77 · TIER 1 (LOWER ASSESSED RISK) · TIER STABILITY 94%

Magdeburg holds a balanced profile. Streets: mean comparable risk 55.8 per km over 1,280 street-km (dimension rank 86/219), with 9.9% of the network in the highest-risk band. Journeys: 210 standardised trips average 1.1 (rank 67/219; 90th percentile 20.3). Access: composite low-stress accessibility 0.529 (rank 57/219); 42% of 229,914 residents are locked to high-stress streets at the central threshold, against 995 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, On-road stress (speed  $\times$  lanes), Track Grade 4/5. Leading protections: Signed  $\leq 30$  km/h zone, Bicycle Designated.

---

98. MAINZ DE · INDEX 59.7 · INTERVAL 88–127 · TIER 2 (MIDDLE) · TIER STABILITY 100%

Mainz holds an uneven profile — strongest on low-stress access (rank 76), weakest on journey risk (rank 144). Streets: mean comparable risk 55.7 per km over 841 street-km (dimension rank 97/219), with 11.6% of the network in the highest-risk band. Journeys: 210 standardised trips average 21.3 (rank 144/219; 90th percentile 38.1). Access: composite low-stress accessibility 0.471 (rank 76/219); 35% of 207,314 residents are locked to high-stress streets at the central threshold, against 712 km of low-stress network. Leading risk drivers on its

streets: Motor-traffic volume, Track Grade 4/5, Heavy-vehicle access. Leading protections: Signed  $\leq 30$  km/h zone, Bicycle Designated.

---

38. MALMÖ SE · INDEX 78.3 · INTERVAL 12–57 · TIER 1 (LOWER ASSESSED RISK) · TIER STABILITY 99%

Malmö holds an uneven profile — strongest on journey risk (rank 8), weakest on street-network risk (rank 113). Streets: mean comparable risk 59.2 per km over 1,095 street-km (dimension rank 113/219), with 9.9% of the network in the highest-risk band. Journeys: 210 standardised trips average -18.5 (rank 8/219; 90th percentile -8.3). Access: composite low-stress accessibility 0.612 (rank 34/219); 31% of 359,132 residents are locked to high-stress streets at the central threshold, against 719 km of low-stress network. Its routed journeys score far below its street average — a working low-risk sub-network the router can exploit (escape-valve rank 6 of 219). Leading risk drivers on its streets: Score cap, Motor-traffic volume, Heavy-vehicle access. Leading protections: Cycleway/Sidewalk, Separated cycle track. Notes: top-ten on at least one dimension.

---

52. MANNHEIM DE · INDEX 73.6 · INTERVAL 40–73 · TIER 1 (LOWER ASSESSED RISK) · TIER STABILITY 99%

Mannheim holds an uneven profile — strongest on low-stress access (rank 27), weakest on street-network risk (rank 94). Streets: mean comparable risk 54.3 per km over 1,298 street-km (dimension rank 94/219), with 12.3% of the network in the highest-risk band. Journeys: 210 standardised trips average 6.1 (rank 86/219; 90th percentile 24.4). Access: composite low-stress accessibility 0.633 (rank 27/219); 7% of 280,913 residents are locked to high-stress streets at the central threshold, against 1,126 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, Track Grade 4/5, On-road stress (speed  $\times$  lanes). Leading protections: Car-Free Zone, Signed  $\leq 30$  km/h zone.

---

60. MARIBOR SI · INDEX 70.1 · INTERVAL 47–73 · TIER 1 (LOWER ASSESSED RISK) · TIER STABILITY 84%

Maribor holds an uneven profile — strongest on journey risk (rank 47), weakest on low-stress access (rank 100). Streets: mean comparable risk 54.0 per km over 909 street-km (dimension rank 52/219), with 6.3% of the network in the highest-risk band. Journeys: 210 standardised trips average -3.8 (rank 47/219; 90th percentile 31.8). Access: composite low-stress accessibility 0.424 (rank 100/219); 25% of 105,163 residents are locked to high-stress streets at the central threshold, against 678 km of low-stress network. Leading risk drivers on its streets: Incline, Track Grade 4/5, Score cap. Leading protections: Cycleway/Sidewalk, Separated cycle track.

---

165. MARSEILLE FR · INDEX 42.4 · INTERVAL 159–171 · TIER 3 (HIGHER ASSESSED RISK) · TIER STABILITY 95%

Marseille holds a balanced profile. Streets: mean comparable risk 61.6 per km over 2,252 street-km (dimension rank 145/219), with 11.8% of the network in the highest-risk band. Journeys: 210 standardised trips average 26.1 (rank 154/219; 90th percentile 50.6). Access: composite low-stress accessibility 0.251 (rank 177/219); 89% of 898,652 residents are locked to high-stress streets at the central threshold, against 1,172 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, Incline, On-road stress (speed  $\times$  lanes). Leading protections: Signed  $\leq 30$  km/h zone, Residential street.

198. MESSINA IT · INDEX 23.8 · INTERVAL 169–209 · TIER 3 (HIGHER ASSESSED RISK) · TIER STABILITY 100%

Messina holds an uneven profile — strongest on street-network risk (rank 123), weakest on low-stress access (rank 215). Streets: mean comparable risk 61.0 per km over 947 street-km (dimension rank 123/219), with 9.3% of the network in the highest-risk band. Journeys: 210 standardised trips average 71.4 (rank 213/219; 90th percentile 92.9). Access: composite low-stress accessibility 0.113 (rank 215/219); 87% of 222,663 residents are locked to high-stress streets at the central threshold, against 635 km of low-stress network. Its routed journeys score close to (or above) its street average — the router finds little refuge from the ordinary street network (escape-valve rank 218 of 219). Leading risk drivers on its streets: Incline, Motor-traffic volume, On-road stress (speed × lanes). Leading protections: Track Grade 2, Residential street.

---

177. MILAN IT · INDEX 36.6 · INTERVAL 162–184 · TIER 3 (HIGHER ASSESSED RISK) · TIER STABILITY 97%

Milan holds an uneven profile — strongest on journey risk (rank 120), weakest on low-stress access (rank 192). Streets: mean comparable risk 67.1 per km over 2,146 street-km (dimension rank 181/219), with 16.3% of the network in the highest-risk band. Journeys: 210 standardised trips average 15.2 (rank 120/219; 90th percentile 36.9). Access: composite low-stress accessibility 0.207 (rank 192/219); 93% of 1,161,421 residents are locked to high-stress streets at the central threshold, against 769 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, On-road stress (speed × lanes), 50 km/h limit. Leading protections: Cycleway/Sidewalk, Separated cycle track.

---

30. MONTPELLIER FR · INDEX 82.4 · INTERVAL 25–42 · TIER 1 (LOWER ASSESSED RISK) · TIER STABILITY 100%

Montpellier holds a balanced profile. Streets: mean comparable risk 51.6 per km over 897 street-km (dimension rank 44/219), with 7.0% of the network in the highest-risk band. Journeys: 210 standardised trips average -0.7 (rank 55/219; 90th percentile 13.6). Access: composite low-stress accessibility 0.656 (rank 22/219); 0% of 304,163 residents are locked to high-stress streets at the central threshold, against 769 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, Score cap, Heavy-vehicle access. Leading protections: Signed ≤30 km/h zone, Cycleway/Sidewalk.

---

21. MUNICH DE · INDEX 86.9 · INTERVAL 20–33 · TIER 1 (LOWER ASSESSED RISK) · TIER STABILITY 100%

Munich holds a balanced profile. Streets: mean comparable risk 49.1 per km over 3,108 street-km (dimension rank 22/219), with 4.6% of the network in the highest-risk band. Journeys: 210 standardised trips average -6.7 (rank 41/219; 90th percentile 2.3). Access: composite low-stress accessibility 0.639 (rank 25/219); 13% of 1,485,388 residents are locked to high-stress streets at the central threshold, against 2,913 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, Score cap, On-road stress (speed × lanes). Leading protections: Bicycle Designated, Signed ≤30 km/h zone.

---

73. MURCIA ES · INDEX 67.0 · INTERVAL 61–85 · TIER 1 (LOWER ASSESSED RISK) · TIER STABILITY 59%

Murcia holds an uneven profile — strongest on street-network risk (rank 56), weakest on journey risk (rank 98). Streets: mean comparable risk 55.0 per km over 2,974 street-km (dimension rank 56/219), with 5.5% of the network in the highest-risk band. Journeys: 210 standardised trips average 8.9 (rank 98/219; 90th percentile 48.2). Access: composite low-stress accessibility 0.447 (rank 89/219); 10% of 584,240 residents are locked to high-stress streets at the central threshold, against 3,303 km of low-stress network. Leading risk drivers on its streets: Incline, Motor-traffic volume, Heavy-vehicle access. Leading protections: Signed  $\leq 30$  km/h zone, Cycleway/Sidewalk. Notes: straddles a tier boundary under respecification.

---

36. MÁLAGA ES · INDEX 78.6 · INTERVAL 30–50 · TIER 1 (LOWER ASSESSED RISK) · TIER STABILITY 100%

Málaga holds an uneven profile — strongest on street-network risk (rank 28), weakest on journey risk (rank 69). Streets: mean comparable risk 50.0 per km over 1,764 street-km (dimension rank 28/219), with 5.6% of the network in the highest-risk band. Journeys: 210 standardised trips average 1.6 (rank 69/219; 90th percentile 37.7). Access: composite low-stress accessibility 0.555 (rank 46/219); 8% of 683,468 residents are locked to high-stress streets at the central threshold, against 2,818 km of low-stress network. Leading risk drivers on its streets: Incline, Heavy-vehicle access, Motor-traffic volume. Leading protections: Signed  $\leq 30$  km/h zone, Track Grade 2.

---

125. MÖNCHEGLADBACH DE · INDEX 55.8 · INTERVAL 106–142 · TIER 2 (MIDDLE) · TIER STABILITY 100%

Mönchengladbach holds an uneven profile — strongest on low-stress access (rank 86), weakest on journey risk (rank 145). Streets: mean comparable risk 57.5 per km over 1,343 street-km (dimension rank 134/219), with 14.3% of the network in the highest-risk band. Journeys: 210 standardised trips average 21.4 (rank 145/219; 90th percentile 39.5). Access: composite low-stress accessibility 0.456 (rank 86/219); 49% of 244,618 residents are locked to high-stress streets at the central threshold, against 953 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, Track Grade 4/5, On-road stress (speed  $\times$  lanes). Leading protections: Signed  $\leq 30$  km/h zone, Bicycle Designated.

---

37. MÜNSTER DE · INDEX 78.5 · INTERVAL 34–49 · TIER 1 (LOWER ASSESSED RISK) · TIER STABILITY 100%

Münster holds a balanced profile. Streets: mean comparable risk 51.7 per km over 1,700 street-km (dimension rank 41/219), with 6.6% of the network in the highest-risk band. Journeys: 210 standardised trips average -0.5 (rank 57/219; 90th percentile 17.3). Access: composite low-stress accessibility 0.578 (rank 42/219); 37% of 321,103 residents are locked to high-stress streets at the central threshold, against 1,366 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, On-road stress (speed  $\times$  lanes), Track Grade 4/5. Leading protections: Bicycle Designated, Cycle Infra Car-Free.

---

65. NANCY FR · INDEX 69.0 · INTERVAL 38–108 · TIER 1 (LOWER ASSESSED RISK) · TIER STABILITY 76%

Nancy holds an uneven profile — strongest on street-network risk (rank 26), weakest on journey risk (rank 170). Streets: mean comparable risk 48.2 per km over 270 street-km (dimension rank 26/219), with 6.9% of the network in the highest-risk band. Journeys: 210 standardised trips average 31.4 (rank 170/219; 90th percentile 45.6). Access: composite low-stress accessibility 0.545 (rank 53/219); 0% of 108,783 residents are locked to high-stress streets at the central threshold, against 186 km of low-stress network. Leading risk drivers on its streets:

Motor-traffic volume, Heavy-vehicle access, On-road stress (speed × lanes). Leading protections: Signed ≤30 km/h zone, Signed bike network.

---

35. NANTES FR · INDEX 79.0 · INTERVAL 27–52 · TIER 1 (LOWER ASSESSED RISK) · TIER STABILITY 100%

Nantes holds an uneven profile — strongest on street-network risk (rank 18), weakest on journey risk (rank 82). Streets: mean comparable risk 45.9 per km over 948 street-km (dimension rank 18/219), with 7.0% of the network in the highest-risk band. Journeys: 210 standardised trips average 5.5 (rank 82/219; 90th percentile 26.1). Access: composite low-stress accessibility 0.546 (rank 52/219); 0% of 286,121 residents are locked to high-stress streets at the central threshold, against 775 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, Score cap, On-road stress (speed × lanes). Leading protections: Signed ≤30 km/h zone, Signed bike network.

---

188. NAPLES IT · INDEX 29.0 · INTERVAL 173–200 · TIER 3 (HIGHER ASSESSED RISK) · TIER STABILITY 100%

Naples holds an uneven profile — strongest on street-network risk (rank 152), weakest on journey risk (rank 206). Streets: mean comparable risk 62.2 per km over 1,578 street-km (dimension rank 152/219), with 11.8% of the network in the highest-risk band. Journeys: 210 standardised trips average 57.6 (rank 206/219; 90th percentile 68.8). Access: composite low-stress accessibility 0.197 (rank 196/219); 86% of 898,628 residents are locked to high-stress streets at the central threshold, against 437 km of low-stress network. Its routed journeys score close to (or above) its street average — the router finds little refuge from the ordinary street network (escape-valve rank 215 of 219). Leading risk drivers on its streets: Incline, Motor-traffic volume, On-road stress (speed × lanes). Leading protections: Residential street, Pedestrian Zone.

---

159. NICE FR · INDEX 45.2 · INTERVAL 145–165 · TIER 3 (HIGHER ASSESSED RISK) · TIER STABILITY 83%

Nice holds an uneven profile — strongest on journey risk (rank 118), weakest on street-network risk (rank 168). Streets: mean comparable risk 62.3 per km over 905 street-km (dimension rank 168/219), with 16.0% of the network in the highest-risk band. Journeys: 210 standardised trips average 14.7 (rank 118/219; 90th percentile 50.5). Access: composite low-stress accessibility 0.297 (rank 159/219); 89% of 326,754 residents are locked to high-stress streets at the central threshold, against 386 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, Incline, On-road stress (speed × lanes). Leading protections: Signed ≤30 km/h zone, Cycleway/Sidewalk.

---

196. NICOSIA CY · INDEX 25.7 · INTERVAL 193–198 · TIER 3 (HIGHER ASSESSED RISK) · TIER STABILITY 100%

Nicosia holds a balanced profile. Streets: mean comparable risk 69.1 per km over 316 street-km (dimension rank 187/219), with 17.8% of the network in the highest-risk band. Journeys: 210 standardised trips average 39.0 (rank 184/219; 90th percentile 67.2). Access: composite low-stress accessibility 0.193 (rank 199/219); 97% of 43,726 residents are locked to high-stress streets at the central threshold, against 62 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, On-road stress (speed × lanes), 50 km/h limit. Leading protections: Cycleway/Sidewalk, Separated cycle track.

99. NUREMBERG DE · INDEX 59.7 · INTERVAL 95–116 · TIER 2 (MIDDLE) · TIER STABILITY 100%

Nuremberg holds a balanced profile. Streets: mean comparable risk 57.1 per km over 1,780 street-km (dimension rank 103/219), with 10.8% of the network in the highest-risk band. Journeys: 210 standardised trips average 15.6 (rank 122/219; 90th percentile 35.1). Access: composite low-stress accessibility 0.444 (rank 90/219); 55% of 486,520 residents are locked to high-stress streets at the central threshold, against 1,352 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, On-road stress (speed × lanes), 50 km/h limit. Leading protections: Signed ≤30 km/h zone, Bicycle Designated.

---

74. OBERHAUSEN DE · INDEX 66.3 · INTERVAL 52–103 · TIER 2 (MIDDLE) · TIER STABILITY 49%

Oberhausen holds an uneven profile — strongest on street-network risk (rank 39), weakest on journey risk (rank 141). Streets: mean comparable risk 49.1 per km over 806 street-km (dimension rank 39/219), with 9.1% of the network in the highest-risk band. Journeys: 210 standardised trips average 20.3 (rank 141/219; 90th percentile 38.2). Access: composite low-stress accessibility 0.464 (rank 80/219); 42% of 192,982 residents are locked to high-stress streets at the central threshold, against 509 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, 50 km/h limit, On-road stress (speed × lanes). Leading protections: Signed ≤30 km/h zone, Bicycle Designated. Notes: straddles a tier boundary under respecification.

---

66. ODENSE DK · INDEX 68.6 · INTERVAL 23–89 · TIER 1 (LOWER ASSESSED RISK) · TIER STABILITY 73%

Odense holds an uneven profile — strongest on journey risk (rank 14), weakest on street-network risk (rank 122). Streets: mean comparable risk 59.9 per km over 1,983 street-km (dimension rank 122/219), with 10.6% of the network in the highest-risk band. Journeys: 210 standardised trips average -14.5 (rank 14/219; 90th percentile 0.1). Access: composite low-stress accessibility 0.466 (rank 78/219); 91% of 193,803 residents are locked to high-stress streets at the central threshold, against 842 km of low-stress network. Its routed journeys score far below its street average — a working low-risk sub-network the router can exploit (escape-valve rank 11 of 219). Leading risk drivers on its streets: On-road stress (speed × lanes), Score cap, 50 km/h limit. Leading protections: Cycleway/Sidewalk, Separated cycle track.

---

195. ORADEA RO · INDEX 26.4 · INTERVAL 172–205 · TIER 3 (HIGHER ASSESSED RISK) · TIER STABILITY 99%

Oradea holds an uneven profile — strongest on journey risk (rank 119), weakest on street-network risk (rank 209). Streets: mean comparable risk 74.6 per km over 881 street-km (dimension rank 209/219), with 24.4% of the network in the highest-risk band. Journeys: 210 standardised trips average 14.8 (rank 119/219; 90th percentile 58.8). Access: composite low-stress accessibility 0.219 (rank 190/219); 97% of 198,591 residents are locked to high-stress streets at the central threshold, against 240 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, On-road stress (speed × lanes), 50 km/h limit. Leading protections: Cycleway/Sidewalk, Signed ≤30 km/h zone.

---

106. ORLÉANS FR · INDEX 58.9 · INTERVAL 98–124 · TIER 2 (MIDDLE) · TIER STABILITY 100%

Orléans holds an uneven profile — strongest on low-stress access (rank 88), weakest on journey risk (rank 134). Streets: mean comparable risk 58.1 per km over 459 street-km (dimension rank 105/219), with 9.8% of the

network in the highest-risk band. Journeys: 210 standardised trips average 18.4 (rank 134/219; 90th percentile 38.8). Access: composite low-stress accessibility 0.448 (rank 88/219); 13% of 103,906 residents are locked to high-stress streets at the central threshold, against 217 km of low-stress network. Leading risk drivers on its streets: On-road stress (speed × lanes), Motor-traffic volume, Score cap. Leading protections: Cycleway/Sidewalk, Separated cycle track.

---

62. OSLO NO · INDEX 69.9 · INTERVAL 40–81 · TIER 1 (LOWER ASSESSED RISK) · TIER STABILITY 80%

Oslo holds an uneven profile — strongest on street-network risk (rank 33), weakest on low-stress access (rank 125). Streets: mean comparable risk 51.9 per km over 3,010 street-km (dimension rank 33/219), with 4.4% of the network in the highest-risk band. Journeys: 210 standardised trips average -0.9 (rank 54/219; 90th percentile 24.1). Access: composite low-stress accessibility 0.383 (rank 125/219); 19% of 698,949 residents are locked to high-stress streets at the central threshold, against 3,455 km of low-stress network. Leading risk drivers on its streets: Incline, Score cap, Motor-traffic volume. Leading protections: Cycleway/Sidewalk, Separated cycle track.

---

173. OSTRAVA CZ · INDEX 39.3 · INTERVAL 145–181 · TIER 3 (HIGHER ASSESSED RISK) · TIER STABILITY 89%

Ostrava holds an uneven profile — strongest on journey risk (rank 93), weakest on street-network risk (rank 184). Streets: mean comparable risk 68.8 per km over 1,656 street-km (dimension rank 184/219), with 15.9% of the network in the highest-risk band. Journeys: 210 standardised trips average 8.3 (rank 93/219; 90th percentile 32.5). Access: composite low-stress accessibility 0.236 (rank 183/219); 83% of 273,406 residents are locked to high-stress streets at the central threshold, against 734 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, On-road stress (speed × lanes), 50 km/h limit. Leading protections: Cycleway/Sidewalk, Separated cycle track.

---

54. OULU FI · INDEX 71.8 · INTERVAL 10–81 · TIER 1 (LOWER ASSESSED RISK) · TIER STABILITY 88%

Oulu holds an uneven profile — strongest on journey risk (rank 6), weakest on street-network risk (rank 128). Streets: mean comparable risk 63.4 per km over 3,312 street-km (dimension rank 128/219), with 7.1% of the network in the highest-risk band. Journeys: 210 standardised trips average -22.1 (rank 6/219; 90th percentile -0.1). Access: composite low-stress accessibility 0.497 (rank 63/219); 48% of 210,303 residents are locked to high-stress streets at the central threshold, against 3,183 km of low-stress network. Its routed journeys score far below its street average — a working low-risk sub-network the router can exploit (escape-valve rank 1 of 219). Leading risk drivers on its streets: Score cap, On-road stress (speed × lanes), Motor-traffic volume. Leading protections: Cycleway/Sidewalk, Separated cycle track. Notes: top-ten on at least one dimension.

---

124. OVIEDO ES · INDEX 55.9 · INTERVAL 66–157 · TIER 2 (MIDDLE) · TIER STABILITY 70%

Oviedo holds an uneven profile — strongest on street-network risk (rank 35), weakest on journey risk (rank 188). Streets: mean comparable risk 51.6 per km over 865 street-km (dimension rank 35/219), with 5.5% of the network in the highest-risk band. Journeys: 210 standardised trips average 41.4 (rank 188/219; 90th percentile 58.3). Access: composite low-stress accessibility 0.386 (rank 122/219); 7% of 218,267 residents are locked to high-stress streets at the central threshold, against 763 km of low-stress network. Its routed journeys score

close to (or above) its street average — the router finds little refuge from the ordinary street network (escape-valve rank 210 of 219). Leading risk drivers on its streets: Incline, Heavy-vehicle access, Motor-traffic volume. Leading protections: Signed  $\leq 30$  km/h zone, Pedestrian Zone.

---

180. **PADUA IT** · INDEX 34.7 · INTERVAL 176–187 · TIER 3 (HIGHER ASSESSED RISK) · TIER STABILITY 100%

Padua holds a balanced profile. Streets: mean comparable risk 68.1 per km over 1,017 street-km (dimension rank 183/219), with 16.1% of the network in the highest-risk band. Journeys: 210 standardised trips average 28.8 (rank 163/219; 90th percentile 48.6). Access: composite low-stress accessibility 0.268 (rank 171/219); 98% of 209,164 residents are locked to high-stress streets at the central threshold, against 278 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, On-road stress (speed  $\times$  lanes), 50 km/h limit. Leading protections: Bicycle Designated, Cycle Infra Car-Free.

---

190. **PALERMO IT** · INDEX 28.7 · INTERVAL 186–196 · TIER 3 (HIGHER ASSESSED RISK) · TIER STABILITY 100%

Palermo holds a balanced profile. Streets: mean comparable risk 67.0 per km over 1,415 street-km (dimension rank 179/219), with 15.5% of the network in the highest-risk band. Journeys: 210 standardised trips average 32.4 (rank 172/219; 90th percentile 76.9). Access: composite low-stress accessibility 0.150 (rank 210/219); 98% of 603,577 residents are locked to high-stress streets at the central threshold, against 439 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, Incline, On-road stress (speed  $\times$  lanes). Leading protections: Residential street, Cycleway/Sidewalk.

---

50. **PALMA ES** · INDEX 73.8 · INTERVAL 48–59 · TIER 1 (LOWER ASSESSED RISK) · TIER STABILITY 100%

Palma holds a balanced profile. Streets: mean comparable risk 54.3 per km over 1,298 street-km (dimension rank 63/219), with 8.3% of the network in the highest-risk band. Journeys: 210 standardised trips average -1.7 (rank 52/219; 90th percentile 28.0). Access: composite low-stress accessibility 0.538 (rank 54/219); 9% of 503,810 residents are locked to high-stress streets at the central threshold, against 1,056 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, Incline, Heavy-vehicle access. Leading protections: Signed  $\leq 30$  km/h zone, Cycleway/Sidewalk.

---

53. **PAMPLONA ES** · INDEX 73.4 · INTERVAL 45–72 · TIER 1 (LOWER ASSESSED RISK) · TIER STABILITY 99%

Pamplona holds an uneven profile — strongest on low-stress access (rank 38), weakest on street-network risk (rank 100). Streets: mean comparable risk 56.1 per km over 341 street-km (dimension rank 100/219), with 11.3% of the network in the highest-risk band. Journeys: 210 standardised trips average -0.5 (rank 60/219; 90th percentile 21.9). Access: composite low-stress accessibility 0.602 (rank 38/219); 1% of 195,988 residents are locked to high-stress streets at the central threshold, against 333 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, Score cap, Heavy-vehicle access. Leading protections: Cycleway/Sidewalk, Separated cycle track.

---

11. **PARIS FR** · INDEX 93.0 · INTERVAL 5–16 · TIER 1 (LOWER ASSESSED RISK) · TIER STABILITY 100%

Paris holds a balanced profile. Streets: mean comparable risk 46.6 per km over 1,419 street-km (dimension rank 16/219), with 6.0% of the network in the highest-risk band. Journeys: 210 standardised trips average -10.9 (rank 25/219; 90th percentile 2.5). Access: composite low-stress accessibility 0.755 (rank 5/219); 0% of 2,374,638 residents are locked to high-stress streets at the central threshold, against 1,619 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, Score cap, Heavy-vehicle access. Leading protections: Signed  $\leq 30$  km/h zone, Cycleway/Sidewalk. Notes: top-ten on at least one dimension.

---

141. PARMA IT · INDEX 51.0 · INTERVAL 67–166 · TIER 2 (MIDDLE) · TIER STABILITY 56%

Parma holds an uneven profile — strongest on journey risk (rank 38), weakest on street-network risk (rank 186). Streets: mean comparable risk 67.0 per km over 1,270 street-km (dimension rank 186/219), with 20.2% of the network in the highest-risk band. Journeys: 210 standardised trips average -8.3 (rank 38/219; 90th percentile 15.4). Access: composite low-stress accessibility 0.388 (rank 120/219); 76% of 182,953 residents are locked to high-stress streets at the central threshold, against 452 km of low-stress network. Its routed journeys score far below its street average — a working low-risk sub-network the router can exploit (escape-valve rank 9 of 219). Leading risk drivers on its streets: Track Grade 4/5, Motor-traffic volume, Heavy-vehicle access. Leading protections: Cycleway/Sidewalk, Separated cycle track. Notes: straddles a tier boundary under respecification.

---

155. PESCARA IT · INDEX 47.1 · INTERVAL 128–162 · TIER 3 (HIGHER ASSESSED RISK) · TIER STABILITY 61%

Pescara holds an uneven profile — strongest on journey risk (rank 89), weakest on street-network risk (rank 169). Streets: mean comparable risk 63.9 per km over 479 street-km (dimension rank 169/219), with 14.3% of the network in the highest-risk band. Journeys: 210 standardised trips average 7.0 (rank 89/219; 90th percentile 47.8). Access: composite low-stress accessibility 0.288 (rank 164/219); 99% of 117,486 residents are locked to high-stress streets at the central threshold, against 166 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, On-road stress (speed  $\times$  lanes), 50 km/h limit. Leading protections: Cycleway/Sidewalk, Separated cycle track. Notes: straddles a tier boundary under respecification.

---

212. PITESTI RO · INDEX 10.9 · INTERVAL 207–213 · TIER 3 (HIGHER ASSESSED RISK) · TIER STABILITY 100%

Pitesti holds a balanced profile. Streets: mean comparable risk 77.6 per km over 349 street-km (dimension rank 214/219), with 24.9% of the network in the highest-risk band. Journeys: 210 standardised trips average 57.6 (rank 207/219; 90th percentile 75.5). Access: composite low-stress accessibility 0.230 (rank 186/219); 98% of 135,915 residents are locked to high-stress streets at the central threshold, against 89 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, On-road stress (speed  $\times$  lanes), 50 km/h limit. Leading protections: Signed  $\leq 30$  km/h zone, Residential street.

---

218. PLOIESTI RO · INDEX 1.6 · INTERVAL 215–219 · TIER 3 (HIGHER ASSESSED RISK) · TIER STABILITY 100%

Ploiesti holds a balanced profile. Streets: mean comparable risk 75.2 per km over 589 street-km (dimension rank 212/219), with 26.3% of the network in the highest-risk band. Journeys: 210 standardised trips average 76.0 (rank 218/219; 90th percentile 92.9). Access: composite low-stress accessibility 0.124 (rank 219/219); 99% of 185,677 residents are locked to high-stress streets at the central threshold, against 56 km of low-stress

network. Its routed journeys score close to (or above) its street average — the router finds little refuge from the ordinary street network (escape-valve rank 216 of 219). Leading risk drivers on its streets: Motor-traffic volume, On-road stress (speed × lanes), 50 km/h limit. Leading protections: Residential street, Signed ≤30 km/h zone.

---

172. **PLOVDIV BG** · INDEX 39.7 · INTERVAL 110–187 · TIER 3 (HIGHER ASSESSED RISK) · TIER STABILITY 75%

Plovdiv holds an uneven profile — strongest on journey risk (rank 51), weakest on street-network risk (rank 197). Streets: mean comparable risk 69.4 per km over 946 street-km (dimension rank 197/219), with 20.8% of the network in the highest-risk band. Journeys: 210 standardised trips average -1.9 (rank 51/219; 90th percentile 29.1). Access: composite low-stress accessibility 0.251 (rank 178/219); 99% of 296,828 residents are locked to high-stress streets at the central threshold, against 217 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, On-road stress (speed × lanes), 50 km/h limit. Leading protections: Cycleway/Sidewalk, Separated cycle track.

---

183. **PORTO PT** · INDEX 32.0 · INTERVAL 169–196 · TIER 3 (HIGHER ASSESSED RISK) · TIER STABILITY 100%

Porto holds an uneven profile — strongest on street-network risk (rank 146), weakest on journey risk (rank 201). Streets: mean comparable risk 61.8 per km over 765 street-km (dimension rank 146/219), with 11.6% of the network in the highest-risk band. Journeys: 210 standardised trips average 49.3 (rank 201/219; 90th percentile 66.8). Access: composite low-stress accessibility 0.194 (rank 197/219); 98% of 201,129 residents are locked to high-stress streets at the central threshold, against 191 km of low-stress network. Its routed journeys score close to (or above) its street average — the router finds little refuge from the ordinary street network (escape-valve rank 206 of 219). Leading risk drivers on its streets: On-road stress (speed × lanes), 50 km/h limit, Motor-traffic volume. Leading protections: Cycleway/Sidewalk, Separated cycle track.

---

121. **POZNAŃ PL** · INDEX 56.1 · INTERVAL 106–141 · TIER 2 (MIDDLE) · TIER STABILITY 99%

Poznań holds an uneven profile — strongest on low-stress access (rank 95), weakest on street-network risk (rank 153). Streets: mean comparable risk 62.3 per km over 2,267 street-km (dimension rank 153/219), with 12.4% of the network in the highest-risk band. Journeys: 210 standardised trips average 9.7 (rank 103/219; 90th percentile 31.9). Access: composite low-stress accessibility 0.433 (rank 95/219); 64% of 510,987 residents are locked to high-stress streets at the central threshold, against 1,283 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, On-road stress (speed × lanes), 50 km/h limit. Leading protections: Signed ≤30 km/h zone, Cycleway/Sidewalk.

---

153. **PRAGUE CZ** · INDEX 47.5 · INTERVAL 143–158 · TIER 3 (HIGHER ASSESSED RISK) · TIER STABILITY 61%

Prague holds a balanced profile. Streets: mean comparable risk 59.8 per km over 4,626 street-km (dimension rank 141/219), with 12.5% of the network in the highest-risk band. Journeys: 210 standardised trips average 17.2 (rank 132/219; 90th percentile 42.4). Access: composite low-stress accessibility 0.278 (rank 169/219); 59% of 1,334,017 residents are locked to high-stress streets at the central threshold, against 3,005 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, On-road stress (speed × lanes), 50 km/h

limit. Leading protections: Signed  $\leq 30$  km/h zone, Cycleway/Sidewalk. Notes: straddles a tier boundary under respecification.

---

168. PRATO IT · INDEX 40.7 · INTERVAL 125–185 · TIER 3 (HIGHER ASSESSED RISK) · TIER STABILITY 79%

Prato holds an uneven profile — strongest on journey risk (rank 71), weakest on street-network risk (rank 195). Streets: mean comparable risk 71.8 per km over 722 street-km (dimension rank 195/219), with 17.8% of the network in the highest-risk band. Journeys: 210 standardised trips average 2.4 (rank 71/219; 90th percentile 23.9). Access: composite low-stress accessibility 0.294 (rank 161/219); 94% of 192,494 residents are locked to high-stress streets at the central threshold, against 327 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, Heavy-vehicle access, On-road stress (speed  $\times$  lanes). Leading protections: Cycleway/Sidewalk, Separated cycle track.

---

166. RADOM PL · INDEX 42.4 · INTERVAL 150–175 · TIER 3 (HIGHER ASSESSED RISK) · TIER STABILITY 91%

Radom holds an uneven profile — strongest on journey risk (rank 116), weakest on street-network risk (rank 182). Streets: mean comparable risk 66.6 per km over 1,133 street-km (dimension rank 182/219), with 17.1% of the network in the highest-risk band. Journeys: 210 standardised trips average 14.3 (rank 116/219; 90th percentile 43.7). Access: composite low-stress accessibility 0.318 (rank 154/219); 89% of 198,991 residents are locked to high-stress streets at the central threshold, against 423 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, On-road stress (speed  $\times$  lanes), 50 km/h limit. Leading protections: Cycleway/Sidewalk, Separated cycle track.

---

145. REIMS FR · INDEX 50.1 · INTERVAL 133–160 · TIER 2 (MIDDLE) · TIER STABILITY 53%

Reims holds an uneven profile — strongest on street-network risk (rank 120), weakest on journey risk (rank 167). Streets: mean comparable risk 58.1 per km over 786 street-km (dimension rank 120/219), with 12.2% of the network in the highest-risk band. Journeys: 210 standardised trips average 30.3 (rank 167/219; 90th percentile 49.7). Access: composite low-stress accessibility 0.383 (rank 126/219); 25% of 147,981 residents are locked to high-stress streets at the central threshold, against 323 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, On-road stress (speed  $\times$  lanes), 50 km/h limit. Leading protections: Signed  $\leq 30$  km/h zone, Cycleway/Sidewalk. Notes: straddles a tier boundary under respecification.

---

75. RENNES FR · INDEX 66.1 · INTERVAL 61–100 · TIER 2 (MIDDLE) · TIER STABILITY 53%

Rennes holds an uneven profile — strongest on street-network risk (rank 49), weakest on journey risk (rank 127). Streets: mean comparable risk 51.0 per km over 750 street-km (dimension rank 49/219), with 8.9% of the network in the highest-risk band. Journeys: 210 standardised trips average 16.5 (rank 127/219; 90th percentile 34.8). Access: composite low-stress accessibility 0.463 (rank 82/219); 1% of 207,978 residents are locked to high-stress streets at the central threshold, against 499 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, Score cap, Heavy-vehicle access. Leading protections: Signed  $\leq 30$  km/h zone, Cycleway/Sidewalk. Notes: straddles a tier boundary under respecification.

---

105. REYKJAVIK IS · INDEX 59.0 · INTERVAL 89–129 · TIER 2 (MIDDLE) · TIER STABILITY 99%

Reykjavik holds an uneven profile — strongest on low-stress access (rank 81), weakest on street-network risk (rank 150). Streets: mean comparable risk 61.6 per km over 1,089 street-km (dimension rank 150/219), with 12.2% of the network in the highest-risk band. Journeys: 210 standardised trips average 7.4 (rank 90/219; 90th percentile 28.6). Access: composite low-stress accessibility 0.463 (rank 81/219); 6% of 109,562 residents are locked to high-stress streets at the central threshold, against 912 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, Incline, Heavy-vehicle access. Leading protections: Bicycle Designated, Signed  $\leq 30$  km/h zone.

---

140. RIGA LV · INDEX 51.0 · INTERVAL 120–150 · TIER 2 (MIDDLE) · TIER STABILITY 67%

Riga holds an uneven profile — strongest on journey risk (rank 101), weakest on low-stress access (rank 162). Streets: mean comparable risk 59.5 per km over 2,988 street-km (dimension rank 140/219), with 12.7% of the network in the highest-risk band. Journeys: 210 standardised trips average 9.1 (rank 101/219; 90th percentile 45.0). Access: composite low-stress accessibility 0.294 (rank 162/219); 85% of 564,574 residents are locked to high-stress streets at the central threshold, against 1,400 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, On-road stress (speed  $\times$  lanes), Heavy-vehicle access. Leading protections: Signed  $\leq 30$  km/h zone, Cycleway/Sidewalk.

---

185. ROME IT · INDEX 31.5 · INTERVAL 176–193 · TIER 3 (HIGHER ASSESSED RISK) · TIER STABILITY 100%

Rome holds an uneven profile — strongest on journey risk (rank 151), weakest on low-stress access (rank 214). Streets: mean comparable risk 66.1 per km over 8,943 street-km (dimension rank 176/219), with 14.8% of the network in the highest-risk band. Journeys: 210 standardised trips average 24.1 (rank 151/219; 90th percentile 57.7). Access: composite low-stress accessibility 0.125 (rank 214/219); 97% of 2,596,788 residents are locked to high-stress streets at the central threshold, against 2,829 km of low-stress network. Leading risk drivers on its streets: On-road stress (speed  $\times$  lanes), Incline, 50 km/h limit. Leading protections: Residential street, Cycleway/Sidewalk.

---

95. ROSTOCK DE · INDEX 60.0 · INTERVAL 80–123 · TIER 2 (MIDDLE) · TIER STABILITY 88%

Rostock holds an uneven profile — strongest on street-network risk (rank 57), weakest on journey risk (rank 142). Streets: mean comparable risk 52.9 per km over 1,137 street-km (dimension rank 57/219), with 7.8% of the network in the highest-risk band. Journeys: 210 standardised trips average 20.8 (rank 142/219; 90th percentile 34.1). Access: composite low-stress accessibility 0.384 (rank 124/219); 78% of 190,579 residents are locked to high-stress streets at the central threshold, against 732 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, Track Grade 4/5, Heavy-vehicle access. Leading protections: Bicycle Designated, Signed  $\leq 30$  km/h zone.

---

16. ROTTERDAM NL · INDEX 90.8 · INTERVAL 14–24 · TIER 1 (LOWER ASSESSED RISK) · TIER STABILITY 100%

Rotterdam holds a balanced profile. Streets: mean comparable risk 48.8 per km over 1,860 street-km (dimension rank 21/219), with 4.0% of the network in the highest-risk band. Journeys: 210 standardised trips average -12.5 (rank 19/219; 90th percentile 8.8). Access: composite low-stress accessibility 0.667 (rank 16/219); 13% of 602,272 residents are locked to high-stress streets at the central threshold, against 2,306 km of low-

stress network. Leading risk drivers on its streets: Score cap, On-road stress (speed × lanes), Heavy-vehicle access. Leading protections: Cycleway/Sidewalk, Separated cycle track.

---

**130. ROUEN FR** · INDEX 54.5 · INTERVAL 90–159 · TIER 2 (MIDDLE) · TIER STABILITY 91%

Rouen holds an uneven profile — strongest on low-stress access (rank 60), weakest on journey risk (rank 178). Streets: mean comparable risk 58.4 per km over 384 street-km (dimension rank 135/219), with 13.3% of the network in the highest-risk band. Journeys: 210 standardised trips average 35.2 (rank 178/219; 90th percentile 54.5). Access: composite low-stress accessibility 0.516 (rank 60/219); 2% of 110,003 residents are locked to high-stress streets at the central threshold, against 224 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, Heavy-vehicle access, On-road stress (speed × lanes). Leading protections: Signed ≤30 km/h zone, Pedestrian Zone.

---

**138. RZESZÓW PL** · INDEX 51.0 · INTERVAL 135–150 · TIER 2 (MIDDLE) · TIER STABILITY 68%

Rzeszów holds a balanced profile. Streets: mean comparable risk 60.5 per km over 1,433 street-km (dimension rank 148/219), with 13.3% of the network in the highest-risk band. Journeys: 210 standardised trips average 16.1 (rank 125/219; 90th percentile 51.7). Access: composite low-stress accessibility 0.360 (rank 131/219); 91% of 210,050 residents are locked to high-stress streets at the central threshold, against 528 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, On-road stress (speed × lanes), 50 km/h limit. Leading protections: Bicycle Designated, Service Road.

---

**46. SABADELL ES** · INDEX 75.5 · INTERVAL 24–69 · TIER 1 (LOWER ASSESSED RISK) · TIER STABILITY 99%

Sabadell holds an uneven profile — strongest on low-stress access (rank 15), weakest on journey risk (rank 117). Streets: mean comparable risk 54.5 per km over 380 street-km (dimension rank 64/219), with 8.2% of the network in the highest-risk band. Journeys: 210 standardised trips average 14.6 (rank 117/219; 90th percentile 46.7). Access: composite low-stress accessibility 0.674 (rank 15/219); 0% of 256,346 residents are locked to high-stress streets at the central threshold, against 458 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, Incline, Score cap. Leading protections: Signed ≤30 km/h zone, Cycleway/Sidewalk.

---

**18. SAN SEBASTIÁN ES** · INDEX 88.8 · INTERVAL 17–29 · TIER 1 (LOWER ASSESSED RISK) · TIER STABILITY 100%

San Sebastián holds a balanced profile. Streets: mean comparable risk 47.5 per km over 490 street-km (dimension rank 19/219), with 5.3% of the network in the highest-risk band. Journeys: 210 standardised trips average -8.8 (rank 36/219; 90th percentile 9.0). Access: composite low-stress accessibility 0.649 (rank 23/219); 0% of 206,288 residents are locked to high-stress streets at the central threshold, against 581 km of low-stress network. Leading risk drivers on its streets: Incline, Motor-traffic volume, Score cap. Leading protections: Cycleway/Sidewalk, Separated cycle track.

---

**194. SATU MARE RO** · INDEX 26.8 · INTERVAL 170–208 · TIER 3 (HIGHER ASSESSED RISK) · TIER STABILITY 100%

Satu Mare holds an uneven profile — strongest on journey risk (rank 128), weakest on street-network risk (rank 216). Streets: mean comparable risk 82.4 per km over 398 street-km (dimension rank 216/219), with 26.3% of the network in the highest-risk band. Journeys: 210 standardised trips average 16.9 (rank 128/219; 90th percentile 47.6). Access: composite low-stress accessibility 0.302 (rank 157/219); 96% of 96,200 residents are locked to high-stress streets at the central threshold, against 90 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, On-road stress (speed × lanes), 50 km/h limit. Leading protections: Cycleway/Sidewalk, Separated cycle track.

---

39. SEVILLE ES · INDEX 78.0 · INTERVAL 24–55 · TIER 1 (LOWER ASSESSED RISK) · TIER STABILITY 100%

Seville holds an uneven profile — strongest on journey risk (rank 16), weakest on street-network risk (rank 109). Streets: mean comparable risk 58.6 per km over 1,310 street-km (dimension rank 109/219), with 10.0% of the network in the highest-risk band. Journeys: 210 standardised trips average -13.9 (rank 16/219; 90th percentile -1.2). Access: composite low-stress accessibility 0.628 (rank 29/219); 4% of 753,142 residents are locked to high-stress streets at the central threshold, against 1,181 km of low-stress network. Its routed journeys score far below its street average — a working low-risk sub-network the router can exploit (escape-valve rank 14 of 219). Leading risk drivers on its streets: Motor-traffic volume, Score cap, Heavy-vehicle access. Leading protections: Cycleway/Sidewalk, Separated cycle track.

---

201. SIBIU RO · INDEX 21.7 · INTERVAL 198–201 · TIER 3 (HIGHER ASSESSED RISK) · TIER STABILITY 100%

Sibiu holds a balanced profile. Streets: mean comparable risk 71.0 per km over 565 street-km (dimension rank 198/219), with 19.5% of the network in the highest-risk band. Journeys: 210 standardised trips average 46.3 (rank 196/219; 90th percentile 77.1). Access: composite low-stress accessibility 0.208 (rank 191/219); 95% of 137,639 residents are locked to high-stress streets at the central threshold, against 189 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, On-road stress (speed × lanes), 50 km/h limit. Leading protections: Cycleway/Sidewalk, Signed ≤30 km/h zone.

---

191. SINTRA PT · INDEX 28.2 · INTERVAL 181–198 · TIER 3 (HIGHER ASSESSED RISK) · TIER STABILITY 100%

Sintra holds an uneven profile — strongest on street-network risk (rank 167), weakest on low-stress access (rank 212). Streets: mean comparable risk 66.2 per km over 2,116 street-km (dimension rank 167/219), with 11.5% of the network in the highest-risk band. Journeys: 210 standardised trips average 43.2 (rank 193/219; 90th percentile 75.8). Access: composite low-stress accessibility 0.144 (rank 212/219); 95% of 366,087 residents are locked to high-stress streets at the central threshold, against 1,053 km of low-stress network. Leading risk drivers on its streets: Incline, Motor-traffic volume, On-road stress (speed × lanes). Leading protections: Residential street, Cycleway/Sidewalk.

---

184. SOFIA BG · INDEX 31.8 · INTERVAL 182–189 · TIER 3 (HIGHER ASSESSED RISK) · TIER STABILITY 100%

Sofia holds a balanced profile. Streets: mean comparable risk 67.5 per km over 3,586 street-km (dimension rank 175/219), with 12.9% of the network in the highest-risk band. Journeys: 210 standardised trips average 33.1 (rank 174/219; 90th percentile 58.8). Access: composite low-stress accessibility 0.189 (rank 201/219); 90% of 782,516 residents are locked to high-stress streets at the central threshold, against 1,083 km of low-stress

network. Leading risk drivers on its streets: On-road stress (speed × lanes), Motor-traffic volume, 50 km/h limit. Leading protections: Residential street, Service Road.

---

211. SPLIT HR · INDEX 11.9 · INTERVAL 203–214 · TIER 3 (HIGHER ASSESSED RISK) · TIER STABILITY 100%

Split holds an uneven profile — strongest on low-stress access (rank 174), weakest on journey risk (rank 215). Streets: mean comparable risk 73.8 per km over 402 street-km (dimension rank 207/219), with 22.0% of the network in the highest-risk band. Journeys: 210 standardised trips average 85.6 (rank 215/219; 90th percentile 115.7). Access: composite low-stress accessibility 0.257 (rank 174/219); 97% of 86,044 residents are locked to high-stress streets at the central threshold, against 105 km of low-stress network. Its routed journeys score close to (or above) its street average — the router finds little refuge from the ordinary street network (escape-valve rank 219 of 219). Leading risk drivers on its streets: Motor-traffic volume, On-road stress (speed × lanes), 50 km/h limit. Leading protections: Pedestrian Zone, Service Road.

---

88. STAVANGER NO · INDEX 61.3 · INTERVAL 75–117 · TIER 2 (MIDDLE) · TIER STABILITY 78%

Stavanger holds an uneven profile — strongest on journey risk (rank 62), weakest on low-stress access (rank 136). Streets: mean comparable risk 58.0 per km over 1,533 street-km (dimension rank 87/219), with 7.6% of the network in the highest-risk band. Journeys: 203 standardised trips average 0.6 (rank 62/219; 90th percentile 26.9). Access: composite low-stress accessibility 0.352 (rank 136/219); 65% of 155,900 residents are locked to high-stress streets at the central threshold, against 961 km of low-stress network. Leading risk drivers on its streets: Incline, On-road stress (speed × lanes), Score cap. Leading protections: Cycleway/Sidewalk, Separated cycle track.

---

1. STOCKHOLM SE · INDEX 98.0 · INTERVAL 1–7 · TIER 1 (LOWER ASSESSED RISK) · TIER STABILITY 100%

Stockholm holds a balanced profile. Streets: mean comparable risk 44.2 per km over 2,227 street-km (dimension rank 1/219), with 1.5% of the network in the highest-risk band. Journeys: 210 standardised trips average -14.4 (rank 15/219; 90th percentile 9.0). Access: composite low-stress accessibility 0.774 (rank 1/219); 0% of 993,829 residents are locked to high-stress streets at the central threshold, against 2,858 km of low-stress network. Leading risk drivers on its streets: Score cap, Incline, Heavy-vehicle access. Leading protections: Cycleway/Sidewalk, Separated cycle track. Notes: top-ten overall; top-ten on at least one dimension.

---

34. STRASBOURG FR · INDEX 79.1 · INTERVAL 22–55 · TIER 1 (LOWER ASSESSED RISK) · TIER STABILITY 99%

Strasbourg holds an uneven profile — strongest on journey risk (rank 18), weakest on street-network risk (rank 115). Streets: mean comparable risk 58.9 per km over 788 street-km (dimension rank 115/219), with 10.6% of the network in the highest-risk band. Journeys: 210 standardised trips average -13.2 (rank 18/219; 90th percentile -1.1). Access: composite low-stress accessibility 0.664 (rank 20/219); 1% of 273,459 residents are locked to high-stress streets at the central threshold, against 656 km of low-stress network. Its routed journeys score far below its street average — a working low-risk sub-network the router can exploit (escape-valve rank 15 of 219). Leading risk drivers on its streets: Score cap, Motor-traffic volume, On-road stress (speed × lanes). Leading protections: Cycleway/Sidewalk, Separated cycle track.

133. STUTTGART DE · INDEX 53.5 · INTERVAL 113–152 · TIER 2 (MIDDLE) · TIER STABILITY 87%

Stuttgart holds an uneven profile — strongest on street-network risk (rank 96), weakest on journey risk (rank 169). Streets: mean comparable risk 56.0 per km over 1,875 street-km (dimension rank 96/219), with 10.8% of the network in the highest-risk band. Journeys: 210 standardised trips average 30.8 (rank 169/219; 90th percentile 44.4). Access: composite low-stress accessibility 0.404 (rank 113/219); 30% of 585,807 residents are locked to high-stress streets at the central threshold, against 1,961 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, Incline, On-road stress (speed × lanes). Leading protections: Car-Free Zone, Signed ≤30 km/h zone.

---

148. SZCZECIN PL · INDEX 48.9 · INTERVAL 137–159 · TIER 3 (HIGHER ASSESSED RISK) · TIER STABILITY 60%

Szczecin holds an uneven profile — strongest on journey risk (rank 115), weakest on street-network risk (rank 162). Streets: mean comparable risk 63.6 per km over 1,687 street-km (dimension rank 162/219), with 13.6% of the network in the highest-risk band. Journeys: 210 standardised trips average 14.4 (rank 115/219; 90th percentile 50.8). Access: composite low-stress accessibility 0.356 (rank 134/219); 81% of 383,763 residents are locked to high-stress streets at the central threshold, against 868 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, On-road stress (speed × lanes), Track Grade 4/5. Leading protections: Cycleway/Sidewalk, Separated cycle track. Notes: straddles a tier boundary under respecification.

---

68. SZEGED HU · INDEX 68.3 · INTERVAL 63–75 · TIER 1 (LOWER ASSESSED RISK) · TIER STABILITY 74%

Szeged holds a balanced profile. Streets: mean comparable risk 55.7 per km over 884 street-km (dimension rank 66/219), with 6.9% of the network in the highest-risk band. Journeys: 210 standardised trips average 4.4 (rank 76/219; 90th percentile 32.1). Access: composite low-stress accessibility 0.471 (rank 75/219); 19% of 167,859 residents are locked to high-stress streets at the central threshold, against 554 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, On-road stress (speed × lanes), Score cap. Leading protections: Signed ≤30 km/h zone, Cycleway/Sidewalk.

---

103. TALLINN EE · INDEX 59.3 · INTERVAL 53–130 · TIER 2 (MIDDLE) · TIER STABILITY 67%

Tallinn holds an uneven profile — strongest on journey risk (rank 31), weakest on low-stress access (rank 145). Streets: mean comparable risk 60.9 per km over 1,911 street-km (dimension rank 136/219), with 10.8% of the network in the highest-risk band. Journeys: 210 standardised trips average -9.4 (rank 31/219; 90th percentile 14.9). Access: composite low-stress accessibility 0.335 (rank 145/219); 83% of 357,104 residents are locked to high-stress streets at the central threshold, against 1,054 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, Score cap, On-road stress (speed × lanes). Leading protections: Cycleway/Sidewalk, Separated cycle track.

---

12. TAMPERE FI · INDEX 92.5 · INTERVAL 4–17 · TIER 1 (LOWER ASSESSED RISK) · TIER STABILITY 100%

Tampere holds a balanced profile. Streets: mean comparable risk 49.7 per km over 2,254 street-km (dimension rank 15/219), with 2.5% of the network in the highest-risk band. Journeys: 210 standardised trips average -30.6

(rank 4/219; 90th percentile -22.0). Access: composite low-stress accessibility 0.642 (rank 24/219); 3% of 227,142 residents are locked to high-stress streets at the central threshold, against 2,555 km of low-stress network. Its routed journeys score far below its street average — a working low-risk sub-network the router can exploit (escape-valve rank 4 of 219). Leading risk drivers on its streets: Score cap, Incline, MTB  $\geq 1$ . Leading protections: Cycleway/Sidewalk, Separated cycle track. Notes: top-ten on at least one dimension.

---

**197. TARGU MURES RO** · INDEX 23.9 · INTERVAL 189–201 · TIER 3 (HIGHER ASSESSED RISK) · TIER STABILITY 100%

Targu Mures holds a balanced profile. Streets: mean comparable risk 71.6 per km over 402 street-km (dimension rank 203/219), with 21.8% of the network in the highest-risk band. Journeys: 210 standardised trips average 43.0 (rank 192/219; 90th percentile 67.7). Access: composite low-stress accessibility 0.269 (rank 170/219); 92% of 124,537 residents are locked to high-stress streets at the central threshold, against 185 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, On-road stress (speed  $\times$  lanes), 50 km/h limit. Leading protections: Signed  $\leq 30$  km/h zone, Residential street.

---

**101. TERRASSA ES** · INDEX 59.6 · INTERVAL 85–126 · TIER 2 (MIDDLE) · TIER STABILITY 100%

Terrassa holds an uneven profile — strongest on low-stress access (rank 68), weakest on journey risk (rank 129). Streets: mean comparable risk 60.3 per km over 499 street-km (dimension rank 124/219), with 10.2% of the network in the highest-risk band. Journeys: 210 standardised trips average 17.0 (rank 129/219; 90th percentile 48.4). Access: composite low-stress accessibility 0.490 (rank 68/219); 2% of 234,350 residents are locked to high-stress streets at the central threshold, against 573 km of low-stress network. Leading risk drivers on its streets: Incline, Motor-traffic volume, On-road stress (speed  $\times$  lanes). Leading protections: Signed  $\leq 30$  km/h zone, Cycleway/Sidewalk.

---

**7. THE HAGUE NL** · INDEX 95.0 · INTERVAL 4–18 · TIER 1 (LOWER ASSESSED RISK) · TIER STABILITY 100%

The Hague holds a balanced profile. Streets: mean comparable risk 46.6 per km over 1,100 street-km (dimension rank 6/219), with 2.0% of the network in the highest-risk band. Journeys: 210 standardised trips average -11.4 (rank 23/219; 90th percentile 7.1). Access: composite low-stress accessibility 0.698 (rank 8/219); 4% of 508,908 residents are locked to high-stress streets at the central threshold, against 1,443 km of low-stress network. Leading risk drivers on its streets: Score cap, Motor-traffic volume, On-road stress (speed  $\times$  lanes). Leading protections: Cycleway/Sidewalk, Separated cycle track. Notes: top-ten overall; top-ten on at least one dimension.

---

**170. THESSALONIKI GR** · INDEX 39.9 · INTERVAL 134–187 · TIER 3 (HIGHER ASSESSED RISK) · TIER STABILITY 79%

Thessaloniki holds an uneven profile — strongest on street-network risk (rank 76), weakest on low-stress access (rank 198). Streets: mean comparable risk 60.6 per km over 9,099 street-km (dimension rank 76/219), with 3.0% of the network in the highest-risk band. Journeys: 210 standardised trips average 44.9 (rank 194/219; 90th percentile 61.6). Access: composite low-stress accessibility 0.194 (rank 198/219); 93% of 1,081,946 residents are locked to high-stress streets at the central threshold, against 5,704 km of low-stress network. Leading risk

drivers on its streets: Incline, On-road stress (speed × lanes), Heavy-vehicle access. Leading protections: Residential street, Track Grade 1.

---

33. TILBURG NL · INDEX 79.6 · INTERVAL 31–46 · TIER 1 (LOWER ASSESSED RISK) · TIER STABILITY 100%

Tilburg holds an uneven profile — strongest on low-stress access (rank 28), weakest on street-network risk (rank 77). Streets: mean comparable risk 54.0 per km over 957 street-km (dimension rank 77/219), with 10.7% of the network in the highest-risk band. Journeys: 210 standardised trips average -9.3 (rank 32/219; 90th percentile 5.9). Access: composite low-stress accessibility 0.630 (rank 28/219); 4% of 218,558 residents are locked to high-stress streets at the central threshold, against 960 km of low-stress network. Leading risk drivers on its streets: Score cap, Motor-traffic volume, On-road stress (speed × lanes). Leading protections: Cycleway/Sidewalk, Separated cycle track.

---

202. TIMISOARA RO · INDEX 19.7 · INTERVAL 200–204 · TIER 3 (HIGHER ASSESSED RISK) · TIER STABILITY 100%

Timisoara holds a balanced profile. Streets: mean comparable risk 73.8 per km over 896 street-km (dimension rank 201/219), with 18.2% of the network in the highest-risk band. Journeys: 210 standardised trips average 45.2 (rank 195/219; 90th percentile 76.8). Access: composite low-stress accessibility 0.186 (rank 203/219); 98% of 286,276 residents are locked to high-stress streets at the central threshold, against 205 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, On-road stress (speed × lanes), 50 km/h limit. Leading protections: Bicycle Designated, Cycleway/Sidewalk.

---

96. TORUŃ PL · INDEX 59.9 · INTERVAL 85–127 · TIER 2 (MIDDLE) · TIER STABILITY 96%

Toruń holds an uneven profile — strongest on journey risk (rank 77), weakest on street-network risk (rank 151). Streets: mean comparable risk 63.4 per km over 1,252 street-km (dimension rank 151/219), with 10.4% of the network in the highest-risk band. Journeys: 210 standardised trips average 4.9 (rank 77/219; 90th percentile 24.6). Access: composite low-stress accessibility 0.468 (rank 77/219); 97% of 190,856 residents are locked to high-stress streets at the central threshold, against 528 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, On-road stress (speed × lanes), 50 km/h limit. Leading protections: Bicycle Designated, Cycleway/Sidewalk.

---

139. TOULON FR · INDEX 51.0 · INTERVAL 118–155 · TIER 2 (MIDDLE) · TIER STABILITY 65%

Toulon holds an uneven profile — strongest on street-network risk (rank 88), weakest on journey risk (rank 159). Streets: mean comparable risk 58.6 per km over 534 street-km (dimension rank 88/219), with 6.8% of the network in the highest-risk band. Journeys: 210 standardised trips average 27.8 (rank 159/219; 90th percentile 57.1). Access: composite low-stress accessibility 0.322 (rank 151/219); 94% of 160,097 residents are locked to high-stress streets at the central threshold, against 234 km of low-stress network. Leading risk drivers on its streets: Incline, On-road stress (speed × lanes), 50 km/h limit. Leading protections: Residential street, Cycleway/Sidewalk. Notes: straddles a tier boundary under respecification.

---

31. TOULOUSE FR · INDEX 82.1 · INTERVAL 27–44 · TIER 1 (LOWER ASSESSED RISK) · TIER STABILITY 100%

Toulouse holds a balanced profile. Streets: mean comparable risk 49.2 per km over 1,717 street-km (dimension rank 27/219), with 6.1% of the network in the highest-risk band. Journeys: 210 standardised trips average 0.7 (rank 64/219; 90th percentile 14.6). Access: composite low-stress accessibility 0.611 (rank 36/219); 7% of 503,163 residents are locked to high-stress streets at the central threshold, against 1,277 km of low-stress network. Leading risk drivers on its streets: Score cap, Motor-traffic volume, Heavy-vehicle access. Leading protections: Signed  $\leq 30$  km/h zone, Cycleway/Sidewalk.

---

42. TOURS FR · INDEX 76.7 · INTERVAL 32–65 · TIER 1 (LOWER ASSESSED RISK) · TIER STABILITY 99%

Tours holds an uneven profile — strongest on street-network risk (rank 31), weakest on journey risk (rank 121). Streets: mean comparable risk 48.8 per km over 606 street-km (dimension rank 31/219), with 7.4% of the network in the highest-risk band. Journeys: 210 standardised trips average 15.6 (rank 121/219; 90th percentile 33.2). Access: composite low-stress accessibility 0.612 (rank 35/219); 15% of 130,355 residents are locked to high-stress streets at the central threshold, against 403 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, Heavy-vehicle access, Primary-road exposure. Leading protections: Signed  $\leq 30$  km/h zone, Signed bike network.

---

189. TRIESTE IT · INDEX 28.8 · INTERVAL 188–194 · TIER 3 (HIGHER ASSESSED RISK) · TIER STABILITY 100%

Trieste holds a balanced profile. Streets: mean comparable risk 67.8 per km over 769 street-km (dimension rank 180/219), with 14.9% of the network in the highest-risk band. Journeys: 210 standardised trips average 39.0 (rank 183/219; 90th percentile 87.5). Access: composite low-stress accessibility 0.197 (rank 195/219); 94% of 187,765 residents are locked to high-stress streets at the central threshold, against 434 km of low-stress network. Leading risk drivers on its streets: Incline, Motor-traffic volume, On-road stress (speed  $\times$  lanes). Leading protections: Cycleway/Sidewalk, Car-Free Zone.

---

61. TRONDHEIM NO · INDEX 70.1 · INTERVAL 46–73 · TIER 1 (LOWER ASSESSED RISK) · TIER STABILITY 84%

Trondheim holds an uneven profile — strongest on journey risk (rank 40), weakest on low-stress access (rank 94). Streets: mean comparable risk 56.0 per km over 1,899 street-km (dimension rank 65/219), with 6.5% of the network in the highest-risk band. Journeys: 210 standardised trips average -7.0 (rank 40/219; 90th percentile 19.4). Access: composite low-stress accessibility 0.434 (rank 94/219); 11% of 205,269 residents are locked to high-stress streets at the central threshold, against 2,251 km of low-stress network. Leading risk drivers on its streets: Incline, Score cap, Track Grade 4/5. Leading protections: Cycleway/Sidewalk, Separated cycle track.

---

137. TURIN IT · INDEX 51.4 · INTERVAL 24–167 · TIER 2 (MIDDLE) · TIER STABILITY 44%

Turin holds an uneven profile — strongest on journey risk (rank 12), weakest on street-network risk (rank 194). Streets: mean comparable risk 69.6 per km over 1,742 street-km (dimension rank 194/219), with 20.1% of the network in the highest-risk band. Journeys: 210 standardised trips average -14.9 (rank 12/219; 90th percentile 2.3). Access: composite low-stress accessibility 0.393 (rank 118/219); 90% of 869,949 residents are locked to high-stress streets at the central threshold, against 785 km of low-stress network. Its routed journeys score far below its street average — a working low-risk sub-network the router can exploit (escape-valve rank 2 of 219). Leading risk drivers on its streets: Motor-traffic volume, On-road stress (speed  $\times$  lanes), 50 km/h

limit. Leading protections: Cycleway/Sidewalk, Separated cycle track. Notes: straddles a tier boundary under respecification.

---

19. TURKU FI · INDEX 88.4 · INTERVAL 6–24 · TIER 1 (LOWER ASSESSED RISK) · TIER STABILITY 100%

Turku holds a balanced profile. Streets: mean comparable risk 52.9 per km over 1,557 street-km (dimension rank 29/219), with 2.6% of the network in the highest-risk band. Journeys: 210 standardised trips average -25.9 (rank 5/219; 90th percentile -16.6). Access: composite low-stress accessibility 0.610 (rank 37/219); 25% of 190,529 residents are locked to high-stress streets at the central threshold, against 1,492 km of low-stress network. Its routed journeys score far below its street average — a working low-risk sub-network the router can exploit (escape-valve rank 5 of 219). Leading risk drivers on its streets: Score cap, Incline, On-road stress (speed × lanes). Leading protections: Cycleway/Sidewalk, Separated cycle track. Notes: top-ten on at least one dimension.

---

85. UPPSALA SE · INDEX 62.5 · INTERVAL 18–131 · TIER 2 (MIDDLE) · TIER STABILITY 47%

Uppsala holds an uneven profile — strongest on journey risk (rank 11), weakest on street-network risk (rank 172). Streets: mean comparable risk 64.7 per km over 3,938 street-km (dimension rank 172/219), with 15.0% of the network in the highest-risk band. Journeys: 210 standardised trips average -16.2 (rank 11/219; 90th percentile 22.7). Access: composite low-stress accessibility 0.463 (rank 83/219); 21% of 225,093 residents are locked to high-stress streets at the central threshold, against 2,654 km of low-stress network. Its routed journeys score far below its street average — a working low-risk sub-network the router can exploit (escape-valve rank 3 of 219). Leading risk drivers on its streets: Incline, Heavy-vehicle access, On-road stress (speed × lanes). Leading protections: Cycleway/Sidewalk, Separated cycle track. Notes: straddles a tier boundary under respecification.

---

14. UTRECHT NL · INDEX 91.8 · INTERVAL 11–22 · TIER 1 (LOWER ASSESSED RISK) · TIER STABILITY 100%

Utrecht holds a balanced profile. Streets: mean comparable risk 48.0 per km over 1,070 street-km (dimension rank 14/219), with 3.7% of the network in the highest-risk band. Journeys: 210 standardised trips average -13.2 (rank 17/219; 90th percentile 2.6). Access: composite low-stress accessibility 0.667 (rank 18/219); 5% of 351,191 residents are locked to high-stress streets at the central threshold, against 1,074 km of low-stress network. Leading risk drivers on its streets: Score cap, On-road stress (speed × lanes), Heavy-vehicle access. Leading protections: Cycleway/Sidewalk, Separated cycle track.

---

41. VALENCIA ES · INDEX 77.8 · INTERVAL 15–67 · TIER 1 (LOWER ASSESSED RISK) · TIER STABILITY 98%

Valencia holds an uneven profile — strongest on low-stress access (rank 12), weakest on street-network risk (rank 137). Streets: mean comparable risk 60.3 per km over 1,126 street-km (dimension rank 137/219), with 11.7% of the network in the highest-risk band. Journeys: 210 standardised trips average -10.5 (rank 28/219; 90th percentile 5.3). Access: composite low-stress accessibility 0.691 (rank 12/219); 3% of 874,661 residents are locked to high-stress streets at the central threshold, against 1,103 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, Score cap, Heavy-vehicle access. Leading protections: Cycleway/Sidewalk, Separated cycle track.

43. VALLADOLID ES · INDEX 76.4 · INTERVAL 37–60 · TIER 1 (LOWER ASSESSED RISK) · TIER STABILITY 100%

Valladolid holds an uneven profile — strongest on low-stress access (rank 30), weakest on street-network risk (rank 102). Streets: mean comparable risk 57.1 per km over 763 street-km (dimension rank 102/219), with 10.4% of the network in the highest-risk band. Journeys: 210 standardised trips average -6.4 (rank 42/219; 90th percentile 13.5). Access: composite low-stress accessibility 0.627 (rank 30/219); 1% of 324,563 residents are locked to high-stress streets at the central threshold, against 830 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, Score cap, Heavy-vehicle access. Leading protections: Cycleway/Sidewalk, Signed  $\leq 30$  km/h zone.

---

114. VALLETTA MT · INDEX 57.9 · INTERVAL 65–151 · TIER 2 (MIDDLE) · TIER STABILITY 74%

Valletta holds an uneven profile — strongest on street-network risk (rank 37), weakest on journey risk (rank 186). Streets: mean comparable risk 54.9 per km over 21 street-km (dimension rank 37/219), with 1.5% of the network in the highest-risk band. Journeys: 210 standardised trips average 40.7 (rank 186/219; 90th percentile 49.3). Access: composite low-stress accessibility 0.421 (rank 103/219); 0% of 954 residents are locked to high-stress streets at the central threshold, against 8 km of low-stress network. Leading risk drivers on its streets: On-road stress (speed  $\times$  lanes), 50 km/h limit, Incline. Leading protections: Pedestrian Zone, Signed  $\leq 30$  km/h zone.

---

6. VANTAA FI · INDEX 96.2 · INTERVAL 2–10 · TIER 1 (LOWER ASSESSED RISK) · TIER STABILITY 100%

Vantaa holds a balanced profile. Streets: mean comparable risk 49.1 per km over 1,426 street-km (dimension rank 13/219), with 1.6% of the network in the highest-risk band. Journeys: 210 standardised trips average -25.6 (rank 2/219; 90th percentile -1.9). Access: composite low-stress accessibility 0.697 (rank 10/219); 3% of 240,437 residents are locked to high-stress streets at the central threshold, against 2,142 km of low-stress network. Its routed journeys score far below its street average — a working low-risk sub-network the router can exploit (escape-valve rank 10 of 219). Leading risk drivers on its streets: Score cap, Incline, MTB  $\geq 1$ . Leading protections: Cycleway/Sidewalk, Separated cycle track. Notes: top-ten overall; top-ten on at least one dimension.

---

178. VARNA BG · INDEX 36.3 · INTERVAL 169–183 · TIER 3 (HIGHER ASSESSED RISK) · TIER STABILITY 100%

Varna holds a balanced profile. Streets: mean comparable risk 64.2 per km over 1,401 street-km (dimension rank 157/219), with 11.8% of the network in the highest-risk band. Journeys: 210 standardised trips average 34.0 (rank 176/219; 90th percentile 75.1). Access: composite low-stress accessibility 0.220 (rank 189/219); 86% of 241,288 residents are locked to high-stress streets at the central threshold, against 498 km of low-stress network. Leading risk drivers on its streets: On-road stress (speed  $\times$  lanes), Incline, Motor-traffic volume. Leading protections: Residential street, Service Road.

---

117. VENICE IT · INDEX 57.2 · INTERVAL 62–142 · TIER 2 (MIDDLE) · TIER STABILITY 74%

Venice holds an uneven profile — strongest on journey risk (rank 39), weakest on street-network risk (rank 158). Streets: mean comparable risk 65.9 per km over 1,029 street-km (dimension rank 158/219), with 10.1% of the network in the highest-risk band. Journeys: 165 standardised trips average -7.8 (rank 39/219; 90th percentile 25.5). Access: composite low-stress accessibility 0.367 (rank 130/219); 68% of 242,184 residents are locked to high-stress streets at the central threshold, against 474 km of low-stress network. Its routed journeys score far below its street average — a working low-risk sub-network the router can exploit (escape-valve rank 12 of 219). Leading risk drivers on its streets: Score cap, On-road stress (speed × lanes), Motor-traffic volume. Leading protections: Cycleway/Sidewalk, Separated cycle track. Notes: only 165/210 journeys routable (water/topology).

---

157. VERONA IT · INDEX 46.4 · INTERVAL 124–166 · TIER 3 (HIGHER ASSESSED RISK) · TIER STABILITY 65%

Verona holds an uneven profile — strongest on journey risk (rank 81), weakest on street-network risk (rank 174). Streets: mean comparable risk 66.0 per km over 1,286 street-km (dimension rank 174/219), with 14.6% of the network in the highest-risk band. Journeys: 210 standardised trips average 5.4 (rank 81/219; 90th percentile 38.0). Access: composite low-stress accessibility 0.298 (rank 158/219); 70% of 251,660 residents are locked to high-stress streets at the central threshold, against 614 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, On-road stress (speed × lanes), Heavy-vehicle access. Leading protections: Cycleway/Sidewalk, Separated cycle track. Notes: straddles a tier boundary under respecification.

---

49. VIENNA AT · INDEX 74.0 · INTERVAL 36–77 · TIER 1 (LOWER ASSESSED RISK) · TIER STABILITY 96%

Vienna holds an uneven profile — strongest on journey risk (rank 22), weakest on street-network risk (rank 129). Streets: mean comparable risk 53.8 per km over 3,523 street-km (dimension rank 129/219), with 18.1% of the network in the highest-risk band. Journeys: 210 standardised trips average -11.7 (rank 22/219; 90th percentile 5.3). Access: composite low-stress accessibility 0.596 (rank 39/219); 11% of 1,901,218 residents are locked to high-stress streets at the central threshold, against 3,106 km of low-stress network. Leading risk drivers on its streets: Legal Dismount, Motor-traffic volume, Score cap. Leading protections: Cycleway/Sidewalk, Signed ≤30 km/h zone.

---

90. VIGO ES · INDEX 61.0 · INTERVAL 76–127 · TIER 2 (MIDDLE) · TIER STABILITY 86%

Vigo holds an uneven profile — strongest on street-network risk (rank 54), weakest on journey risk (rank 156). Streets: mean comparable risk 52.1 per km over 1,171 street-km (dimension rank 54/219), with 8.5% of the network in the highest-risk band. Journeys: 210 standardised trips average 26.4 (rank 156/219; 90th percentile 54.4). Access: composite low-stress accessibility 0.435 (rank 93/219); 1% of 295,956 residents are locked to high-stress streets at the central threshold, against 1,039 km of low-stress network. Leading risk drivers on its streets: Incline, Motor-traffic volume, Heavy-vehicle access. Leading protections: Signed ≤30 km/h zone, Residential street.

---

156. VILNIUS LT · INDEX 46.6 · INTERVAL 149–162 · TIER 3 (HIGHER ASSESSED RISK) · TIER STABILITY 70%

Vilnius holds a balanced profile. Streets: mean comparable risk 61.6 per km over 3,462 street-km (dimension rank 130/219), with 9.3% of the network in the highest-risk band. Journeys: 210 standardised trips average 22.7

(rank 148/219; 90th percentile 47.6). Access: composite low-stress accessibility 0.281 (rank 167/219); 82% of 466,507 residents are locked to high-stress streets at the central threshold, against 1,643 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, On-road stress (speed × lanes), Incline. Leading protections: Bicycle Designated, Signed ≤30 km/h zone.

---

**45. VITORIA-GASTEIZ ES** · INDEX 75.7 · INTERVAL 39–56 · TIER 1 (LOWER ASSESSED RISK) · TIER STABILITY 100%

Vitoria-Gasteiz holds an uneven profile — strongest on journey risk (rank 34), weakest on street-network risk (rank 82). Streets: mean comparable risk 57.2 per km over 641 street-km (dimension rank 82/219), with 7.9% of the network in the highest-risk band. Journeys: 210 standardised trips average -9.0 (rank 34/219; 90th percentile 10.5). Access: composite low-stress accessibility 0.565 (rank 44/219); 5% of 270,907 residents are locked to high-stress streets at the central threshold, against 1,113 km of low-stress network. Leading risk drivers on its streets: Incline, Track Grade 4/5, Score cap. Leading protections: Cycleway/Sidewalk, Separated cycle track.

---

**111. WARSAW PL** · INDEX 58.2 · INTERVAL 67–137 · TIER 2 (MIDDLE) · TIER STABILITY 79%

Warsaw holds an uneven profile — strongest on journey risk (rank 43), weakest on street-network risk (rank 159). Streets: mean comparable risk 64.0 per km over 5,462 street-km (dimension rank 159/219), with 12.4% of the network in the highest-risk band. Journeys: 210 standardised trips average -6.2 (rank 43/219; 90th percentile 9.0). Access: composite low-stress accessibility 0.400 (rank 115/219); 91% of 1,750,470 residents are locked to high-stress streets at the central threshold, against 2,350 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, On-road stress (speed × lanes), 50 km/h limit. Leading protections: Cycleway/Sidewalk, Separated cycle track.

---

**176. WIESBADEN DE** · INDEX 37.1 · INTERVAL 165–188 · TIER 3 (HIGHER ASSESSED RISK) · TIER STABILITY 100%

Wiesbaden holds an uneven profile — strongest on low-stress access (rank 147), weakest on journey risk (rank 191). Streets: mean comparable risk 63.5 per km over 1,159 street-km (dimension rank 171/219), with 16.0% of the network in the highest-risk band. Journeys: 210 standardised trips average 42.9 (rank 191/219; 90th percentile 61.2). Access: composite low-stress accessibility 0.334 (rank 147/219); 74% of 269,254 residents are locked to high-stress streets at the central threshold, against 1,124 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, Track Grade 4/5, Incline. Leading protections: Car-Free Zone, Signed ≤30 km/h zone.

---

**80. WROCŁAW PL** · INDEX 65.0 · INTERVAL 66–99 · TIER 2 (MIDDLE) · TIER STABILITY 66%

Wrocław holds an uneven profile — strongest on journey risk (rank 61), weakest on street-network risk (rank 121). Streets: mean comparable risk 58.6 per km over 2,666 street-km (dimension rank 121/219), with 11.8% of the network in the highest-risk band. Journeys: 210 standardised trips average -0.5 (rank 61/219; 90th percentile 13.8). Access: composite low-stress accessibility 0.480 (rank 70/219); 94% of 596,147 residents are locked to high-stress streets at the central threshold, against 1,374 km of low-stress network. Leading risk

drivers on its streets: Motor-traffic volume, On-road stress (speed × lanes), Track Grade 4/5. Leading protections: Signed ≤30 km/h zone, Bicycle Designated. Notes: straddles a tier boundary under respecification.

---

**161. WUPPERTAL DE · INDEX 43.8 · INTERVAL 147–177 · TIER 3 (HIGHER ASSESSED RISK) · TIER STABILITY 88%**

Wuppertal holds an uneven profile — strongest on street-network risk (rank 117), weakest on journey risk (rank 187). Streets: mean comparable risk 58.4 per km over 1,462 street-km (dimension rank 117/219), with 11.3% of the network in the highest-risk band. Journeys: 210 standardised trips average 41.2 (rank 187/219; 90th percentile 82.8). Access: composite low-stress accessibility 0.321 (rank 152/219); 44% of 328,209 residents are locked to high-stress streets at the central threshold, against 1,129 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, Incline, On-road stress (speed × lanes). Leading protections: Signed ≤30 km/h zone, Car-Free Zone.

---

**151. ZAGREB HR · INDEX 47.7 · INTERVAL 136–164 · TIER 3 (HIGHER ASSESSED RISK) · TIER STABILITY 52%**

Zagreb holds an uneven profile — strongest on street-network risk (rank 119), weakest on low-stress access (rank 181). Streets: mean comparable risk 61.1 per km over 2,595 street-km (dimension rank 119/219), with 8.6% of the network in the highest-risk band. Journeys: 210 standardised trips average 15.8 (rank 123/219; 90th percentile 45.5). Access: composite low-stress accessibility 0.242 (rank 181/219); 91% of 525,428 residents are locked to high-stress streets at the central threshold, against 1,162 km of low-stress network. Leading risk drivers on its streets: Incline, Motor-traffic volume, On-road stress (speed × lanes). Leading protections: Bicycle Designated, Cycleway/Sidewalk. Notes: straddles a tier boundary under respecification.

---

**47. ZARAGOZA ES · INDEX 75.0 · INTERVAL 44–56 · TIER 1 (LOWER ASSESSED RISK) · TIER STABILITY 100%**

Zaragoza holds a balanced profile. Streets: mean comparable risk 56.1 per km over 1,679 street-km (dimension rank 61/219), with 5.0% of the network in the highest-risk band. Journeys: 210 standardised trips average -0.6 (rank 58/219; 90th percentile 25.0). Access: composite low-stress accessibility 0.552 (rank 49/219); 12% of 749,796 residents are locked to high-stress streets at the central threshold, against 2,848 km of low-stress network. Leading risk drivers on its streets: Incline, Motor-traffic volume, Heavy-vehicle access. Leading protections: Cycleway/Sidewalk, Signed ≤30 km/h zone.

---

**135. ZURICH CH · INDEX 52.3 · INTERVAL 120–153 · TIER 2 (MIDDLE) · TIER STABILITY 77%**

Zurich holds an uneven profile — strongest on street-network risk (rank 98), weakest on journey risk (rank 165). Streets: mean comparable risk 55.3 per km over 1,052 street-km (dimension rank 98/219), with 12.0% of the network in the highest-risk band. Journeys: 210 standardised trips average 29.0 (rank 165/219; 90th percentile 41.2). Access: composite low-stress accessibility 0.375 (rank 129/219); 32% of 404,189 residents are locked to high-stress streets at the central threshold, against 852 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, Incline, On-road stress (speed × lanes). Leading protections: Signed ≤30 km/h zone, Car-Free Zone.

---

**93. ŁÓDŹ PL · INDEX 60.4 · INTERVAL 88–113 · TIER 2 (MIDDLE) · TIER STABILITY 93%**

Łódź holds an uneven profile — strongest on journey risk (rank 73), weakest on low-stress access (rank 117). Streets: mean comparable risk 59.2 per km over 2,949 street-km (dimension rank 108/219), with 9.2% of the network in the highest-risk band. Journeys: 210 standardised trips average 3.8 (rank 73/219; 90th percentile 23.6). Access: composite low-stress accessibility 0.394 (rank 117/219); 93% of 645,433 residents are locked to high-stress streets at the central threshold, against 1,093 km of low-stress network. Leading risk drivers on its streets: Motor-traffic volume, On-road stress (speed × lanes), 50 km/h limit. Leading protections: Cycleway/Sidewalk, Separated cycle track.

## 9. Robustness, sensitivity, and audits

Beyond the 2,000 respecifications, three targeted audits were run before publication. (1) *Formulation sensitivity*: the three adopted street-set rules (bypass exclusion, mandatory-sidepath treatment, threshold-averaged access) move the median city 6 places of 219 against the unmodified formulation (rank correlation 0.98), change 20 tier memberships, improve confident placement from 173 to 191, and leave the injury criterion unchanged — a correction of identified measurement artifacts, not a re-ranking. (2) *The first place*: Stockholm's lead was audited against tagging completeness (94% explicit speed limits), boundary composition (street density comparable to Copenhagen's), and its unusually low modelled traffic load; even tripling that load to the peer median leaves it on the podium. Its one-point margin over Helsinki is inside both intervals and is not a claim. (3) *A case study in artifact-hunting*: an inversion between two Romanian cities (Timișoara below Bucharest) was traced to three artifacts — an in-boundary ring road, undetected mandatory-sidepath carriageways, and a threshold cliff — whose corrections became the adopted formulation.

## 10. Limitations

- Junction design is under-measured; the validated severity signal is mid-block.
- The traffic layer is a congestion-calibrated model behaving as a road-class proxy, not a volume measurement.
- Country-adjusted outcome validation is inconclusive at  $n = 17$  panel cities in five countries; between-country comparisons inherit national reporting differences.
- Cities are administrative municipalities of very different sizes; the Canary Islands lie outside coverage; Romania is over-represented by design history.
- OSM completeness varies; corrections (legal defaults, sidepath law, parallel-track detection) reduce but cannot eliminate tagging-culture effects.
- Weather, theft, bicycle parking, modal share and culture are not measured at all.

## 11. Reproducibility

Built from a single pinned export of the production risk model with the traffic layer extended to all indexed cities (verified byte-identical on the prior baseline), via the pipeline slice → parallel-track detection → comparability corrections → three pillars → normalisation → 2,000 respecifications. Route sampling used the live read-only endpoint, throttled. Component weights are proprietary; no result in this report requires them. The public summary of this report is published as *Streets, Not Reputations*; the underlying tiers release and its criterion analysis are published separately.

### KEY REFERENCES

Copenhagenize Index 2025 (EIT Urban Mobility edition) · PeopleForBikes Bicycle Network Analysis · Furth et al., Level of Traffic Stress · Winters et al. 2013, Bike Score validation · Beecham et al. 2023 · Hardinghaus et al. 2021 · Elvik & Goel, safety-in-numbers · OECD/JRC Handbook on Constructing Composite Indicators · Eurostat Urban Audit (urb\_cpop1, urb\_ctrans) · WorldPop 2020 · OpenStreetMap contributors.